Complex Analysis I

Lecture notes following Stein–Shakarchi, Ch. 1–7 (through the prime number theorem)

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Chapter 1. Preliminaries to Complex Analysis

Overview

Complex numbers, the topology of $\mathbb{C}$, holomorphic functions (Cauchy–Riemann equations, power series), and curve integrals: if f has a primitive, $\int_\gamma f(z)\,dz = 0$ for every closed γ.

1 Complex numbers and the complex plane

1.1 Basic properties

A complex number is $z = x + iy$, $x, y \in \mathbb{R}$, $i^2 = -1$, with **real part** $x = \mathrm{Re}(z)$ and **imaginary part** $y = \mathrm{Im}(z)$; z is **purely imaginary** if $\mathrm{Re}\, z = 0$. $\mathbb{C}$ is identified with $\mathbb{R}^2$ via $z \leftrightarrow (x, y)$, with **real axis** and **imaginary axis**.

Treating all numbers as real and using $i^2 = -1$: if $z_1 = x_1 + iy_1$, $z_2 = x_2 + iy_2$,

$$z_1 + z_2 = (x_1 + x_2) + i(y_1 + y_2), \qquad z_1 z_2 = (x_1 x_2 - y_1 y_2) + i(x_1 y_2 + y_1 x_2).$$

These operations are commutative, associative and distributive. Addition is vector addition; multiplication by i is rotation by $\pi/2$.

The **absolute value** of $z = x + iy$ is $|z| = (x^2 + y^2)^{1/2}$, the euclidean distance from 0 to (x, y). The triangle inequality $|z + w| \le |z| + |w|$ holds, and also $|\mathrm{Re}(z)| \le |z|$, $|\mathrm{Im}(z)| \le |z|$ and

$$\big||z| - |w|\big| \le |z - w|,$$

which follows from $|z| \le |z - w| + |w|$ and $|w| \le |z - w| + |z|$.

The **complex conjugate** of $z = x + iy$ is $\overline{z} = x - iy$ (reflection across the real axis). z is real iff $z = \overline{z}$, purely imaginary iff $z = -\overline{z}$, and

$$\mathrm{Re}(z) = \frac{z + \overline{z}}{2}, \qquad \mathrm{Im}(z) = \frac{z - \overline{z}}{2i}, \qquad |z|^2 = z\overline{z}, \qquad \frac{1}{z} = \frac{\overline{z}}{|z|^2}\ (z \neq 0).$$

Any $z \neq 0$ has a **polar form** $z = re^{i\theta}$, $r = |z| > 0$, $e^{i\theta} = \cos\theta + i\sin\theta$; θ is the **argument** $\arg z$ (defined mod 2π), the counterclockwise angle from the positive real axis. If $z = re^{i\theta}$, $w = se^{i\varphi}$, then $zw = rse^{i(\theta+\varphi)}$: multiplication is a rotation composed with a dilation.

1.2 Convergence

$\{z_n\}$ **converges** to w, $w = \lim z_n$, if $|z_n - w| \to 0$, i.e. the points converge in $\mathbb{R}^2$, equivalently the real and imaginary parts converge. $\{z_n\}$ is a **Cauchy sequence** if $|z_n - z_m| \to 0$ as $n, m \to \infty$. $\{z_n\}$ is Cauchy iff its real and imaginary parts are, and $\mathbb{R}$ is complete; hence:

Theorem 1.1

$\mathbb{C}$, the complex numbers, is complete.

1.3 Sets in the complex plane

For $z_0 \in \mathbb{C}$, $r > 0$: the **open disc** $D_r(z_0) = \{|z - z_0| < r\}$, the **closed disc** $\overline{D}_r(z_0) = \{|z - z_0| \le r\}$, the circle $C_r(z_0) = \{|z - z_0| = r\}$, and the unit disc $\mathbb{D} = \{|z| < 1\}$.

For $\Omega \subset \mathbb{C}$: z_0 is an **interior point** if $D_r(z_0) \subset \Omega$ for some $r > 0$, and the **interior** is the set of these. Ω is **open** if all its points are interior, **closed** if $\Omega^c = \mathbb{C} - \Omega$ is open. z is a **limit point** if $z_n \to z$ for some $z_n \in \Omega$, $z_n \neq z$; closed sets are those containing their limit points. The **closure** $\overline{\Omega}$ adds the limit points; the **boundary** $\partial\Omega$ is closure minus interior. Ω is **bounded** if $|z| < M$ on Ω, with $\text{diam}(\Omega) = \sup_{z,w\in\Omega}|z - w|$, and **compact** if closed and bounded.

Theorem 1.2

The set $\Omega \subset \mathbb{C}$ is compact if and only if every sequence $\{z_n\} \subset \Omega$ has a subsequence that converges to a point in Ω.

Proof [Supplement]. If Ω is closed and bounded, Bolzano–Weierstrass (twice) gives a convergent subsequence, with limit in Ω since Ω is closed. Conversely, if Ω were unbounded, $z_n \in \Omega$ with $|z_n| > n$ has no convergent subsequence; if not closed, a limit point $z \notin \Omega$ is the limit of $z_n \in \Omega$, and every subsequence converges to $z \notin \Omega$. ■

An **open covering** of Ω is a family of open sets $\{U_\alpha\}$ with $\Omega \subset \bigcup_\alpha U_\alpha$.

Theorem 1.3

A set Ω is compact if and only if every open covering of Ω has a finite subcovering.

Proof [Supplement]. Suppose Ω is compact and the open covering $\{U_\alpha\}$ has no finite subcovering. Put Ω in a square Q_0 and quarter it; for some quarter Q_1, $\Omega \cap Q_1$ is not finitely covered. Repeating gives nested Q_n, $\text{diam}(Q_n) \to 0$, with $\Omega \cap Q_n$ not finitely covered. Points $z_n \in \Omega \cap Q_n$ form a Cauchy sequence with limit $w \in \Omega$, $w \in U_\alpha$ for some α; as U_α is open, $Q_n \subset U_\alpha$ for large n, a contradiction. Conversely, if every open covering has a finite subcovering, the discs $D_n(0)$ show Ω is bounded, and a limit point $z \notin \Omega$ would make $\{w : |w - z| > 1/n\}$ a covering with no finite subcovering. ■

Proposition 1.4

If $\Omega_1 \supset \Omega_2 \supset \cdots$ are non-empty compact sets with $\text{diam}(\Omega_n) \to 0$, there is a unique $w \in \mathbb{C}$ with $w \in \Omega_n$ for all n.

Proof. Choose $z_n \in \Omega_n$; $\text{diam}(\Omega_n) \to 0$ makes $\{z_n\}$ Cauchy, with limit w. Each Ω_n is closed and contains z_m for $m \geq n$, so $w \in \Omega_n$. A second such $w' \neq w$ would contradict $\text{diam}(\Omega_n) \to 0$. ■

An open Ω is **connected** if it is not a union of two disjoint non-empty open sets; a connected open set is a **region**. A closed F is connected if it is not a union of two disjoint non-empty closed sets. An open Ω is connected iff any two of its points can be joined by a curve in Ω.

2 Functions on the complex plane

2.1 Continuous functions

f on Ω is **continuous** at z_0 if for every $\epsilon > 0$ there is $\delta > 0$ with $|f(z) - f(z_0)| < \epsilon$ for $z \in \Omega$, $|z - z_0| < \delta$; equivalently $z_n \to z_0$ implies $f(z_n) \to f(z_0)$. Sums and products of continuous functions are continuous; f is continuous iff it is continuous in (x, y); then $|f|$ is continuous. f **attains a maximum** at z_0 if $|f(z)| \leq |f(z_0)|$ on Ω (reversed for a minimum).

Theorem 2.1

A continuous function on a compact set Ω is bounded and attains a maximum and minimum on Ω.

(The proof is the same as for functions of a real variable.)

2.2 Holomorphic functions

Definition (holomorphic function)

Let Ω be open and f complex-valued on Ω. f is **holomorphic at** $z_0 \in \Omega$ if the quotient

$$\frac{f(z_0+h)-f(z_0)}{h} \tag{1}$$

converges as $h \to 0$ ($h \in \mathbb{C}-\{0\}$, $z_0+h \in \Omega$); the limit is the **derivative** $f'(z_0)$. f is **holomorphic on** Ω if holomorphic at every point; on a closed C, if holomorphic on an open set containing C; **entire** if holomorphic on $\mathbb{C}$.

Here h approaches 0 from any direction. (Also: *regular*, *complex differentiable*.) Holomorphic functions turn out to be infinitely differentiable and *analytic*, unlike real-differentiable ones.

Example 1

$f(z)=z$ is holomorphic on any open set with $f'(z)=1$; any polynomial $p(z)=a_0+a_1z+\cdots+a_nz^n$ is entire, with $p'(z)=a_1+\cdots+na_nz^{n-1}$ (by Proposition 2.2).

Example 2

$1/z$ is holomorphic on any open set not containing 0, with $f'(z)=-1/z^2$.

Example 3

$f(z)=\overline{z}$ is not holomorphic: $\dfrac{f(z_0+h)-f(z_0)}{h}=\dfrac{\overline{h}}{h}$, which has no limit as $h\to 0$ (it is 1 for real h and -1 for purely imaginary h).

Clearly f is holomorphic at z_0 iff there is $a \in \mathbb{C}$ with

$$f(z_0+h)-f(z_0)-ah=h\psi(h), \tag{2}$$

where $\psi(h)\to 0$ as $h\to 0$; then $a=f'(z_0)$. So holomorphic functions are continuous. As for one real variable, using (2):

Proposition 2.2

If f and g are holomorphic in Ω, then:
(i) $f+g$ is holomorphic in Ω and $(f+g)'=f'+g'$.
(ii) fg is holomorphic in Ω and $(fg)'=f'g+fg'$.
(iii) If $g(z_0)\neq 0$, then f/g is holomorphic at z_0 and $(f/g)'=\dfrac{f'g-fg'}{g^2}$. Moreover, if $f:\Omega\to U$ and $g:U\to\mathbb{C}$ are holomorphic, the chain rule holds: $(g\circ f)'(z)=g'(f(z))f'(z)$ for all $z\in\Omega$.

Proof [Supplement](chain rule). Write $f(z+h)-f(z)=f'(z)h+h\psi_1(h)$ and $g(w+k)-g(w)=g'(w)k+k\psi_2(k)$ with $\psi_j\to 0$, $w=f(z)$. With $k=f(z+h)-f(z)\to 0$, $g(f(z+h))-g(f(z))=g'(w)k+k\psi_2(k)=g'(w)f'(z)h+h[g'(w)\psi_1(h)+(f'(z)+\psi_1(h))\psi_2(k)]$, and the bracket tends to 0. (i)–(iii) are similar. ∎

Complex-valued functions as mappings. $\overline{z}$ corresponds to the linear map $F:(x,y)\mapsto(x,-y)$, real-differentiable to all orders: real differentiability does not imply holomorphy. To $f=u+iv$ associate $F(x,y)=(u(x,y),v(x,y))$. F is **differentiable** at $P_0=(x_0,y_0)$ if there is a linear $J:\mathbb{R}^2\to\mathbb{R}^2$ with

$$\frac{|F(P_0+H)-F(P_0)-J(H)|}{|H|}\to 0 \quad \text{as } |H|\to 0,\ H\in\mathbb{R}^2; \tag{3}$$

then J is the Jacobian $J_F=\begin{pmatrix}\partial u/\partial x & \partial u/\partial y\\ \partial v/\partial x & \partial v/\partial y\end{pmatrix}$. If f is holomorphic at z_0, real $h=h_1$ in (1) gives $f'(z_0)=\frac{\partial f}{\partial x}(z_0)$ and $h=ih_2$ gives $f'(z_0)=\frac{1}{i}\frac{\partial f}{\partial y}(z_0)$. So $\frac{\partial f}{\partial x}=\frac{1}{i}\frac{\partial f}{\partial y}$; real and imaginary parts give the **Cauchy–Riemann equations**

$$\frac{\partial u}{\partial x}=\frac{\partial v}{\partial y}\qquad\text{and}\qquad\frac{\partial u}{\partial y}=-\frac{\partial v}{\partial x}.$$

Define the operators

$$\frac{\partial}{\partial z}=\frac{1}{2}\Big(\frac{\partial}{\partial x}+\frac{1}{i}\frac{\partial}{\partial y}\Big),\qquad \frac{\partial}{\partial \overline{z}}=\frac{1}{2}\Big(\frac{\partial}{\partial x}-\frac{1}{i}\frac{\partial}{\partial y}\Big).$$

Proposition 2.3

If f is holomorphic at z_0, then

$$\frac{\partial f}{\partial \overline{z}}(z_0)=0\qquad\text{and}\qquad f'(z_0)=\frac{\partial f}{\partial z}(z_0)=2\frac{\partial u}{\partial z}(z_0).$$

Also, if we write $F(x,y)=f(z)$, then F is differentiable in the sense of real variables, and $\det J_F(x_0,y_0)=|f'(z_0)|^2$.

Proof. The Cauchy–Riemann equations are equivalent to $\partial f/\partial\overline{z}=0$. Then $f'(z_0)=\frac{1}{2}(\frac{\partial f}{\partial x}(z_0)+\frac{1}{i}\frac{\partial f}{\partial y}(z_0))=\frac{\partial f}{\partial z}(z_0)$ and $\partial f/\partial z=2\partial u/\partial z$. For differentiability of F: with $H=(h_1,h_2)$, $h=h_1+ih_2$, the Cauchy–Riemann equations give

$$J_F(x_0,y_0)(H)=\Big(\frac{\partial u}{\partial x}-i\frac{\partial u}{\partial y}\Big)(h_1+ih_2)=f'(z_0)h,$$

identifying complex numbers with pairs of reals. Finally, by the Cauchy–Riemann equations,

$$\det J_F(x_0,y_0)=\frac{\partial u}{\partial x}\frac{\partial v}{\partial y}-\frac{\partial v}{\partial x}\frac{\partial u}{\partial y}=\Big(\frac{\partial u}{\partial x}\Big)^2+\Big(\frac{\partial u}{\partial y}\Big)^2=\Big|2\frac{\partial u}{\partial z}\Big|^2=|f'(z_0)|^2. \tag{4}$$

∎

The converse:

Theorem 2.4

Let $f=u+iv$ on an open Ω. If u,v are continuously differentiable and satisfy the Cauchy–Riemann equations on Ω, then f is holomorphic on Ω and $f'(z)=\partial f/\partial z$.

Proof. Write

$$u(x+h_1,y+h_2)-u(x,y)=\frac{\partial u}{\partial x}h_1+\frac{\partial u}{\partial y}h_2+|h|\psi_1(h), \qquad v(x+h_1,y+h_2)-v(x,y)=\frac{\partial v}{\partial x}h_1+\frac{\partial v}{\partial y}h_2+|h|\psi_2(h),$$

with $\psi_j(h)\to 0$ as $|h|\to 0$, $h=h_1+ih_2$ (differentiability follows from continuity of the partials). Using the Cauchy–Riemann equations,

$$f(z+h)-f(z)=\Big(\frac{\partial u}{\partial x}-i\frac{\partial u}{\partial y}\Big)(h_1+ih_2)+|h|\psi(h),$$

with $\psi=\psi_1+i\psi_2\to 0$. So f is holomorphic and $f'(z)=2\partial u/\partial z=\partial f/\partial z$. ∎

2.3 Power series

The complex exponential $e^z=\sum_{n\geq 0}z^n/n!$ converges absolutely for every z ($\sum|z|^n/n!=e^{|z|}$), uniformly on every disc. It is entire with termwise derivative $(e^z)'=\sum nz^{n-1}/n!=e^z$ (shown below). The geometric series $\sum z^n$ converges absolutely only for $|z|<1$, with sum $1/(1-z)$, since $\sum_{n=0}^{N}z^n=\frac{1-z^{N+1}}{1-z}$.

A **power series** is an expansion $\sum_{n=0}^{\infty}a_nz^n$ with $a_n\in\mathbb{C}$ (5). If it converges absolutely at z_0, it converges for all $|z|\leq|z_0|$.

Theorem 2.5

Given a power series $\sum_{n=0}^{\infty}a_nz^n$, there exists $0\leq R\leq\infty$ such that:
(i) If $|z|<R$ the series converges absolutely.
(ii) If $|z|>R$ the series diverges. Moreover, if we use the convention that $1/0=\infty$ and $1/\infty=0$, then R is given by Hadamard's formula

$$1/R=\limsup|a_n|^{1/n}.$$

R is the **radius of convergence** and $|z|<R$ the **disc of convergence**; $R=\infty$ for e^z and $R=1$ for the geometric series.

Proof. Let $L=1/R\neq 0,\infty$ (these cases are easy). If $|z|<R$, choose $\epsilon>0$ with $(L+\epsilon)|z|=r<1$; for large n, $|a_n|^{1/n}\leq L+\epsilon$, so $|a_n||z|^n\leq r^n$ and the series converges. If $|z|>R$, similarly a subsequence of terms is unbounded, so the series diverges. ∎

Remark

On the boundary $|z|=R$ either convergence or divergence can occur.

Further entire power series are

$$\cos z=\sum_{n=0}^{\infty}(-1)^n\frac{z^{2n}}{(2n)!}, \qquad \sin z=\sum_{n=0}^{\infty}(-1)^n\frac{z^{2n+1}}{(2n+1)!},$$

agreeing with the usual functions for real z, and related to the exponential by the **Euler formulas**

$$\cos z=\frac{e^{iz}+e^{-iz}}{2}, \qquad \sin z=\frac{e^{iz}-e^{-iz}}{2i}.$$

Theorem 2.6

The power series $f(z) = \sum_{n\geq 0} a_n z^n$ is holomorphic in its disc of convergence, and $f'(z) = \sum_{n\geq 0} n a_n z^{n-1}$, with the same radius of convergence.

Proof. Since $\lim n^{1/n} = 1$, $\limsup|a_n|^{1/n} = \limsup|na_n|^{1/n}$, so $\sum a_n z^n$ and $\sum n a_n z^n$, hence $\sum n a_n z^{n-1}$, have the same radius of convergence (Hadamard's formula).

Let $g(z) = \sum n a_n z^{n-1}$, let R be the radius of convergence, and $|z_0| < r < R$. Write $f(z) = S_N(z) + E_N(z)$ with $S_N(z) = \sum_{n=0}^{N} a_n z^n$, $E_N(z) = \sum_{n=N+1}^{\infty} a_n z^n$. If $|z_0 + h| < r$,

$$\frac{f(z_0+h)-f(z_0)}{h} - g(z_0) = \Big(\frac{S_N(z_0+h)-S_N(z_0)}{h} - S_N'(z_0)\Big) + (S_N'(z_0) - g(z_0)) + \frac{E_N(z_0+h)-E_N(z_0)}{h}.$$

Since $a^n - b^n = (a-b)(a^{n-1} + a^{n-2}b + \cdots + b^{n-1})$,

$$\Big|\frac{E_N(z_0+h)-E_N(z_0)}{h}\Big| \leq \sum_{n=N+1}^{\infty} |a_n| \Big|\frac{(z_0+h)^n - z_0^n}{h}\Big| \leq \sum_{n=N+1}^{\infty} |a_n| n r^{n-1},$$

the tail of a convergent series (g converges absolutely on $|z| < R$). Given $\epsilon > 0$, choose N_1 making this $< \epsilon$ for $N > N_1$, and N_2 with $|S_N'(z_0) - g(z_0)| < \epsilon$ for $N > N_2$. Fix $N > \max(N_1, N_2)$; as S_N is a polynomial, $\big|\frac{S_N(z_0+h)-S_N(z_0)}{h} - S_N'(z_0)\big| < \epsilon$ for $|h| < \delta$. Then $\big|\frac{f(z_0+h)-f(z_0)}{h} - g(z_0)\big| < 3\epsilon$ for $|h| < \delta$. ∎

Corollary 2.7

A power series is infinitely complex differentiable in its disc of convergence, the derivatives being termwise.

A power series centered at z_0 is $f(z) = \sum a_n (z - z_0)^n$; with $g(w) = \sum a_n w^n$, $f(z) = g(z - z_0)$, so everything about g transfers, e.g. $f'(z) = \sum n a_n (z - z_0)^{n-1}$.

Definition (analytic function)

f on an open Ω is **analytic** at z_0 if $f(z) = \sum a_n (z - z_0)^n$ near z_0 for a power series with positive radius; analytic on Ω if so at every point.

By Theorem 2.6, analytic functions are holomorphic; the converse (Chapter 2) justifies using the two terms interchangeably.

3 Integration along curves

A **parametrized curve** is a map $z : [a, b] \to \mathbb{C}$. It is **smooth** if z' exists, is continuous and $z'(t) \neq 0$ on $[a, b]$, one-sided at the end-points:

$$z'(a) = \lim_{h\to 0,\, h>0} \frac{z(a+h)-z(a)}{h}, \qquad z'(b) = \lim_{h\to 0,\, h<0} \frac{z(b+h)-z(b)}{h}.$$

It is **piecewise-smooth** if z is continuous and smooth on each $[a_k, a_{k+1}]$ for some $a = a_0 < \cdots < a_n = b$. Parametrizations $z : [a, b] \to \mathbb{C}$, $\tilde{z} : [c, d] \to \mathbb{C}$ are **equivalent** if $\tilde{z}(s) = z(t(s))$ for a C^1 bijection $t : [c, d] \to [a, b]$ with $t'(s) > 0$ (orientation preserved). The class determines a curve $\gamma \subset \mathbb{C}$, the image with its orientation; γ^- is reversed, e.g. $z^-(t) = z(b + a - t)$. The **end-points** $z(a)$, $z(b)$ are parametrization-independent. γ is **closed** if $z(a) = z(b)$, **simple** if $z(t) \neq z(s)$ for $s \neq t$ (for closed curves, except $\{s, t\} = \{a, b\}$). A *curve* means a piecewise-smooth curve.

$C_r(z_0)$ has the **positive** (counterclockwise) **orientation** $z(t) = z_0 + re^{it}$, $t \in [0, 2\pi]$, and the negative one $z_0 + re^{-it}$; C denotes a positively oriented circle.

Definition (integral along a curve; length)

For a smooth curve γ parametrized by $z : [a, b] \to \mathbb{C}$ and f continuous on γ,

$$\int_\gamma f(z)\,dz = \int_a^b f(z(t))z'(t)\,dt.$$

For piecewise-smooth γ, $\int_\gamma f(z)\,dz = \sum_{k=0}^{n-1} \int_{a_k}^{a_{k+1}} f(z(t))z'(t)\,dt$. The **length** is $\text{length}(\gamma) = \int_a^b |z'(t)|\,dt$, summed over the smooth parts.

These are independent of the parametrization: if $\tilde{z}$ is equivalent to z, by change of variables and the chain rule

$$\int_a^b f(z(t))z'(t)\,dt = \int_c^d f(z(t(s)))z'(t(s))t'(s)\,ds = \int_c^d f(\tilde{z}(s))\tilde{z}'(s)\,ds.$$

Proposition 3.1

Integration of continuous functions over curves satisfies the following properties:
(i) It is linear: if $\alpha, \beta \in \mathbb{C}$, then $\int_\gamma (\alpha f(z) + \beta g(z))\,dz = \alpha \int_\gamma f(z)\,dz + \beta \int_\gamma g(z)\,dz$.
(ii) If γ^- is γ with the reverse orientation, then $\int_\gamma f(z)\,dz = -\int_{\gamma^-} f(z)\,dz$.
(iii) One has the inequality $\left|\int_\gamma f(z)\,dz\right| \le \sup_{z\in\gamma}|f(z)| \cdot \text{length}(\gamma)$.

Proof. (i) follows from linearity of the Riemann integral.

(ii) [Supplement]: with $z^-(t) = z(a+b-t)$, $\int_a^b f(z^-(t))(z^-)'(t)\,dt = -\int_a^b f(z(a+b-t))z'(a+b-t)\,dt = -\int_a^b f(z(s))z'(s)\,ds$. (iii): $\left|\int_\gamma f(z)\,dz\right| \le \sup_{t\in[a,b]}|f(z(t))| \int_a^b |z'(t)|\,dt \le \sup_{z\in\gamma}|f(z)| \cdot \text{length}(\gamma)$. ■

A **primitive** for f on Ω is a function F holomorphic on Ω with $F'(z) = f(z)$ for all $z \in \Omega$.

Theorem 3.2

If a continuous function f has a primitive F in Ω, and γ is a curve in Ω that begins at w_1 and ends at w_2, then $\int_\gamma f(z)\,dz = F(w_2) - F(w_1)$.

Proof. If γ is smooth with parametrization $z : [a, b] \to \mathbb{C}$, $z(a) = w_1$, $z(b) = w_2$, then by the chain rule and the fundamental theorem of calculus

$$\int_\gamma f(z)\,dz = \int_a^b F'(z(t))z'(t)\,dt = \int_a^b \frac{d}{dt}F(z(t))\,dt = F(z(b)) - F(z(a)).$$

If γ is piecewise-smooth we get a telescoping sum: $\sum_{k=0}^{n-1}[F(z(a_{k+1})) - F(z(a_k))] = F(z(b)) - F(z(a))$. ■

Corollary 3.3

If γ is a closed curve in an open set Ω, and f is continuous and has a primitive in Ω, then $\oint_\gamma f(z)\,dz = 0$.

Immediate, as the end-points coincide. E.g. $1/z$ has no primitive in $\mathbb{C} - \{0\}$: for the unit circle $z(t) = e^{it}$,

$$\oint_C f(z)\,dz = \int_0^{2\pi} \frac{ie^{it}}{e^{it}}\,dt = 2\pi i \neq 0.$$

This innocent calculation lies at the heart of the theory.

Corollary 3.4

If f is holomorphic in a region Ω and $f' = 0$, then f is constant.

Proof. Fix $w_0 \in \Omega$ and join it to w by a curve γ (Ω connected). By Theorem 3.2, $f(w) - f(w_0) = \int_\gamma f'(z)\,dz = 0$. ■

Chapter 2. Cauchy's Theorem and Its Applications

Overview

Loosely, Cauchy's theorem says that if f is holomorphic in an open set Ω and $\gamma \subset \Omega$ is a closed curve whose interior is also in Ω, then

$$\int_\gamma f(z)\,dz = 0. \tag{1}$$

Defining the "interior" of a general curve is not easy, so we use "toy contours" with obvious interiors. It suffices that f have a primitive (Corollary 3.3), and primitives come from *Goursat's theorem*: $\int_T f(z)\,dz = 0$ for a triangle T with interior in Ω. This leads to the *Cauchy integral formula*

$$f(z) = \frac{1}{2\pi i}\int_C \frac{f(\zeta)}{\zeta - z}\,d\zeta \qquad (z \text{ inside } C),$$

and hence to regularity (derivatives of all orders, false for real variables), power series, analytic continuation, Liouville's theorem, the fundamental theorem of algebra and Morera's theorem.

1 Goursat's theorem

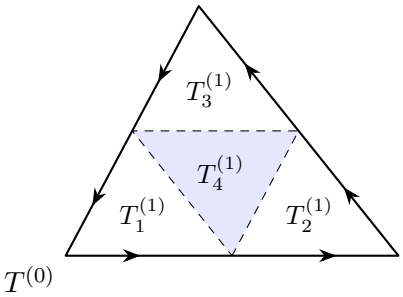


Figure. *Goursat's subdivision: dashed interior edges are traversed twice in opposite directions and cancel, so* $\int_{\partial T^{(0)}} f = \sum_{j=1}^4 \int_{\partial T_j^{(1)}} f$.

By Corollary 3.3, a primitive forces $\int_\gamma f = 0$ for closed γ; conversely, vanishing integrals over certain curves produce a primitive. Goursat's theorem needs only the existence of f', not its continuity.

Theorem 1.1

If Ω is an open set in $\mathbb{C}$, and $T \subset \Omega$ a triangle whose interior is also contained in Ω, then

$$\int_T f(z)\,dz = 0$$

whenever f is holomorphic in Ω.

Proof. Let $T^{(0)}$ be the positively oriented triangle, with diameter $d^{(0)}$ and perimeter $p^{(0)}$. Joining the midpoints of the sides gives four similar triangles $T_1^{(1)}, \dots, T_4^{(1)}$, consistently oriented; interior sides cancel, so

$$\int_{T^{(0)}} f(z)\,dz = \int_{T_1^{(1)}} f + \int_{T_2^{(1)}} f + \int_{T_3^{(1)}} f + \int_{T_4^{(1)}} f. \tag{2}$$

For some j, $\left|\int_{T^{(0)}} f\right| \le 4\left|\int_{T_j^{(1)}} f\right|$, for otherwise (2) would be contradicted. Call this triangle $T^{(1)}$; its diameter and perimeter are $d^{(1)} = \frac{1}{2}d^{(0)}$, $p^{(1)} = \frac{1}{2}p^{(0)}$. Repeating, we obtain triangles

$T^{(0)}, T^{(1)}, \dots, T^{(n)}, \dots$ with

$$\Big|\int_{T^{(0)}} f(z)\,dz\Big| \le 4^n \Big|\int_{T^{(n)}} f(z)\,dz\Big|, \qquad d^{(n)} = 2^{-n} d^{(0)}, \quad p^{(n)} = 2^{-n} p^{(0)}.$$

The solid closed triangles $\mathcal{T}^{(n)}$ bounded by $T^{(n)}$ are nested compact sets with diameters $\to 0$, so by Proposition 1.4 they share a unique point z_0. Since f is holomorphic at z_0,

$$f(z) = f(z_0) + f'(z_0)(z - z_0) + \psi(z)(z - z_0), \qquad \psi(z) \to 0 \text{ as } z \to z_0.$$

The constant $f(z_0)$ and the linear function $f'(z_0)(z - z_0)$ have primitives, so by Corollary 3.3

$$\int_{T^{(n)}} f(z)\,dz = \int_{T^{(n)}} \psi(z)(z - z_0)\,dz. \tag{3}$$

z_0 is in the solid triangle $\mathcal{T}^{(n)}$ and z on its boundary, so $|z - z_0| \le d^{(n)}$, and by Proposition 3.1(iii)

$$\Big|\int_{T^{(n)}} f(z)\,dz\Big| \le \epsilon_n d^{(n)} p^{(n)} = \epsilon_n 4^{-n} d^{(0)} p^{(0)}, \qquad \epsilon_n = \sup_{z \in T^{(n)}} |\psi(z)| \to 0.$$

Hence $\left|\int_{T^{(0)}} f\right| \le 4^n \left|\int_{T^{(n)}} f\right| \le \epsilon_n d^{(0)} p^{(0)}$, and letting $n \to \infty$ concludes the proof. ∎

Corollary 1.2

If f is holomorphic in an open set Ω that contains a rectangle R and its interior, then $\int_R f(z)\,dz = 0$.

Proof. Split R by a diagonal into two triangles T_1, T_2, oriented consistently; then $\int_R f = \int_{T_1} f + \int_{T_2} f = 0$. ∎

2 Local existence of primitives and Cauchy's theorem in a disc

Motivation. Integrating along polygonal paths turns Goursat's theorem into a primitive on a disc.

Theorem 2.1

A holomorphic function in an open disc has a primitive in that disc.

Proof. Translate so that D is centered at 0. For $z \in D$ let γ_z be the polygonal path from 0 horizontally to $\mathrm{Re}(z)$ and then vertically to z, and define

$$F(z) = \int_{\gamma_z} f(w)\,dw.$$

For $z, z + h \in D$, in $F(z + h) - F(z) = \int_{\gamma_{z+h}} f - \int_{\gamma_z} f$ the common segments cancel; completing a square and a triangle and applying Goursat for triangles and rectangles leaves

$$F(z + h) - F(z) = \int_{\eta} f(w)\,dw,$$

η the segment from z to $z + h$. Since f is continuous at z, $f(w) = f(z) + \psi(w)$ with $\psi(w) \to 0$ as $w \to z$. Hence

$$F(z + h) - F(z) = f(z) \int_{\eta} dw + \int_{\eta} \psi(w)\,dw. \tag{4}$$

The constant 1 has primitive w, so the first integral is h (Theorem 3.2); and $\left|\int_\eta \psi(w)\,dw\right| \le \sup_{w \in \eta} |\psi(w)|\,|h|$. The supremum tends to 0 as $h \to 0$, so (4) gives $\lim_{h \to 0} \frac{F(z+h) - F(z)}{h} = f(z)$. ∎

So holomorphic functions have primitives *locally*; the same holds on "toy contours" below.

Theorem 2.2 (Cauchy's theorem for a disc)

If f is holomorphic in a disc, then $\int_\gamma f(z)\,dz = 0$ for any closed curve γ in that disc.

Proof. f has a primitive (Theorem 2.1); apply Corollary 3.3. ∎

Corollary 2.3

Suppose f is holomorphic in an open set containing the circle C and its interior. Then $\int_C f(z)\,dz = 0$.

Proof. Let D be the disc bounded by C. There is a slightly larger disc $D' \supset \overline{D}$ on which f is holomorphic; apply Cauchy's theorem in D'. ∎

Definition (toy contour)

A **toy contour** is a closed curve with obvious interior for which the construction of Theorem 2.1 works near the curve and its interior; its **positive orientation** keeps the interior on the left.

Circles, triangles and rectangles are toy contours, as is the **keyhole** Γ: a large and a small almost-complete circle joined by a narrow corridor. If f is holomorphic near Γ and its interior, it is holomorphic inside a slightly larger keyhole Γ' with $\overline{\operatorname{int}\Gamma} \subset \operatorname{int}\Gamma'$. Fix $z_0 \in \operatorname{int}\Gamma'$ and join it to z by a path γ_z of horizontal and vertical segments in $\operatorname{int}\Gamma'$; by Corollary 1.2 $F(z) = \int_{\gamma_z} f$ does not depend on the path, F is a primitive, and $\int_\Gamma f = 0$. In general, for a toy contour γ,

$$\int_\gamma f(z)\,dz = 0$$

whenever f is holomorphic in an open set containing γ and its interior. Other examples used later: rectangular and multiple keyholes, semicircle, indented semicircle, sector, parallelogram. Appendix B proves the Jordan curve theorem for piecewise-smooth curves and Cauchy's theorem in that setting.

3 Evaluation of some integrals

We show how some integrals may be evaluated using Cauchy's theorem; a systematic approach via residues is in Chapter 3.

Example 1

If $\xi \in \mathbb{R}$, then

$$e^{-\pi\xi^2} = \int_{-\infty}^{\infty} e^{-\pi x^2} e^{-2\pi i x\xi}\,dx, \tag{5}$$

i.e. $e^{-\pi x^2}$ is its own Fourier transform. For $\xi = 0$ this is $\int_{-\infty}^{\infty} e^{-\pi x^2}\,dx = 1$. Let $\xi > 0$, $f(z) = e^{-\pi z^2}$ (entire), and γ_R the positively oriented rectangle with vertices R, $R + i\xi$, $-R + i\xi$, $-R$. By Cauchy's theorem

$$\int_{\gamma_R} f(z)\,dz = 0. \tag{6}$$

The integral over the real segment is $\int_{-R}^{R} e^{-\pi x^2}\,dx \to 1$. The integral over the right vertical side is $I(R) = \int_0^\xi f(R+iy)\,i\,dy = \int_0^\xi e^{-\pi(R^2+2iRy-y^2)}\,i\,dy$, and $|I(R)| \le Ce^{-\pi R^2} \to 0$ (ξ fixed);

similarly for the left side. The top side contributes

$$\int_R^{-R} e^{-\pi(x+i\xi)^2}\,dx = -e^{\pi\xi^2}\int_{-R}^{R} e^{-\pi x^2}e^{-2\pi i x\xi}\,dx.$$

Letting $R\to\infty$, (6) gives $0 = 1 - e^{\pi\xi^2}\int_{-\infty}^{\infty} e^{-\pi x^2}e^{-2\pi i x\xi}\,dx$. If $\xi < 0$ use the symmetric rectangle in the lower half-plane. (This *shifting of the contour* has many applications.)

Example 2

$$\int_0^\infty \frac{1-\cos x}{x^2}\,dx = \frac{\pi}{2}.$$

Integrate $f(z) = (1-e^{iz})/z^2$ over the indented semicircle: $[-R,-\epsilon]$, $[\epsilon,R]$, the negatively oriented γ_ϵ^+ and the positively oriented γ_R^+. By Cauchy's theorem

$$\int_{-R}^{-\epsilon}\frac{1-e^{ix}}{x^2}\,dx + \int_{\gamma_\epsilon^+}\frac{1-e^{iz}}{z^2}\,dz + \int_\epsilon^R \frac{1-e^{ix}}{x^2}\,dx + \int_{\gamma_R^+}\frac{1-e^{iz}}{z^2}\,dz = 0.$$

Since $|(1-e^{iz})/z^2| \le 2/|z|^2$ in the upper half-plane, the integral over γ_R^+ tends to 0 as $R\to\infty$, so $\int_{|x|\ge\epsilon}\frac{1-e^{ix}}{x^2}\,dx = -\int_{\gamma_\epsilon^+}\frac{1-e^{iz}}{z^2}\,dz$. Now $f(z) = \frac{-iz}{z^2} + E(z)$ with E bounded near 0, and on γ_ϵ^+, $z = \epsilon e^{i\theta}$, $dz = i\epsilon e^{i\theta}d\theta$, so

$$\int_{\gamma_\epsilon^+}\frac{1-e^{iz}}{z^2}\,dz \to \int_\pi^0 (-i)(i)\,d\theta = -\pi \quad \text{as } \epsilon\to 0.$$

Taking real parts, $\int_{-\infty}^{\infty}\frac{1-\cos x}{x^2}\,dx = \pi$; the integrand is even.

4 Cauchy's integral formulas

Motivation. Applied to $f(\zeta)/(\zeta - z)$, Cauchy's theorem determines f inside a disc from its boundary values, giving infinite differentiability and Taylor series.

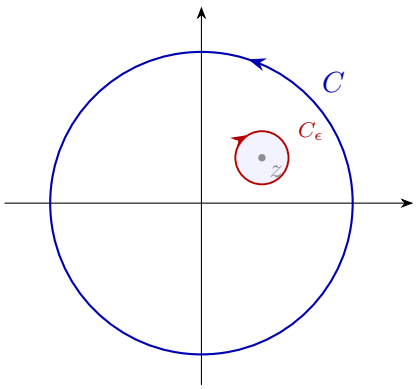


Figure. *$f(\zeta)/(\zeta - z)$ is holomorphic between C and C_ϵ, so both circles give the same integral; as $\epsilon\to 0$ the small one tends to $2\pi i f(z)$.*

Integral representations recover a function on a large set from its values on a smaller one (cf. the Poisson integral in the disc).

Theorem 4.1

Suppose f is holomorphic in an open set containing the closure of a disc D. If

$$f(z) = \frac{1}{2\pi i}\int_C \frac{f(\zeta)}{\zeta - z}\,d\zeta \qquad \text{for any point } z \in D.$$

Proof. Fix $z \in D$ and take the keyhole $\Gamma_{\delta,\epsilon}$ omitting z: C and a circle of radius ϵ about z, joined by a corridor of width δ. $F(\zeta) = f(\zeta)/(\zeta - z)$ is holomorphic away from z, so $\int_{\Gamma_{\delta,\epsilon}} F = 0$. As $\delta \to 0$ the corridor integrals cancel by continuity, leaving C (positive) and C_ϵ (negative). Write

$$F(\zeta) = \frac{f(\zeta) - f(z)}{\zeta - z} + \frac{f(z)}{\zeta - z}. \tag{8}$$

The first term is bounded (as f is holomorphic), so its integral over C_ϵ tends to 0 as $\epsilon \to 0$. And

$$\int_{C_\epsilon} \frac{f(z)}{\zeta - z}\,d\zeta = -f(z)\int_0^{2\pi} \frac{\epsilon i e^{-it}}{\epsilon e^{-it}}\,dt = -f(z)2\pi i.$$

So in the limit $0 = \int_C \frac{f(\zeta)}{\zeta - z}\,d\zeta - 2\pi i f(z)$. ∎

Remark

The same holds for other toy contours: for a positively oriented rectangle R, $f(z) = \frac{1}{2\pi i}\int_R \frac{f(\zeta)}{\zeta - z}\,d\zeta$ inside R (rectangular keyhole), and the integral vanishes for z outside R.

Corollary 4.2

If f is holomorphic in an open set Ω, then f has infinitely many complex derivatives in Ω. Moreover, if $C \subset \Omega$ is a circle whose interior is also contained in Ω, then

$$f^{(n)}(z) = \frac{n!}{2\pi i}\int_C \frac{f(\zeta)}{(\zeta - z)^{n+1}}\,d\zeta$$

for all z in the interior of C.

Proof. Induction on n; $n = 0$ is the Cauchy integral formula. Suppose f has up to $n-1$ complex derivatives with $f^{(n-1)}(z) = \frac{(n-1)!}{2\pi i}\int_C \frac{f(\zeta)}{(\zeta - z)^n}\,d\zeta$. For small h,

$$\frac{f^{(n-1)}(z+h) - f^{(n-1)}(z)}{h} = \frac{(n-1)!}{2\pi i}\int_C f(\zeta)\frac{1}{h}\Big[\frac{1}{(\zeta - z - h)^n} - \frac{1}{(\zeta - z)^n}\Big]d\zeta. \tag{9}$$

Since $A^n - B^n = (A-B)[A^{n-1} + A^{n-2}B + \cdots + AB^{n-2} + B^{n-1}]$, with $A = 1/(\zeta - z - h)$, $B = 1/(\zeta - z)$ the bracket in (9) is

$$\frac{h}{(\zeta - z - h)(\zeta - z)}[A^{n-1} + A^{n-2}B + \cdots + AB^{n-2} + B^{n-1}].$$

For small h, $z + h$ and z stay at a finite distance from C, so as $h \to 0$ the quotient converges to

$$\frac{(n-1)!}{2\pi i}\int_C f(\zeta)\frac{1}{(\zeta - z)^2}\frac{n}{(\zeta - z)^{n-1}}\,d\zeta = \frac{n!}{2\pi i}\int_C \frac{f(\zeta)}{(\zeta - z)^{n+1}}\,d\zeta.$$

∎

The formulas of Theorem 4.1 and Corollary 4.2 are the **Cauchy integral formulas**.

Corollary 4.3 (Cauchy inequalities)

If f is holomorphic in an open set that contains the closure of a disc D centered at z_0 and of radius R, then

$$|f^{(n)}(z_0)| \leq \frac{n!\,\|f\|_C}{R^n},$$

where $\|f\|_C = \sup_{z\in C}|f(z)|$ denotes the supremum of $|f|$ on the boundary circle C.

Proof. By the Cauchy integral formula,

$$|f^{(n)}(z_0)| = \Big|\frac{n!}{2\pi i}\int_C \frac{f(\zeta)}{(\zeta - z_0)^{n+1}}\,d\zeta\Big| = \frac{n!}{2\pi}\Big|\int_0^{2\pi} \frac{f(z_0 + Re^{i\theta})}{(Re^{i\theta})^{n+1}} Rie^{i\theta}\,d\theta\Big| \leq \frac{n!}{2\pi}\frac{\|f\|_C}{R^n}2\pi.$$

■

Theorem 4.4

Suppose f is holomorphic in an open set Ω. If D is a disc centered at z_0 and whose closure is contained in Ω, then f has a power series expansion at z_0

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n$$

for all $z \in D$, and the coefficients are given by $a_n = \dfrac{f^{(n)}(z_0)}{n!}$ for all $n \geq 0$.

Proof. Fix $z \in D$. By the Cauchy integral formula

$$f(z) = \frac{1}{2\pi i}\int_C \frac{f(\zeta)}{\zeta - z}\,d\zeta, \tag{10}$$

C the boundary of D. Write

$$\frac{1}{\zeta - z} = \frac{1}{\zeta - z_0 - (z - z_0)} = \frac{1}{\zeta - z_0}\cdot\frac{1}{1 - \left(\frac{z - z_0}{\zeta - z_0}\right)}. \tag{11}$$

Since $\zeta \in C$ and $z \in D$ is fixed, there is $0 < r < 1$ with $\left|\frac{z-z_0}{\zeta - z_0}\right| < r$, so

$$\frac{1}{1 - \left(\frac{z - z_0}{\zeta - z_0}\right)} = \sum_{n=0}^{\infty}\left(\frac{z - z_0}{\zeta - z_0}\right)^n, \tag{12}$$

uniformly for $\zeta \in C$. Interchanging sum and integral in (10)–(12),

$$f(z) = \sum_{n=0}^{\infty}\Big(\frac{1}{2\pi i}\int_C \frac{f(\zeta)}{(\zeta - z_0)^{n+1}}\,d\zeta\Big)(z - z_0)^n.$$

The formula for a_n follows from the Cauchy integral formulas for derivatives (or by differentiating the series). ■

This again shows infinite differentiability. The expansion converges in *any* disc centered at z_0 with closure in Ω; an entire function's expansion about 0 converges on all of $\mathbb{C}$.

Corollary 4.5 (Liouville's theorem)

If f is entire and bounded, then f is constant.

Proof. It suffices to show $f' = 0$ ($\mathbb{C}$ is connected; Corollary 3.4). For each $z_0 \in \mathbb{C}$ and $R > 0$ the Cauchy inequalities give $|f'(z_0)| \leq B/R$, B a bound for f. Let $R \to \infty$. ■

Corollary 4.6

Every non-constant polynomial $P(z) = a_n z^n + \cdots + a_0$ with complex coefficients has a root in $\mathbb{C}$.

Proof. If P has no roots, $1/P$ is a bounded entire function: with $a_n \neq 0$, $\frac{P(z)}{z^n} = a_n + (\frac{a_{n-1}}{z} + \cdots + \frac{a_0}{z^n})$, the bracket $\to 0$, so $|P(z)| \geq c|z|^n$ for $|z| > R$ ($c = |a_n|/2$), and P is continuous without roots on $|z| \leq R$, hence bounded below there. Liouville makes $1/P$ constant, a contradiction. ■

Corollary 4.7

Every polynomial $P(z) = a_n z^n + \cdots + a_0$ of degree $n \geq 1$ has precisely n roots in $\mathbb{C}$. If these roots are denoted by $w_1, \ldots, w_n$, then P can be factored as $P(z) = a_n(z - w_1)(z - w_2)\cdots(z - w_n)$.

Proof. P has a root w_1. Writing $z = (z - w_1) + w_1$ and expanding, $P(z) = b_n(z - w_1)^n + \cdots + b_1(z - w_1) + b_0$ with $b_n = a_n$; $P(w_1) = 0$ forces $b_0 = 0$, so $P(z) = (z - w_1)Q(z)$ with Q of degree $n - 1$. By induction $P(z) = c(z - w_1)\cdots(z - w_n)$, and comparing coefficients of z^n, $c = a_n$. ■

Finally, *analytic continuation*: a holomorphic function is determined by its values on arbitrarily small sets (Ω connected).

Theorem 4.8

Suppose f is a holomorphic function in a region Ω that vanishes on a sequence of distinct points with a limit point in Ω. Then f is identically 0.

Proof. Let $z_0 \in \Omega$ be a limit point of zeros w_k. First, $f \equiv 0$ near z_0: on a disc $D \subset \Omega$ about z_0, $f(z) = \sum a_n(z - z_0)^n$; otherwise with m the least index with $a_m \neq 0$, $f(z) = a_m(z - z_0)^m(1 + g(z - z_0))$, $g \to 0$, and $f(w_k) \neq 0$ for $w_k \neq z_0$ close to z_0, a contradiction.

Let U be the interior of $\{f = 0\}$, open and non-empty. U is closed in Ω: if $U \ni z_n \to z \in \Omega$, then $f(z) = 0$ and f vanishes near z by the first part. So $\Omega = U \cup (\Omega - U)$ with both open, and $U = \Omega$ by connectedness. ■

Corollary 4.9

If f, g are holomorphic in a region Ω and $f = g$ on a non-empty open subset (or on a sequence of distinct points with limit point in Ω), then $f = g$ on Ω.

If f, F are analytic in regions $\Omega \subset \Omega'$ and agree on Ω, F is an **analytic continuation** of f into Ω'; by the corollary it is unique.

5 Further applications

Motivation. Morera's converse, uniform limits, parameter integrals, reflection and polynomial approximation all follow from the integral formula.

5.1 Morera's theorem

Theorem 5.1 (Morera)

Suppose f is a continuous function in the open disc D such that for any triangle T contained in D, $\int_T f(z)\,dz = 0$. Then f is holomorphic.

Proof. By the proof of Theorem 2.1, f has a primitive F in D with $F' = f$. By regularity (Corollary 4.2), F is indefinitely (hence twice) complex differentiable, so f is holomorphic. ■

5.2 Sequences of holomorphic functions

Theorem 5.2

If $\{f_n\}_{n=1}^{\infty}$ is a sequence of holomorphic functions that converges uniformly to a function f in every compact subset of Ω, then f is holomorphic in Ω.

Proof. For a disc D, $\overline{D} \subset \Omega$, and a triangle $T \subset D$, $\int_T f_n = 0$ (Goursat). Uniform convergence makes f continuous and $\int_T f = \lim \int_T f_n = 0$, so f is holomorphic in D by Morera. ■

False for real variables: every continuous function on $[0,1]$ is a uniform limit of polynomials (Weierstrass).

Theorem 5.3

Under the hypotheses of the previous theorem, the sequence of derivatives $\{f_n'\}_{n=1}^{\infty}$ converges uniformly to f' on every compact subset of Ω.

Proof. Assume uniform convergence on Ω. Let $\Omega_\delta = \{z \in \Omega : \overline{D_\delta(z)} \subset \Omega\}$; it suffices that $f_n' \to f'$ uniformly on each Ω_δ, which follows by applying to $F = f_n - f$ the inequality

$$\sup_{z\in\Omega_\delta} |F'(z)| \le \frac{1}{\delta} \sup_{\zeta\in\Omega} |F(\zeta)|, \tag{13}$$

valid for F holomorphic in Ω. For $z \in \Omega_\delta$, $\overline{D_\delta(z)} \subset \Omega$ and by the Cauchy integral formula

$$|F'(z)| = \Big|\frac{1}{2\pi i}\int_{C_\delta(z)} \frac{F(\zeta)}{(\zeta - z)^2}\,d\zeta\Big| \le \frac{1}{2\pi} \sup_{\zeta\in\Omega}|F(\zeta)| \frac{1}{\delta^2} 2\pi\delta = \frac{1}{\delta}\sup_{\zeta\in\Omega}|F(\zeta)|.$$

■

Likewise $f_n^{(k)} \to f^{(k)}$ uniformly on compacta for every k. Theorem 5.2 constructs holomorphic functions as series $F(z) = \sum_{n=1}^{\infty} f_n(z)$ (14) converging uniformly on compacta, e.g. the zeta function (Chapter 6).

5.3 Holomorphic functions defined in terms of integrals

After a linear change of variables we take the interval $[0,1]$.

Theorem 5.4

Let $F(z,s)$ be defined for $(z,s) \in \Omega \times [0,1]$ where Ω is an open set in $\mathbb{C}$. Suppose F satisfies the following properties:
(i) $F(z,s)$ is holomorphic in z for each s.

(ii) F is continuous on $\Omega \times [0,1]$. Then the function f defined on Ω by $f(z) = \int_0^1 F(z,s)\,ds$ is holomorphic.

Proof. Use Riemann sums $f_n(z) = \frac{1}{n}\sum_{k=1}^n F(z,k/n)$, holomorphic by (i). F is uniformly continuous on $\overline{D}\times[0,1]$ ($\overline{D} \subset \Omega$), so for $\epsilon > 0$ there is $\delta > 0$ with $\sup_{z\in\overline{D}}|F(z,s_1)-F(z,s_2)| < \epsilon$ for $|s_1-s_2| < \delta$. For $n > 1/\delta$ and $z \in D$,

$$|f_n(z) - f(z)| = \Big|\sum_{k=1}^n \int_{(k-1)/n}^{k/n} [F(z,k/n) - F(z,s)]\,ds\Big| \le \sum_{k=1}^n \frac{\epsilon}{n} = \epsilon.$$

So $f_n \to f$ uniformly on D, and f is holomorphic in D by Theorem 5.2; hence in Ω. ∎

5.4 Schwarz reflection principle

Holomorphic functions are rigid (some cannot be continued beyond a disc; vanishing on a segment forces $f \equiv 0$), yet they extend by reflection. Let Ω be open and symmetric about $\mathbb{R}$, $\Omega^{\pm}$ its parts in the upper and lower half-planes, $I = \Omega \cap \mathbb{R}$; $\Omega = \Omega^+ \cup I \cup \Omega^-$.

Theorem 5.5 (Symmetry principle)

If f^+ and f^- are holomorphic functions in Ω^+ and Ω^- respectively, that extend continuously to I and $f^+(x) = f^-(x)$ for all $x \in I$, then the function f defined on Ω by

$$f(z) = \begin{cases} f^+(z) & z \in \Omega^+, \\ f^+(z) = f^-(z) & z \in I, \\ f^-(z) & z \in \Omega^- \end{cases}$$

is holomorphic on all of Ω.

Proof. f is continuous; holomorphy at I follows from Morera on a disc $D \subset \Omega$ centered on I. If a triangle $T \subset D$ misses I, $\int_T f = 0$. If an edge or vertex of T lies on I and the rest in the upper half-disc, raise it by ϵ to get T_ϵ with $\int_{T_\epsilon} f = 0$, and let $\epsilon \to 0$. If the interior of T meets I, split T into triangles with an edge or vertex on I. ∎

Theorem 5.6 (Schwarz reflection principle)

If f is holomorphic in Ω^+, extends continuously to I and is real on I, then there is F holomorphic in Ω with $F = f$ on Ω^+.

Proof. Set $F(z) = \overline{f(\overline{z})}$ on Ω^-. Near $\overline{z_0} \in \Omega^+$, $f(\overline{z}) = \sum a_n(\overline{z} - \overline{z_0})^n$, so $F(z) = \sum \overline{a_n}(z - z_0)^n$ is holomorphic in Ω^-. As f is real on I, F extends continuously to I and agrees with f; apply the symmetry principle. ∎

5.5 Runge's approximation theorem

Which compact $K \subset \mathbb{C}$ allow every function holomorphic near K to be a uniform limit of polynomials on K? Discs do (power series). $K = C$ does not: $\int_C 1/z = 2\pi i$ but $\int_C p = 0$ for polynomials p. The right condition is that K^c be connected (necessity by a variant of this example).

Theorem 5.7 (Runge)

Any function holomorphic in a neighborhood of a compact set K can be approximated uniformly on K by rational functions whose singularities are in K^c. If K^c is connected, any function holomorphic in a neighborhood of K can be approximated uniformly on K by polynomials.

The second part follows by "pushing the singularities to infinity". The key is a representation from the Cauchy formula for squares:

Lemma 5.8

Suppose f is holomorphic in an open set Ω, and $K \subset \Omega$ is compact. Then there exist finitely many segments $\gamma_1, \dots, \gamma_N$ in $\Omega - K$ such that

$$f(z) = \sum_{n=1}^{N} \frac{1}{2\pi i} \int_{\gamma_n} \frac{f(\zeta)}{\zeta - z}\, d\zeta \qquad \text{for all } z \in K. \tag{15}$$

Proof. Let $d = c \cdot d(K, \Omega^c)$, $c < 1/\sqrt{2}$, and take a grid of closed axis-parallel squares of side d. Let $\mathcal{Q} = \{Q_1, \dots, Q_M\}$ be those meeting K, with positively oriented boundaries ∂Q_m, and $\gamma_1, \dots, \gamma_N$ the sides not shared by two squares of $\mathcal{Q}$. By the choice of d, each γ_n lies in Ω and misses K (else it would be shared). For $z \in K$ off the square boundaries, $z \in Q_j$ for some j, and by Cauchy's theorem and formula

$$\frac{1}{2\pi i} \int_{\partial Q_m} \frac{f(\zeta)}{\zeta - z}\, d\zeta = \begin{cases} f(z) & m = j, \\ 0 & m \neq j, \end{cases} \qquad \text{so} \qquad f(z) = \sum_{m=1}^{M} \frac{1}{2\pi i} \int_{\partial Q_m} \frac{f(\zeta)}{\zeta - z}\, d\zeta.$$

Shared sides are traversed once each way and cancel, giving (15) for such z; as the γ_n lie in K^c, continuity extends (15) to all of K. ∎

Lemma 5.9

For any segment $\gamma \subset \Omega - K$, $\int_\gamma f(\zeta)/(\zeta - z)\, d\zeta$ is a uniform limit on K of rational functions with singularities on γ.

Proof. $\int_\gamma \frac{f(\zeta)}{\zeta - z}\, d\zeta = \int_0^1 \frac{f(\gamma(t))\gamma'(t)}{\gamma(t) - z}\, dt$; as γ misses K, the integrand is uniformly continuous on $K \times [0,1]$, and as in Theorem 5.4 its Riemann sums, rational functions with singularities on γ, converge uniformly on K. ∎

Lemmas 5.8 and 5.9 give the first part of Theorem 5.7. A rational function singular only at z_0 is a polynomial in $1/(z - z_0)$, so the second part follows from:

Lemma 5.10

If K^c is connected and $z_0 \notin K$, then the function $1/(z - z_0)$ can be approximated uniformly on K by polynomials.

Proof. Choose z_1 outside a large open disc D centered at 0 containing K. Then

$$\frac{1}{z - z_1} = -\frac{1}{z_1} \frac{1}{1 - z/z_1} = -\sum_{n=0}^{\infty} \frac{z^n}{z_1^{n+1}},$$

uniformly on K, so $1/(z - z_1)$ and its powers are uniform limits of polynomials on K. It remains to approximate $1/(z - z_0)$ uniformly on K by polynomials in $1/(z - z_1)$. As K^c is connected, there is

a curve $\gamma \subset K^c$ from z_0 to z_1; let $\rho = \frac{1}{2}d(K,\gamma) > 0$ and choose $w_0 = z_0, w_1, \ldots, w_\ell = z_1$ on γ with $|w_j - w_{j+1}| < \rho$.

Claim: if $w, w' \in \gamma$ with $|w - w'| < \rho$, then $1/(z-w)$ is a uniform limit on K of polynomials in $1/(z-w')$. Indeed

$$\frac{1}{z-w} = \frac{1}{z-w'} \cdot \frac{1}{1 - \frac{w-w'}{z-w'}} = \sum_{n=0}^{\infty} \frac{(w-w')^n}{(z-w')^{n+1}},$$

converging uniformly on K, as $|z - w'| \geq 2\rho$ gives $|w - w'|/|z - w'| < 1/2$. Applying the claim successively, $1/(z-w_0)$ is a uniform limit of polynomials in $1/(z-w_1)$, hence in $1/(z-w_2)$, ..., hence in $1/(z-w_\ell) = 1/(z-z_1)$. ■

Chapter 3. Meromorphic Functions and the Logarithm

Overview

Analytic functions are essentially characterized by their singularities (Riemann): holomorphic functions by their zeros, meromorphic ones by zeros and poles. We study removable singularities, poles and essential singularities; poles lead to the *calculus of residues*, built on $\int_C dz/z = 2\pi i$. The multivaluedness of $\int dz/z = \log z$ gives the *argument principle*, the open mapping theorem and the maximum principle; homotopies and simply connected domains define the logarithm.

1 Zeros and poles

A **point singularity** (**isolated singularity**) of f is a point z_0 where f is defined nearby but not at z_0. For $f(z) = z$ on the punctured plane, 0 is removable; $1/z$ grows to infinity at 0 (a pole); $e^{1/z}$ blows up along the positive reals, tends to 0 along the negative reals and oscillates boundedly along the imaginary axis.

z_0 is a **zero** of f if $f(z_0) = 0$. By analytic continuation, the zeros of a non-trivial holomorphic function are isolated.

Theorem 1.1

Suppose f is holomorphic in a connected open Ω, $f(z_0) = 0$, $f \not\equiv 0$. Then there are a neighborhood $U \subset \Omega$ of z_0, a non-vanishing holomorphic g on U and a unique $n \geq 1$ with $f(z) = (z - z_0)^n g(z)$ on U.

Proof. $f \not\equiv 0$ near z_0 (Theorem 4.8); with $f(z) = \sum_k a_k (z - z_0)^k$ near z_0 and n least with $a_n \neq 0$, $f(z) = (z - z_0)^n [a_n + a_{n+1}(z - z_0) + \cdots] = (z - z_0)^n g(z)$, $g(z_0) = a_n \neq 0$.

Uniqueness: if also $f(z) = (z - z_0)^m h(z)$ with $h(z_0) \neq 0$ and $m > n$, then $g(z) = (z - z_0)^{m-n} h(z)$ and letting $z \to z_0$ gives $g(z_0) = 0$, a contradiction; $m < n$ similarly forces $h(z_0) = 0$. So $m = n$ and $h = g$. ∎

f has a **zero of order** n (multiplicity n) at z_0; a zero of order 1 is **simple**. A **deleted neighborhood** of z_0 is $\{z : 0 < |z - z_0| < r\}$. A function f defined in a deleted neighborhood of z_0 has a **pole** at z_0 if $1/f$, defined to be zero at z_0, is holomorphic in a full neighborhood of z_0.

Theorem 1.2

If f has a pole at $z_0 \in \Omega$, then near z_0 there are a non-vanishing holomorphic h and a unique $n \geq 1$ with $f(z) = (z - z_0)^{-n} h(z)$.

Proof. By Theorem 1.1, $1/f(z) = (z - z_0)^n g(z)$ with g holomorphic and non-vanishing near z_0; take $h = 1/g$. ∎

n is the **order** of the pole (it describes the rate of growth); a pole of order 1 is **simple**.

Theorem 1.3

If f has a pole of order n at z_0, then

$$f(z) = \frac{a_{-n}}{(z - z_0)^n} + \frac{a_{-n+1}}{(z - z_0)^{n-1}} + \cdots + \frac{a_{-1}}{z - z_0} + G(z), \tag{1}$$

where G is a holomorphic function in a neighborhood of z_0.

Proof. Expand $h(z) = A_0 + A_1(z-z_0) + \cdots$ in Theorem 1.2; then $f(z) = (z-z_0)^{-n}(A_0 + A_1(z-z_0) + \cdots)$ has the stated form. ∎

$\frac{a_{-n}}{(z-z_0)^n} + \cdots + \frac{a_{-1}}{z-z_0}$ is the **principal part** of f at z_0, and $\operatorname{res}_{z_0} f = a_{-1}$ the **residue**. The other terms have primitives near z_0, so $\frac{1}{2\pi i}\int_C P(z)\,dz = a_{-1}$ for the principal part P and any circle C about z_0. For a simple pole, $\operatorname{res}_{z_0} f = \lim_{z\to z_0}(z-z_0)f(z)$.

Theorem 1.4

If f has a pole of order n at z_0, then

$$\operatorname{res}_{z_0} f = \lim_{z\to z_0} \frac{1}{(n-1)!}\Big(\frac{d}{dz}\Big)^{n-1}(z-z_0)^n f(z).$$

Proof. By (1), $(z-z_0)^n f(z) = a_{-n} + a_{-n+1}(z-z_0) + \cdots + a_{-1}(z-z_0)^{n-1} + G(z)(z-z_0)^n$; differentiate $n-1$ times and let $z \to z_0$. ∎

2 The residue formula

Motivation. Each singularity inside a contour contributes its residue, turning hard real integrals into a few coefficients.

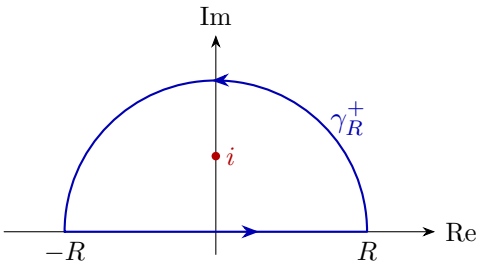


Figure. *The standard semicircular contour for $\int_{-\infty}^{\infty} \frac{dx}{1+x^2}$: only the pole at i lies inside, the arc contributes $O(1/R) \to 0$, and the residue formula gives $2\pi i \cdot \frac{1}{2i} = \pi$.*

Theorem 2.1

Suppose that f is holomorphic in an open set containing a circle C and its interior, except for a pole at z_0 inside C. Then

$$\int_C f(z)\,dz = 2\pi i \operatorname{res}_{z_0} f.$$

Proof. A keyhole avoiding the pole, with corridor width $\to 0$, gives $\int_C f = \int_{C_\epsilon} f$ for the circle C_ϵ of radius ϵ about z_0. By the Cauchy integral formula $\frac{1}{2\pi i}\int_{C_\epsilon} \frac{a_{-1}}{z-z_0}\,dz = a_{-1}$; by Corollary 4.2, $\int_{C_\epsilon} \frac{a_{-k}}{(z-z_0)^k}\,dz = 0$ for $k > 1$; $\int_{C_\epsilon} G = 0$ by Cauchy. By (1), $\int_{C_\epsilon} f = 2\pi i a_{-1}$. ∎

Corollary 2.2

Suppose that f is holomorphic in an open set containing a circle C and its interior, except for poles at the points $z_1, \ldots, z_N$ inside C. Then

$$\int_C f(z)\,dz = 2\pi i \sum_{k=1}^{N} \operatorname{res}_{z_k} f.$$

Proof. A multiple keyhole with a loop around each pole reduces $\int_C$ to a sum over small circles; apply Theorem 2.1. ∎

Corollary 2.3

If f is holomorphic in an open set containing a toy contour γ and its interior, except for poles $z_1, \dots, z_N$ inside γ, then

$$\int_\gamma f(z)\,dz = 2\pi i \sum_{k=1}^{N} \operatorname{res}_{z_k} f.$$

Here γ has positive orientation.

(Choose a keyhole appropriate to the toy contour and reduce to small circles around the poles.) This identity is the **residue formula**.

2.1 Examples

To compute $\int_{-\infty}^{\infty} f(x)\,dx$, choose toy contours γ_R with $\int_{\gamma_R} f \to \int_{-\infty}^{\infty} f(x)\,dx$ and evaluate $\int_{\gamma_R} f$ by residues; the decay of f guides the choice.

Example 1

$$\int_{-\infty}^{\infty} \frac{dx}{1+x^2} = \pi. \tag{2}$$

(Equivalently $\int P_y(x)\,dx = 1$ for the Poisson kernel $P_y(x) = \frac{1}{\pi}\frac{y}{y^2+x^2}$.) $f(z) = 1/((z-i)(z+i))$ has simple poles at $\pm i$, $\operatorname{res}_i f = 1/2i$. With $\gamma_R = [-R, R] \cup C_R^+$, $\int_{\gamma_R} f = \pi$ for large R, and $|f| \le B/|z|^2$ on C_R^+ gives $|\int_{C_R^+} f| \le \frac{B}{R^2}\pi R \to 0$. So $\int_{-\infty}^{\infty} f(x)\,dx = \pi$.

Example 2

An integral that plays an important role in Chapter 6:

$$\int_{-\infty}^{\infty} \frac{e^{ax}}{1+e^x}\,dx = \frac{\pi}{\sin \pi a}, \qquad 0 < a < 1.$$

Let $f(z) = e^{az}/(1+e^z)$ and γ_R the rectangle with sides on $\mathbb{R}$ and on $\operatorname{Im}(z) = 2\pi$, vertices $\pm R$, $\pm R + 2\pi i$. The only zero of the denominator inside is $z = \pi i$. There

$$(z-\pi i) f(z) = e^{az}\frac{z-\pi i}{e^z - e^{\pi i}}, \qquad \lim_{z\to\pi i} \frac{e^z - e^{\pi i}}{z-\pi i} = e^{\pi i} = -1,$$

so f has a simple pole at πi with residue $-e^{a\pi i}$, and

$$\int_{\gamma_R} f = -2\pi i e^{a\pi i}. \tag{3}$$

Let $I_R = \int_{-R}^{R} f(x)\,dx \to I$. The top side contributes $-e^{2\pi i a} I_R$. On $A_R = \{R+it : 0 \le t \le 2\pi\}$, $|\int_{A_R} f| \le \int_0^{2\pi} |\frac{e^{a(R+it)}}{1+e^{R+it}}|dt \le Ce^{(a-1)R} \to 0$ $(a<1)$; the left side is $\le Ce^{-aR} \to 0$ $(a>0)$. So (3) gives $I - e^{2\pi i a} I = -2\pi i e^{a\pi i}$, whence

$$I = -2\pi i \frac{e^{a\pi i}}{1 - e^{2\pi i a}} = \frac{2\pi i}{e^{\pi i a} - e^{-\pi i a}} = \frac{\pi}{\sin \pi a}.$$

Example 3

$1/\cosh \pi x$ is its own Fourier transform: with $\cosh z = \frac{e^z+e^{-z}}{2}$,

$$\int_{-\infty}^{\infty} \frac{e^{-2\pi i x\xi}}{\cosh \pi x}\, dx = \frac{1}{\cosh \pi \xi}.$$

Let $f(z) = e^{-2\pi i z\xi}/\cosh \pi z$ and γ_R the rectangle with vertices $\pm R$, $\pm R+2i$. The denominator vanishes when $e^{2\pi z} = -1$; inside the rectangle the poles are $\alpha = i/2$ and $\beta = 3i/2$. Since

$$(z-\alpha)f(z) = 2e^{-2\pi i z\xi} e^{\pi z} \frac{z-\alpha}{e^{2\pi z} - e^{2\pi \alpha}}, \qquad \lim_{z\to\alpha}(z-\alpha)f(z) = 2e^{-2\pi i \alpha\xi} e^{\pi\alpha} \frac{1}{2\pi e^{2\pi\alpha}} = \frac{e^{\pi\xi}}{\pi i},$$

f has simple poles at α, β with residues $e^{\pi\xi}/(\pi i)$, $-e^{3\pi\xi}/(\pi i)$. On $z = R + iy$, $0 \le y \le 2$: $|e^{-2\pi i z\xi}| \le e^{4\pi|\xi|}$ and $|\cosh \pi z| \ge \frac{1}{2}(e^{\pi R} - e^{-\pi R}) \to \infty$, so the vertical sides vanish in the limit. As $\cosh \pi z$ has period $2i$, the top side gives $-e^{4\pi\xi} I$. The residue formula gives

$$I - e^{4\pi\xi} I = 2\pi i\Big(\frac{e^{\pi\xi}}{\pi i} - \frac{e^{3\pi\xi}}{\pi i}\Big) = -2e^{2\pi\xi}(e^{\pi\xi} - e^{-\pi\xi}),$$

and since $1 - e^{4\pi\xi} = -e^{2\pi\xi}(e^{2\pi\xi} - e^{-2\pi\xi})$,

$$I = 2\frac{e^{\pi\xi} - e^{-\pi\xi}}{e^{2\pi\xi} - e^{-2\pi\xi}} = \frac{2}{e^{\pi\xi} + e^{-\pi\xi}} = \frac{1}{\cosh \pi\xi}.$$

A similar argument gives, for $0 < a < 1$, with $\sinh z = (e^z - e^{-z})/2$,

$$\int_{-\infty}^{\infty} e^{-2\pi i x\xi} \frac{\sin \pi a}{\cosh \pi x + \cos \pi a}\, dx = \frac{2\sinh 2\pi a\xi}{\sinh 2\pi\xi};$$

$a = 1/2$ is the case above. This yields the Poisson kernel for the strip, and is used in the two-squares theorem (Chapter 10).

3 Singularities and meromorphic functions

Motivation. Removable singularities, poles and essential singularities behave very differently; near an essential one f comes close to every value.

Let f be holomorphic in Ω except possibly at $z_0 \in \Omega$. If f can be defined at z_0 so that it becomes holomorphic in all of Ω, z_0 is a **removable singularity**.

Theorem 3.1 (Riemann's theorem on removable singularities)

Suppose that f is holomorphic in an open set Ω except possibly at a point z_0 in Ω. If f is bounded on $\Omega - \{z_0\}$, then z_0 is a removable singularity.

Proof. Take a small disc D centered at z_0 with $\overline{D} \subset \Omega$, boundary circle C positively oriented. We show that for $z \in D$, $z \ne z_0$,

$$f(z) = \frac{1}{2\pi i}\int_C \frac{f(\zeta)}{\zeta - z}\, d\zeta. \tag{4}$$

By Theorem 5.4 the right side is holomorphic in D and equals f for $z \ne z_0$. To prove (4), fix z and shrink the corridors of a multiple keyhole avoiding z and z_0:

$$\int_C \frac{f(\zeta)}{\zeta - z}\, d\zeta + \int_{\gamma_\epsilon} \frac{f(\zeta)}{\zeta - z}\, d\zeta + \int_{\gamma'_\epsilon} \frac{f(\zeta)}{\zeta - z}\, d\zeta = 0,$$

with γ_ϵ, γ'_ϵ negatively oriented circles of radius ϵ about z, z_0. As for the Cauchy formula, $\int_{\gamma_\epsilon} \frac{f(\zeta)}{\zeta - z}\, d\zeta = -2\pi i f(z)$; f is bounded and γ'_ϵ stays away from z, so $|\int_{\gamma'_\epsilon} \frac{f(\zeta)}{\zeta - z}\, d\zeta| \le C\epsilon$. Let $\epsilon \to 0$. ■

Corollary 3.2

Suppose that f has an isolated singularity at the point z_0. Then z_0 is a pole of f if and only if $|f(z)| \to \infty$ as $z \to z_0$.

Proof. If z_0 is a pole, $1/f$ has a zero there, so $|f| \to \infty$. Conversely, if $|f| \to \infty$, $1/f$ is bounded near z_0 with $1/f \to 0$, so it has a removable singularity with value 0: a pole. ■

Isolated singularities are **removable** (f bounded near z_0), **poles** ($|f| \to \infty$), or **essential** (all others, e.g. $e^{1/z}$ at 0), where f behaves erratically:

Theorem 3.3 (Casorati–Weierstrass)

If f is holomorphic in $D_r(z_0) - \{z_0\}$ with an essential singularity at z_0, then $f(D_r(z_0) - \{z_0\})$ is dense in $\mathbb{C}$.

Proof. Otherwise $|f(z) - w| > \delta$ on the punctured disc for some w, $\delta > 0$. Then $g = 1/(f - w)$ is bounded by $1/\delta$, so removable at z_0 (Theorem 3.1). If $g(z_0) \neq 0$, $f - w = 1/g$ is holomorphic at z_0; if $g(z_0) = 0$, $f - w$ has a pole. Either contradicts essentiality. ■

Picard proved much more: under these hypotheses f takes every complex value infinitely many times with at most one exception.

Definition (meromorphic function)

f on an open Ω is **meromorphic** if for a sequence $\{z_0, z_1, \dots\}$ with no limit points in Ω, (i) f is holomorphic in $\Omega - \{z_0, z_1, \dots\}$, and (ii) f has poles at the z_k.

If f is holomorphic for large z, let $F(z) = f(1/z)$. f has a **pole**, **essential singularity** or **removable singularity at infinity** if F has one at 0. A meromorphic f on $\mathbb{C}$ that is holomorphic or has a pole at infinity is **meromorphic in the extended complex plane**. The opening principle in its simplest form:

Theorem 3.4

The meromorphic functions in the extended complex plane are the rational functions.

Proof. $f(1/z)$ is holomorphic in a deleted neighborhood of 0, so f has finitely many poles $z_1, \dots, z_n$. Near z_k, $f = f_k + g_k$ with principal part f_k (a polynomial in $1/(z - z_k)$) and g_k holomorphic. Likewise $f(1/z) = \tilde{f}_\infty(z) + \tilde{g}_\infty(z)$ with $\tilde{f}_\infty$ a polynomial in $1/z$; $f_\infty(z) = \tilde{f}_\infty(1/z)$ is a polynomial. $H = f - f_\infty - \sum_k f_k$ is entire and bounded ($H(1/z)$ is bounded near 0), hence constant by Liouville; so f is rational. ■

Consequently, a rational function is determined up to a multiplicative constant by the locations and multiplicities of its zeros and poles.

The Riemann sphere. In $\mathbb{R}^3$ identify the XY-plane with $\mathbb{C}$, and let $\mathbb{S}$ be the sphere of radius $1/2$ centered at $(0, 0, 1/2)$, with north pole $N = (0, 0, 1)$. For $W \in \mathbb{S} - \{N\}$, the line NW meets the plane in $w = x + iy$, the **stereographic projection** of W; this bijection $\mathbb{S} - \{N\} \to \mathbb{C}$ is

$$x = \frac{X}{1 - Z}, \quad y = \frac{Y}{1 - Z}; \qquad X = \frac{x}{x^2 + y^2 + 1}, \quad Y = \frac{y}{x^2 + y^2 + 1}, \quad Z = \frac{x^2 + y^2}{x^2 + y^2 + 1}.$$

As $|w| \to \infty$, $W \to N$, the "point at infinity". The extended plane is the **Riemann sphere** $\mathbb{S}$ (*one-point compactification* of $\mathbb{C}$), and a meromorphic function on it maps $\mathbb{S}$ to itself, poles to N.

4 The argument principle and applications

$\log f(z) = \log|f(z)| + i \arg f(z)$ is multiple-valued, but f'/f is single-valued, and $\int_\gamma \frac{f'(z)}{f(z)}\,dz$ is the change of $\arg f$ along γ, determined by the zeros and poles inside. Logarithmic derivatives are additive:

$$\frac{(f_1 f_2)'}{f_1 f_2} = \frac{f_1'}{f_1} + \frac{f_2'}{f_2}, \qquad \frac{(\prod_{k=1}^N f_k)'}{\prod_{k=1}^N f_k} = \sum_{k=1}^N \frac{f_k'}{f_k}.$$

At a zero of order n, $f = (z-z_0)^n g$ with $g \neq 0$, so $\frac{f'}{f} = \frac{n}{z-z_0} + G$, $G = g'/g$ holomorphic: a simple pole with residue n. At a pole of order n, $f = (z-z_0)^{-n} h$ and $\frac{f'}{f} = \frac{-n}{z-z_0} + H$. The residue formula gives:

Theorem 4.1 (Argument principle)

Suppose f is meromorphic in an open set containing a circle C and its interior. If f has no poles and never vanishes on C, then

$$\frac{1}{2\pi i}\int_C \frac{f'(z)}{f(z)}\,dz = (\text{number of zeros of } f \text{ inside } C) - (\text{number of poles of } f \text{ inside } C),$$

where the zeros and poles are counted with their multiplicities.

Corollary 4.2

The above theorem holds for toy contours.

Three applications: Rouché's theorem (a continuity statement), the open mapping theorem, and the maximum modulus principle.

Theorem 4.3 (Rouché's theorem)

If f, g are holomorphic in an open set containing a circle C and its interior and $|f| > |g|$ on C, then f and $f+g$ have the same number of zeros inside C.

Proof. Let $f_t = f + tg$, $t \in [0,1]$, and n_t its number of zeros inside C. $|f| > |g|$ on C means $f_t \neq 0$ on C, so $n_t = \frac{1}{2\pi i}\int_C \frac{f_t'(z)}{f_t(z)}\,dz$ by the argument principle. The integrand is jointly continuous in (t,z), so n_t is a continuous integer-valued function on $[0,1]$, hence constant: $n_0 = n_1$. ■

A mapping is **open** if it maps open sets to open sets.

Theorem 4.4 (Open mapping theorem)

If f is holomorphic and non-constant in a region Ω, then f is open.

Proof. Let $w_0 = f(z_0)$; we show all w near w_0 are values. Write $f - w = (f - w_0) + (w_0 - w) = F + G$. Choose $\delta > 0$ with $\overline{D_\delta(z_0)} \subset \Omega$ and $f \neq w_0$ on $|z - z_0| = \delta$ (isolated zeros), and $\epsilon > 0$ with $|f - w_0| \geq \epsilon$ there. For $|w - w_0| < \epsilon$, $|F| > |G|$ on the circle, and Rouché gives $F + G$ a zero inside, as F has one (z_0). ■

The maximum of f in Ω means the maximum of $|f|$ in Ω.

Theorem 4.5 (Maximum modulus principle)

If f is a non-constant holomorphic function in a region Ω, then f cannot attain a maximum in Ω.

Proof. If $|f|$ had a maximum at z_0, the open set $f(D)$, D a small disc about z_0, would contain points of modulus $> |f(z_0)|$. ■

Corollary 4.6

Suppose that Ω is a region with compact closure $\overline{\Omega}$. If f is holomorphic on Ω and continuous on $\overline{\Omega}$ then

$$\sup_{z\in\Omega}|f(z)| \leq \sup_{z\in\overline{\Omega}-\Omega}|f(z)|.$$

Proof. $|f|$ attains its maximum on the compact set $\overline{\Omega}$, and not in Ω if f is non-constant; if f is constant the conclusion is trivial. ■

Remark

Boundedness of Ω is essential: on the open first quadrant, $F(z) = e^{-iz^2}$ is entire with $|F| = 1$ on both boundary half-lines, yet $F(re^{i\pi/4}) = e^{r^2}$ is unbounded.

5 Homotopies and simply connected domains

In which regions does a holomorphic function have a primitive? This global question underlies the general Cauchy theorem and the logarithm.

Definition (homotopic curves)

Curves $\gamma_0, \gamma_1 \subset \Omega$ on $[a, b]$ with common end-points α, β are **homotopic in** Ω if there are curves $\gamma_s \subset \Omega$, $0 \leq s \leq 1$, on $[a, b]$ with $\gamma_s(a) = \alpha$, $\gamma_s(b) = \beta$, equal to γ_0, γ_1 at $s = 0, 1$, and $\gamma_s(t)$ jointly continuous in $(s, t) \in [0, 1] \times [a, b]$.

Loosely, one curve can be deformed into the other continuously without leaving Ω.

Theorem 5.1

If f is holomorphic in Ω, then $\int_{\gamma_0} f(z)\,dz = \int_{\gamma_1} f(z)\,dz$ whenever the two curves γ_0 and γ_1 are homotopic in Ω.

Proof. Close curves with the same end-points give the same integral. $F(s, t) = \gamma_s(t)$ has compact image K. Some $\epsilon > 0$ makes every disc of radius 3ϵ about a point of K lie in Ω: otherwise $z_\ell \in K$, $w_\ell \in \Omega^c$ with $|z_\ell - w_\ell| < 1/\ell$ have a subsequence $z_{\ell_k}, w_{\ell_k} \to z \in K \cap \Omega^c$ (Ω^c closed), a contradiction. By uniform continuity choose δ with $\sup_t|\gamma_{s_1}(t) - \gamma_{s_2}(t)| < \epsilon$ for $|s_1 - s_2| < \delta$.

Fix such s_1, s_2. Choose discs $D_0, \ldots, D_n$ of radius 2ϵ covering both curves and consecutive points $z_0, \ldots, z_{n+1}$ on γ_{s_1}, $w_0, \ldots, w_{n+1}$ on γ_{s_2} with $z_i, z_{i+1}, w_i, w_{i+1} \in D_i$, $z_0 = w_0$, $z_{n+1} = w_{n+1}$ the common end-points. Let F_i be a primitive of f on D_i (Theorem 2.1). On $D_i \cap D_{i+1}$, $F_i - F_{i+1}$ is constant, so $F_{i+1}(z_{i+1}) - F_i(z_{i+1}) = F_{i+1}(w_{i+1}) - F_i(w_{i+1})$, i.e.

$$F_{i+1}(z_{i+1}) - F_{i+1}(w_{i+1}) = F_i(z_{i+1}) - F_i(w_{i+1}). \tag{5}$$

Hence

$$\int_{\gamma_{s_1}} f - \int_{\gamma_{s_2}} f = \sum_{i=0}^{n} [F_i(z_{i+1}) - F_i(z_i)] - \sum_{i=0}^{n} [F_i(w_{i+1}) - F_i(w_i)] = \sum_{i=0}^{n} [F_i(z_{i+1}) - F_i(w_{i+1}) - (F_i(z_i) - F_i(w_i))],$$

which telescopes by (5) to $F_n(z_{n+1}) - F_n(w_{n+1}) - (F_0(z_0) - F_0(w_0)) = 0$. Subdividing $[0,1]$ into intervals of length $< \delta$, finitely many applications take us from γ_0 to γ_1. ■

Definition (simply connected)

A region Ω is **simply connected** if any two curves in Ω with the same end-points are homotopic.

Example 1

A disc is simply connected: $\gamma_s(t) = (1-s)\gamma_0(t) + s\gamma_1(t)$ stays in D. So are rectangles and all open convex sets.

Example 2

The slit plane $\Omega = \mathbb{C} - (-\infty, 0]$ is simply connected: write $\gamma_j(t) = r_j(t)e^{i\theta_j(t)}$ with $r_j > 0$ and θ_j continuous, $|\theta_j(t)| < \pi$, and let $\gamma_s(t) = r_s(t)e^{i\theta_s(t)}$ with $r_s = (1-s)r_0 + sr_1$, $\theta_s = (1-s)\theta_0 + s\theta_1$; then $\gamma_s(t) \in \Omega$.

Example 3

With some effort one shows that the interior of a toy contour is simply connected, by dividing it into several subregions.

Example 4

The punctured plane $\mathbb{C} - \{0\}$ is not simply connected (proved after Corollary 5.3).

Theorem 5.2

Any holomorphic function in a simply connected domain has a primitive.

Proof. Fix $z_0 \in \Omega$ and let $F(z) = \int_\gamma f(w)\,dw$ over any curve $\gamma \subset \Omega$ from z_0 to z (independent of γ by Theorem 5.1). Then $F(z+h) - F(z) = \int_\eta f$ over the segment η from z to $z+h$, and as in Theorem 2.1, $F'(z) = f(z)$. ■

Corollary 5.3

If f is holomorphic in the simply connected region Ω, then $\int_\gamma f(z)\,dz = 0$ for any closed curve γ in Ω.

That $\mathbb{C} - \{0\}$ is not simply connected now follows rigorously since $\int_C dz/z = 2\pi i \neq 0$ over the unit circle.

6 The complex logarithm

For $z = re^{i\theta}$ one wants $\log z = \log r + i\theta$, but θ is only defined mod 2π: a continuous choice exists locally, not globally (winding once around 0 adds $2\pi i$). A single-valued logarithm needs a restricted domain: a **branch** (**sheet**).

Theorem 6.1

Suppose that Ω is simply connected with $1 \in \Omega$, and $0 \notin \Omega$. Then in Ω there is a branch of the logarithm $F(z) = \log_\Omega(z)$ so that (i) F is holomorphic in Ω, (ii) $e^{F(z)} = z$ for all $z \in \Omega$, (iii) $F(r) = \log r$ whenever r is a real number and near 1.

So each branch $\log_\Omega(z)$ extends the standard logarithm.

Proof. Let $\log_\Omega(z) = F(z) = \int_\gamma \frac{dw}{w}$, $\gamma \subset \Omega$ from 1 to z (path-independent). As in Theorem 5.2, $F'(z) = 1/z$: (i). For (ii), $\frac{d}{dz}(ze^{-F(z)}) = (1 - zF'(z))e^{-F(z)} = 0$, so $ze^{-F(z)}$ is constant (Corollary 3.4), equal to 1 at $z = 1$. For (iii), integrate along the real segment: $F(r) = \int_1^r \frac{dx}{x} = \log r$. ■

In the slit plane $\Omega = \mathbb{C} - (-\infty, 0]$ we have the **principal branch** $\log z = \log r + i\theta$ for $z = re^{i\theta}$, $|\theta| < \pi$: integrate along the segment from 1 to r and the arc from r to z,

$$\log z = \int_1^r \frac{dx}{x} + \int_{\text{arc}} \frac{dw}{w} = \log r + \int_0^\theta \frac{ire^{it}}{re^{it}}\, dt = \log r + i\theta.$$

In general $\log(z_1 z_2) \neq \log z_1 + \log z_2$: for $z_1 = z_2 = e^{2\pi i/3}$, $\log z_1 = \log z_2 = 2\pi i/3$, while $z_1 z_2 = e^{-2\pi i/3}$ and $\log(z_1 z_2) = -2\pi i/3$. For the principal branch,

$$\log(1+z) = z - \frac{z^2}{2} + \frac{z^3}{3} - \cdots = -\sum_{n=1}^{\infty} (-1)^n \frac{z^n}{n} \qquad \text{for } |z| < 1, \tag{6}$$

since both sides have derivative $1/(1+z)$ and vanish at $z = 0$.

For Ω simply connected with $1 \in \Omega$, $0 \notin \Omega$, and $\alpha \in \mathbb{C}$, define $z^\alpha = e^{\alpha \log z}$ (branch with $\log 1 = 0$). Then $1^\alpha = 1$, and for $\alpha = 1/n$, $(z^{1/n})^n = \prod_{k=1}^n e^{\frac{1}{n}\log z} = e^{\log z} = z$.

Theorem 6.2

If f is a nowhere vanishing holomorphic function in a simply connected region Ω, then there exists a holomorphic function g on Ω such that $f(z) = e^{g(z)}$.

g may be denoted $\log f(z)$; it determines a "branch" of that logarithm.

Proof. Fix $z_0 \in \Omega$ and let $g(z) = \int_\gamma \frac{f'(w)}{f(w)}\, dw + c_0$ over a path from z_0 to z, with $e^{c_0} = f(z_0)$. As in Theorem 2.1, $g' = f'/f$, so $\frac{d}{dz}(fe^{-g}) = 0$ and $fe^{-g} \equiv f(z_0)e^{-c_0} = 1$. ■

7 Fourier series and harmonic functions

Motivation. On a circle a power series is a Fourier series; this gives the mean value property for harmonic functions.

Let f be holomorphic in $D_R(z_0)$, $f(z) = \sum_{n\geq 0} a_n (z - z_0)^n$.

Theorem 7.1

The coefficients of the power series expansion of f are given by

$$a_n = \frac{1}{2\pi r^n} \int_0^{2\pi} f(z_0 + re^{i\theta}) e^{-in\theta}\, d\theta$$

for all $n \geq 0$ and $0 < r < R$. Moreover,

$$0 = \frac{1}{2\pi r^n} \int_0^{2\pi} f(z_0 + re^{i\theta}) e^{-in\theta}\, d\theta$$

whenever $n < 0$.

Proof. Since $f^{(n)}(z_0) = a_n n!$, the Cauchy integral formula over the circle γ of radius r about z_0 gives, with $\zeta = z_0 + re^{i\theta}$,

$$a_n = \frac{1}{2\pi i} \int_\gamma \frac{f(\zeta)}{(\zeta - z_0)^{n+1}}\, d\zeta = \frac{1}{2\pi i} \int_0^{2\pi} \frac{f(z_0 + re^{i\theta})}{(re^{i\theta})^{n+1}} rie^{i\theta}\, d\theta = \frac{1}{2\pi r^n} \int_0^{2\pi} f(z_0 + re^{i\theta}) e^{-in\theta}\, d\theta.$$

For $n < 0$ the same computation gives $\frac{1}{2\pi r^n} \int_0^{2\pi} f(z_0 + re^{i\theta}) e^{-in\theta}\, d\theta = \frac{1}{2\pi i} \int_\gamma \frac{f(\zeta)}{(\zeta - z_0)^{n+1}}\, d\zeta$, and since $-n > 0$, $f(\zeta)(\zeta - z_0)^{-n-1}$ is holomorphic in the disc; by Cauchy's theorem the integral vanishes. ■

So on a circle the Fourier coefficients of f vanish for $n < 0$ and equal $a_n r^n$ for $n \geq 0$. Since $a_0 = f(z_0)$:

Corollary 7.2 (Mean-value property)

If f is holomorphic in a disc $D_R(z_0)$, then

$$f(z_0) = \frac{1}{2\pi} \int_0^{2\pi} f(z_0 + re^{i\theta})\, d\theta \qquad \text{for any } 0 < r < R.$$

Corollary 7.3

If f is holomorphic in a disc $D_R(z_0)$, and $u = \mathrm{Re}(f)$, then

$$u(z_0) = \frac{1}{2\pi} \int_0^{2\pi} u(z_0 + re^{i\theta})\, d\theta \qquad \text{for any } 0 < r < R.$$

Take real parts: every harmonic u in a disc is the real part of a holomorphic function there.

Chapter 4. The Fourier Transform

Overview

For f on $\mathbb{R}$ with suitable regularity and decay, the Fourier transform and its inversion are

$$\hat{f}(\xi) = \int_{-\infty}^{\infty} f(x) e^{-2\pi i x\xi}\, dx, \qquad f(x) = \int_{-\infty}^{\infty} \hat{f}(\xi) e^{2\pi i x\xi}\, d\xi.$$

Holomorphic extension of f corresponds to very rapid decay of $\hat{f}$. If f continues to a horizontal strip with moderate decrease, $\hat{f}$ decays exponentially, and inversion and Poisson summation follow by contour integration. The *Paley–Wiener theorem*: $\hat{f}$ is supported in $[-M, M]$ iff f extends to an entire function with $|f(z)| \le Ae^{2\pi M|z|}$ (*exponential type*). The technique is shifting the real line within a strip, where $e^{-2\pi iz\xi}$ decays or grows according to the sign of $\xi \operatorname{Im}(z)$.

A function f is of **moderate decrease** if it is continuous and $|f(x)| \le A/(1+x^2)$ for some $A > 0$.

1 The class $\mathfrak{F}$

Book I proved inversion and Poisson summation for $|f(x)| \le \frac{A}{1+x^2}$, $|\hat{f}(\xi)| \le \frac{A'}{1+\xi^2}$ (e.g. $P_y(x) = \frac{1}{\pi}\frac{y}{y^2+x^2}$, $\hat{P}_y(\xi) = e^{-2\pi y|\xi|}$). Here is a class suited to complex analysis.

Definition (the classes $\mathfrak{F}_a$ and $\mathfrak{F}$)

For $a > 0$, $\mathfrak{F}_a$ is the class of functions f such that:
(i) f is holomorphic in the horizontal strip $S_a = \{z \in \mathbb{C} : |\operatorname{Im}(z)| < a\}$;
(ii) there is $A > 0$ with $|f(x+iy)| \le \dfrac{A}{1+x^2}$ for all $x \in \mathbb{R}$ and $|y| < a$. $\mathfrak{F}$ is the class of functions belonging to $\mathfrak{F}_a$ for some a.

So $\mathfrak{F}_a$ consists of holomorphic functions on S_a of moderate decay on horizontal lines, uniformly in $-a < y < a$. Examples: $e^{-\pi z^2} \in \mathfrak{F}_a$ for all a; $\frac{1}{\pi}\frac{c}{c^2+z^2} \in \mathfrak{F}_a$ for $0 < a < c$; $1/\cosh \pi z \in \mathfrak{F}_a$ for $|a| < 1/2$. By the Cauchy integral formulas, $f \in \mathfrak{F}_a$ implies $f^{(n)} \in \mathfrak{F}_b$, $0 < b < a$. (Most results hold with $A/(1+|x|^{1+\epsilon})$.)

2 Action of the Fourier transform on $\mathfrak{F}$

Motivation. Contour shifting in a strip proves inversion and Poisson summation without real-variable convergence issues.

All three theorems are proved by contour integration.

Theorem 2.1

If f belongs to the class $\mathfrak{F}_a$ for some $a > 0$, then $|\hat{f}(\xi)| \le Be^{-2\pi b|\xi|}$ for any $0 \le b < a$.

Proof. $b = 0$ is clear from moderate decrease. Let $0 < b < a$, $\xi > 0$, $g(z) = f(z)e^{-2\pi iz\xi}$, and shift the real line down by b using the rectangle with vertices $-R, R, R - ib, -R - ib$. The left vertical side is bounded by

$$\Big|\int_{-R-ib}^{-R} g(z)\, dz\Big| \le \int_0^b |f(-R-it)e^{-2\pi i(-R-it)\xi}|\, dt \le \int_0^b \frac{A}{R^2} e^{-2\pi t\xi}\, dt = O(1/R^2),$$

and similarly for the right side. By Cauchy's theorem, letting $R \to \infty$,

$$\hat{f}(\xi) = \int_{-\infty}^{\infty} f(x - ib) e^{-2\pi i(x-ib)\xi}\, dx, \tag{1}$$

so $|\hat{f}(\xi)| \le \int_{-\infty}^{\infty} \frac{A}{1+x^2} e^{-2\pi b\xi}\, dx \le B e^{-2\pi b\xi}$. For $\xi < 0$ shift the real line up by b. ■

The further f extends (larger a), the better the decay of $\hat{f}$.

Theorem 2.2

If $f \in \mathfrak{F}$, then the Fourier inversion holds, namely

$$f(x) = \int_{-\infty}^{\infty} \hat{f}(\xi) e^{2\pi i x\xi}\, d\xi \qquad \text{for all } x \in \mathbb{R}.$$

Lemma 2.3

If A is positive and B is real, then $\int_0^{\infty} e^{-(A+iB)\xi}\, d\xi = \dfrac{1}{A+iB}$.

Proof. $|e^{-(A+iB)\xi}| = e^{-A\xi}$, so the integral converges, and $\int_0^R e^{-(A+iB)\xi}\, d\xi = \left[-\frac{e^{-(A+iB)\xi}}{A+iB}\right]_0^R \to \frac{1}{A+iB}$. ■

Proof of Theorem 2.2. Split $\int_{-\infty}^{\infty} \hat{f}(\xi) e^{2\pi i x\xi} d\xi = \int_{-\infty}^{0} + \int_0^{\infty}$. Say $f \in \mathfrak{F}_a$, $0 < b < a$. By (1) and the lemma (the integrals converge absolutely),

$$\begin{aligned}\int_0^{\infty} \hat{f}(\xi) e^{2\pi i x\xi}\, d\xi &= \int_0^{\infty}\int_{-\infty}^{\infty} f(u-ib) e^{-2\pi i(u-ib)\xi} e^{2\pi i x\xi}\, du\, d\xi = \int_{-\infty}^{\infty} f(u-ib) \int_0^{\infty} e^{-2\pi i(u-ib-x)\xi}\, d\xi\, du \\ &= \int_{-\infty}^{\infty} f(u-ib) \frac{1}{2\pi b + 2\pi i(u-x)}\, du = \frac{1}{2\pi i}\int_{-\infty}^{\infty} \frac{f(u-ib)}{u-ib-x}\, du = \frac{1}{2\pi i}\int_{L_1} \frac{f(\zeta)}{\zeta - x}\, d\zeta,\end{aligned}$$

$L_1 = \{u - ib : u \in \mathbb{R}\}$, left to right. Similarly $\int_{-\infty}^{0} \hat{f}(\xi) e^{2\pi i x\xi} d\xi = -\frac{1}{2\pi i}\int_{L_2} \frac{f(\zeta)}{\zeta - x}\, d\zeta$, L_2 the line shifted up by b. For $x \in \mathbb{R}$ let γ_R be the rectangle with vertices $-R \pm ib$, $R \pm ib$; the residue of $f(\zeta)/(\zeta - x)$ at x is $f(x)$, so $f(x) = \frac{1}{2\pi i}\int_{\gamma_R} \frac{f(\zeta)}{\zeta - x}\, d\zeta$. As $R \to \infty$ the vertical sides vanish, and

$$f(x) = \frac{1}{2\pi i}\int_{L_1} \frac{f(\zeta)}{\zeta - x}\, d\zeta - \frac{1}{2\pi i}\int_{L_2} \frac{f(\zeta)}{\zeta - x}\, d\zeta = \int_0^{\infty} \hat{f}(\xi) e^{2\pi i x\xi}\, d\xi + \int_{-\infty}^{0} \hat{f}(\xi) e^{2\pi i x\xi}\, d\xi.$$

■

Theorem 2.4 (Poisson summation formula)

If $f \in \mathfrak{F}$, then $\displaystyle\sum_{n\in\mathbb{Z}} f(n) = \sum_{n\in\mathbb{Z}} \hat{f}(n)$.

Proof. Let $f \in \mathfrak{F}_a$, $0 < b < a$. $f(z)/(e^{2\pi i z} - 1)$ has simple poles at the integers n with residues $f(n)/2\pi i$. The residue formula on the rectangle γ_N with vertices $\pm(N + \frac{1}{2}) \pm ib$ gives

$$\sum_{|n|\le N} f(n) = \int_{\gamma_N} \frac{f(z)}{e^{2\pi i z} - 1}\, dz.$$

Let $N \to \infty$; by moderate decrease the vertical sides contribute 0, and

$$\sum_{n\in\mathbb{Z}} f(n) = \int_{L_1} \frac{f(z)}{e^{2\pi i z} - 1}\, dz - \int_{L_2} \frac{f(z)}{e^{2\pi i z} - 1}\, dz, \tag{2}$$

with L_1, L_2 the lines shifted down and up by b. On L_1, $|e^{2\pi iz}| > 1$ and $\frac{1}{e^{2\pi iz}-1} = e^{-2\pi iz}\sum_{n\geq 0} e^{-2\pi inz}$; on L_2, $\frac{1}{e^{2\pi iz}-1} = -\sum_{n\geq 0} e^{2\pi inz}$. Substituting in (2) and shifting back to the real line (by (1) and its analogue),

$$\sum_{n\in\mathbb{Z}} f(n) = \sum_{n=0}^{\infty}\int_{L_1} f(z)e^{-2\pi i(n+1)z}\,dz + \sum_{n=0}^{\infty}\int_{L_2} f(z)e^{2\pi inz}\,dz = \sum_{n=0}^{\infty}\hat{f}(n+1) + \sum_{n=0}^{\infty}\hat{f}(-n) = \sum_{n\in\mathbb{Z}}\hat{f}(n).$$

■

Consequences. Since $e^{-\pi x^2}$ is its own Fourier transform (Example 1), for $t > 0$ and $a \in \mathbb{R}$ the change of variables $x \mapsto t^{1/2}(x+a)$ shows that $f(x) = e^{-\pi t(x+a)^2}$ has $\hat{f}(\xi) = t^{-1/2}e^{-\pi\xi^2/t}e^{2\pi ia\xi}$. Poisson summation gives

$$\sum_{n=-\infty}^{\infty} e^{-\pi t(n+a)^2} = \sum_{n=-\infty}^{\infty} t^{-1/2}e^{-\pi n^2/t}e^{2\pi ina}. \tag{3}$$

For $a = 0$ this is the transformation law of the theta function $\vartheta(t) = \sum_{n=-\infty}^{\infty} e^{-\pi n^2 t}$ $(t > 0)$:

$$\vartheta(t) = t^{-1/2}\vartheta(1/t) \qquad \text{for } t > 0, \tag{4}$$

used for the functional equation of ζ (Chapter 6) and the Jacobi theta function (Chapter 10). Since $1/\cosh \pi x$ is its own Fourier transform (Example 3), $f(x) = e^{-2\pi iax}/\cosh(\pi x/t)$ has $\hat{f}(\xi) = t/\cosh(\pi(\xi+a)t)$, and

$$\sum_{n=-\infty}^{\infty} \frac{e^{-2\pi ian}}{\cosh(\pi n/t)} = \sum_{n=-\infty}^{\infty} \frac{t}{\cosh(\pi(n+a)t)}, \tag{5}$$

used in Chapter 10 for the two-squares theorem.

3 Paley–Wiener theorem

Motivation. Fourier transforms supported in a bounded interval are exactly those of entire functions of exponential type.

Now we assume no analyticity, but the validity of Fourier inversion when $|f(x)| \leq A/(1+x^2)$ and $|\hat{f}(\xi)| \leq A'/(1+\xi^2)$ (proved in Book I). A partial converse to Theorem 2.1:

Theorem 3.1

Suppose $\hat{f}$ satisfies the decay condition $|\hat{f}(\xi)| \leq Ae^{-2\pi a|\xi|}$ for some constants $a, A > 0$. Then $f(x)$ is the restriction to $\mathbb{R}$ of a function $f(z)$ holomorphic in the strip $S_b = \{z \in \mathbb{C} : |\mathrm{Im}(z)| < b\}$, for any $0 < b < a$.

Proof. Let $f_n(z) = \int_{-n}^{n} \hat{f}(\xi)e^{2\pi i\xi z}\,d\xi$, entire by Theorem 5.4, and $f(z) = \int_{-\infty}^{\infty} \hat{f}(\xi)e^{2\pi i\xi z}\,d\xi$ on S_b; the integral converges absolutely, being majorized by $A\int e^{-2\pi a|\xi|}e^{2\pi b|\xi|}\,d\xi < \infty$ as $b < a$. On S_b, $|f(z) - f_n(z)| \leq A\int_{|\xi|\geq n} e^{-2\pi a|\xi|}e^{2\pi b|\xi|}\,d\xi \to 0$, so f is holomorphic there by Theorem 5.2. ■

Corollary 3.2

If $\hat{f}(\xi) = O(e^{-2\pi a|\xi|})$ for some $a > 0$, and f vanishes in a non-empty open interval, then $f = 0$.

f is analytic in a region containing $\mathbb{R}$, so this follows from Theorem 4.8. In particular f and $\hat{f}$ cannot both have compact support unless $f = 0$.

Theorem 3.3 (Paley–Wiener)

Let f be continuous and of moderate decrease on $\mathbb{R}$. Then f extends to an entire function with $|f(z)| \le Ae^{2\pi M|z|}$ for some $A > 0$ if and only if $\hat{f}$ is supported in $[-M, M]$.

Proof. *($\Leftarrow$)* If $\hat{f}$ is supported in $[-M, M]$, both f and $\hat{f}$ have moderate decrease and $f(x) = \int_{-M}^{M} \hat{f}(\xi) e^{2\pi i \xi x}\, d\xi$. Define $g(z) = \int_{-M}^{M} \hat{f}(\xi) e^{2\pi i \xi z}\, d\xi$; $g = f$ on $\mathbb{R}$, g is entire by Theorem 5.4, and for $z = x + iy$, $|g(z)| \le \int_{-M}^{M} |\hat{f}(\xi)| e^{-2\pi \xi y}\, d\xi \le Ae^{2\pi M|z|}$.

($\Rightarrow$) For $\hat{f}$ supported in $[-M, M]$ the argument above gives the stronger bound $Ae^{2\pi M|y|}$. We first assume that bound plus decay in x, then remove the extra assumptions.

Step 1. Assume f is entire with

$$|f(x + iy)| \le A \frac{e^{2\pi M|y|}}{1 + x^2}. \tag{6}$$

Then $\hat{f}(\xi) = 0$ for $|\xi| > M$. For $\xi > M$, shifting the real line down by $y > 0$ (as in (1)),

$$\hat{f}(\xi) = \int_{-\infty}^{\infty} f(x - iy) e^{-2\pi i (x - iy)\xi}\, dx, \qquad |\hat{f}(\xi)| \le \int_{-\infty}^{\infty} A \frac{e^{2\pi M y - 2\pi y \xi}}{1 + x^2}\, dx \le Ce^{-2\pi y(\xi - M)},$$

and letting $y \to \infty$ gives $\hat{f}(\xi) = 0$. For $\xi < -M$ shift up.

Step 2. Assume only

$$|f(x + iy)| \le Ae^{2\pi M|y|}. \tag{7}$$

Let $\xi > M$ and $f_\epsilon(z) = \dfrac{f(z)}{(1 + i\epsilon z)^2}$. The factor has modulus ≤ 1 in the closed lower half-plane and tends to 1, so $|\hat{f}_\epsilon(\xi) - \hat{f}(\xi)| \le \int |f(x)| \left| \frac{1}{(1+i\epsilon x)^2} - 1 \right| dx \to 0$. Each f_ϵ satisfies $|f_\epsilon(x + iy)| \le A' \frac{e^{2\pi M|y|}}{1+x^2}$ in the lower half-plane, where Step 1 shifts the contour for $\xi > M$; so $\hat{f}_\epsilon(\xi) = 0$, hence $\hat{f}(\xi) = 0$. For $\xi < -M$ use the upper half-plane and $1/(1 - i\epsilon z)^2$.

Step 3. It remains to derive (7). It suffices: if $|f(x)| \le 1$ on $\mathbb{R}$ and $|f(z)| \le e^{2\pi M|z|}$, then $|f(x + iy)| \le e^{2\pi M|y|}$. In the first quadrant Q, $F(z) = f(z)e^{2\pi i M z}$ is bounded by 1 on both axes and $|F(z)| \le Ce^{c|z|}$, so Phragmén–Lindelöf (below, rotated) gives $|F| \le 1$ in Q, i.e. $|f(z)| \le e^{2\pi M y}$. The other quadrants are similar. ∎

The idea of Phragmén and Lindelöf adapts the maximum modulus principle to unbounded regions:

Theorem 3.4 (Phragmén–Lindelöf)

Let F be holomorphic in $S = \{z : -\pi/4 < \arg z < \pi/4\}$ and continuous on $\overline{S}$, with $|F| \le 1$ on ∂S and $|F(z)| \le Ce^{c|z|}$ in S for some $C, c > 0$. Then $|F(z)| \le 1$ for all $z \in S$.

Some growth restriction is needed: $F(z) = e^{z^2}$ is bounded by 1 on ∂S but unbounded on $\mathbb{R}_+$.

Proof. Subdue the "enemy" e^{z^2} by $e^{z^{3/2}}$: for $\epsilon > 0$ let $F_\epsilon(z) = F(z)e^{-\epsilon z^{3/2}}$ (principal branch, $z^{3/2} = r^{3/2}e^{3i\theta/2}$, $-\pi < \theta < \pi$), holomorphic in S, continuous on $\overline{S}$, with $|e^{-\epsilon z^{3/2}}| = e^{-\epsilon r^{3/2} \cos(3\theta/2)}$. For $|\theta| < \pi/4$, $|3\theta/2| < 3\pi/8 < \pi/2$, so $\cos(3\theta/2)$ is bounded below by a positive constant, and F_ϵ decays rapidly at infinity in $\overline{S}$ and is bounded. Let $M = \sup_{\overline{S}} |F_\epsilon|$; if $F \not\equiv 0$ take w_j with $|F_\epsilon(w_j)| \to M$. As $M \ne 0$ and F_ϵ decays, the w_j accumulate at some $w \in \overline{S}$; by the maximum principle $w \in \partial S$, where $|F_\epsilon| \le 1$. So $|F_\epsilon| \le 1$ on $\overline{S}$; let $\epsilon \to 0$. ∎

Another version of the Paley–Wiener idea characterizes Fourier transforms vanishing for $\xi < 0$:

Theorem 3.5

Let f, $\hat{f}$ have moderate decrease. Then $\hat{f}(\xi) = 0$ for all $\xi < 0$ iff f extends to a continuous bounded function on $\{y \geq 0\}$, holomorphic in $\{y > 0\}$.

Proof. If $\hat{f} = 0$ on $\xi < 0$, set $f(z) = \int_0^\infty \hat{f}(\xi) e^{2\pi i \xi z}\, d\xi$ for $y \geq 0$; $|f(z)| \leq A \int_0^\infty \frac{d\xi}{1+\xi^2}$, and $f_n(z) = \int_0^n \hat{f}(\xi) e^{2\pi i \xi z}\, d\xi \to f$ uniformly on the closed half-plane, giving continuity and holomorphy. Conversely, as in Theorem 3.3, let $f_{\epsilon,\delta}(z) = \dfrac{f(z + i\delta)}{(1 - i\epsilon z)^2}$, holomorphic near the closed upper half-plane. Shifting the contour up, $\hat{f}_{\epsilon,\delta}(\xi) = 0$ for $\xi < 0$; letting $\delta \to 0$ and then $\epsilon \to 0$, $\hat{f}(\xi) = 0$ for $\xi < 0$. ∎

Remark

Compare Theorem 7.1: holomorphy in a disc kills Fourier coefficients with $n < 0$; holomorphy in the upper half-plane kills $\hat{f}$ on $\xi < 0$.

4 Selected exercises

Motivation. Four exercises from Stein–Shakarchi, Chapter 4 (book numbers kept).

Exercise 4.1 (Uniqueness of the Fourier transform)

Suppose f is continuous and of moderate decrease, and $\hat{f}(\xi) = 0$ for all $\xi \in \mathbb{R}$. Show that $f = 0$, as follows. For fixed $t \in \mathbb{R}$ let

$$A(z) = \int_{-\infty}^{t} f(x) e^{-2\pi i z(x-t)}\, dx \qquad (\operatorname{Im} z \geq 0),$$
$$B(z) = -\int_{t}^{\infty} f(x) e^{-2\pi i z(x-t)}\, dx \qquad (\operatorname{Im} z \leq 0).$$

(a) Show that $A(\xi) = B(\xi)$ for all $\xi \in \mathbb{R}$.
(b) Show that the function F equal to A in the closed upper half-plane and to B in the lower half-plane is entire and bounded, and that in fact $F = 0$.
(c) Deduce that $\int_{-\infty}^{t} f(x)\, dx = 0$ for all t, and conclude that $f = 0$.

Solution. Write $z = u + iv$, $I = \int_{-\infty}^{\infty} |f(x)|\, dx < \infty$. As $|e^{-2\pi i z(x-t)}| = e^{2\pi v(x-t)}$, the integrand of A is $\leq |f(x)|$ for $v \geq 0$, $x \leq t$, and that of B for $v \leq 0$, $x \geq t$; so $|A|, |B| \leq I$ where defined.

(a) For real ξ both are defined, and

$$A(\xi) - B(\xi) = \int_{-\infty}^{\infty} f(x) e^{-2\pi i \xi (x-t)}\, dx = e^{2\pi i \xi t} \hat{f}(\xi) = 0.$$

(b)

Step 1. $A_n(z) = \int_{t-n}^{t} f(x) e^{-2\pi i z(x-t)}\, dx$ is entire (Theorem 5.4 after $x = t - ns$). For $v \geq 0$, $|A - A_n| \leq \int_{-\infty}^{t-n} |f| \to 0$, so A is continuous on $\{v \geq 0\}$ and holomorphic in $\{v > 0\}$ (Theorem 5.2); likewise B on $\{v \leq 0\}$.

Step 2. By (a) they agree on $\mathbb{R}$, so F is entire by the symmetry principle (Theorem 5.5), and constant by Liouville (Corollary 4.5) as $|F| \leq I$.

Step 3. For $v > 0$,

$$|F(iv)| = |A(iv)| \le \int_{-\infty}^{t} |f(x)| e^{2\pi v(x-t)}\, dx.$$

The integrand tends to 0 for every $x < t$ as $v \to \infty$ and is dominated by $|f|$, so $F(iv) \to 0$ by dominated convergence. Hence $F \equiv 0$.

(c) In particular $0 = F(0) = A(0) = \int_{-\infty}^{t} f(x)\, dx$ for every $t \in \mathbb{R}$. Since f is continuous, the fundamental theorem of calculus gives $f(t) = \frac{d}{dt}\int_{-\infty}^{t} f(x)\, dx = 0$ for all t. ∎

Exercise 4.7 (Poisson summation for $(\tau+z)^{-k}$)

Fix τ with $\operatorname{Im}\tau > 0$ and an integer $k \ge 2$.
(a) Apply the Poisson summation formula to $f(z) = (\tau+z)^{-k}$ to obtain

$$\sum_{n=-\infty}^{\infty} \frac{1}{(\tau+n)^k} = \frac{(-2\pi i)^k}{(k-1)!} \sum_{m=1}^{\infty} m^{k-1} e^{2\pi i m\tau}.$$

(b) Deduce that $\displaystyle\sum_{n=-\infty}^{\infty} \frac{1}{(\tau+n)^2} = \frac{\pi^2}{\sin^2(\pi\tau)}$ when $\operatorname{Im}\tau > 0$.
(c) Show that the formula in (b) holds for every $\tau \in \mathbb{C} \setminus \mathbb{Z}$.

Solution.

(a)

Step 1. ($f \in \mathfrak{F}$.) Let $t = \operatorname{Im}\tau$, $0 < a < t$. The only singularity $-\tau$ has $\operatorname{Im}(-\tau) = -t$, so f is holomorphic in S_a. For $|y| < a$, $|\tau+z| \ge t - a > 0$ and $|\tau+z| \ge |x| - |\tau|$, so $|\tau+z| \ge c(1+|x|)$ with $c = c(\tau, a) > 0$, and

$$|f(x+iy)| \le \frac{c^{-k}}{(1+|x|)^k} \le \frac{c^{-k}}{(1+|x|)^2} \le \frac{2c^{-k}}{1+x^2}.$$

So $f \in \mathfrak{F}_a$, and Theorem 2.4 gives $\sum_n f(n) = \sum_n \hat f(n)$.

Step 2. ($\xi < 0$.) Let $g(z) = (\tau+z)^{-k} e^{-2\pi i z\xi}$; $|e^{-2\pi i z\xi}| \le 1$ for $\operatorname{Im} z \ge 0$. The integral over the boundary of $\{|z| < R,\ \operatorname{Im} z > 0\}$ is 0 (no pole), and the arc contributes at most $\pi R(R - |\tau|)^{-k} \to 0$. So $\hat f(\xi) = 0$.

Step 3. ($\xi = 0$.) $\hat f(0) = \int_{-\infty}^{\infty} (\tau+x)^{-k}\, dx = \left[\frac{(\tau+x)^{1-k}}{1-k}\right]_{-\infty}^{\infty} = 0$, since $k \ge 2$.

Step 4. ($\xi > 0$.) Close the contour in the lower half-plane, where $|e^{-2\pi i z\xi}| \le 1$; the arc is $o(1)$ and the pole $-\tau$ of order k is encircled clockwise. By Theorem 1.4,

$$\operatorname{res}_{z=-\tau} g = \frac{1}{(k-1)!} \frac{d^{k-1}}{dz^{k-1}} e^{-2\pi i z\xi}\Big|_{z=-\tau} = \frac{(-2\pi i\xi)^{k-1}}{(k-1)!} e^{2\pi i\tau\xi},$$

so

$$\hat f(\xi) = -2\pi i \operatorname{res}_{z=-\tau} g = \frac{(-2\pi i)^k}{(k-1)!}\, \xi^{k-1} e^{2\pi i\tau\xi}.$$

Step 5. Summing $\hat f(m)$ over $m \ge 1$ (the terms with $m \le 0$ vanish) gives the formula.

(b) Put $k = 2$ and $q = e^{2\pi i\tau}$, so $|q| = e^{-2\pi t} < 1$. Differentiating the geometric series gives $\sum_{m\ge1} m q^m = q/(1-q)^2$, so the right side of (a) equals $-4\pi^2 q/(1-q)^2$. On the other hand,

$$\sin^2(\pi\tau) = \Big(\frac{e^{i\pi\tau} - e^{-i\pi\tau}}{2i}\Big)^2 = -\frac{q - 2 + q^{-1}}{4} = -\frac{(1-q)^2}{4q},$$

so $\pi^2/\sin^2(\pi\tau) = -4\pi^2 q/(1-q)^2$ as well.

(c) For compact $K \subset \mathbb{C}\setminus\mathbb{Z}$, $K \subset \{|z| \le R\}$, and $|n| \ge 2R$, $|\tau+n|^{-2} \le 4/n^2$ on K, so the series is holomorphic on $\mathbb{C}\setminus\mathbb{Z}$ (Theorem 5.2), as is $\pi^2/\sin^2(\pi\tau)$. They agree on the upper half-plane by (b), hence on the connected $\mathbb{C}\setminus\mathbb{Z}$ (Theorem 4.8). ∎

Exercise 4.8 (Power series of a band-limited function)

Let f be continuous and of moderate decrease, with $\hat f$ supported in $[-M, M]$, and let $f(z) = \sum_{n\ge 0} a_n z^n$ be the entire extension given by Theorem 3.3.

(a) Show that $a_n = \dfrac{(2\pi i)^n}{n!}\displaystyle\int_{-M}^{M} \hat f(\xi)\xi^n\,d\xi$, and hence $\limsup_{n\to\infty}(n!|a_n|)^{1/n} \le 2\pi M$.

(b) Conversely, let $f(z) = \sum_{n\ge 0} a_n z^n$ be any power series with $\limsup_n (n!|a_n|)^{1/n} \le 2\pi M$. Show that f is entire and that for every $\epsilon > 0$ there is A_ϵ with $|f(z)| \le A_\epsilon e^{2\pi(M+\epsilon)|z|}$.

Solution.

(a)

Step 1. By the proof of Theorem 3.3, $f(z) = \int_{-M}^{M} \hat f(\xi) e^{2\pi i\xi z}\,d\xi$. Here $\hat f$ is continuous, being the transform of a function of moderate decrease, so $C := \int_{-M}^{M}|\hat f| < \infty$.

Step 2. For fixed z, $e^{2\pi i\xi z} = \sum_n (2\pi i\xi z)^n/n!$ converges uniformly in $\xi \in [-M, M]$ (Weierstrass M-test with $(2\pi M|z|)^n/n!$). So we may integrate term by term:

$$f(z) = \sum_{n=0}^{\infty} \Big(\frac{(2\pi i)^n}{n!}\int_{-M}^{M} \hat f(\xi)\xi^n\,d\xi\Big) z^n.$$

By uniqueness of power series coefficients this is the stated formula for a_n.

Step 3. Hence $n!|a_n| \le (2\pi)^n M^n C$, so $(n!|a_n|)^{1/n} \le 2\pi M C^{1/n} \to 2\pi M$ (if $C = 0$ then $f = 0$).

(b)

Step 1. Fix $\epsilon > 0$ and put $K = 2\pi(M+\epsilon)$. Since $\limsup(n!|a_n|)^{1/n} < K$, there is N with $n!|a_n| \le K^n$ for $n \ge N$. Let $A_\epsilon = \max\big(1, \max_{n<N} n!|a_n|K^{-n}\big)$. Then $|a_n| \le A_\epsilon K^n/n!$ for all n.

Step 2. Hence $\sum|a_n||z|^n \le A_\epsilon \sum (K|z|)^n/n! = A_\epsilon e^{K|z|} < \infty$ for every z. The radius of convergence is infinite, so f is entire, and $|f(z)| \le A_\epsilon e^{2\pi(M+\epsilon)|z|}$. ∎

Exercise 4.9 (Phragmén–Lindelöf in a sector)

Let S be an open sector with vertex 0 and opening angle π/β, where $\beta > 1/2$. Let F be holomorphic in S and continuous on $\overline{S}$, with $|F| \le 1$ on ∂S and $|F(z)| \le Ce^{c|z|^\alpha}$ in S for some $C, c > 0$ and $0 < \alpha < \beta$. Prove that $|F(z)| \le 1$ for all $z \in S$. (For $\beta = 1$ this is the half-plane case; $\beta = 2$ is Theorem 3.4.)

Solution.

Step 1. (Normalization.) A rotation $z \mapsto e^{i\varphi}z$ preserves all hypotheses, so we may assume $S = \{re^{i\theta} : r > 0,\ |\theta| < \pi/(2\beta)\}$. Since $\pi/(2\beta) < \pi$, $\overline{S}\setminus\{0\}$ misses the negative real axis.

Step 2. (Auxiliary factor.) Fix γ with $\alpha < \gamma < \beta$ and let $z^\gamma = r^\gamma e^{i\gamma\theta}$ be the principal branch, holomorphic on S and continuous on $\overline{S}$. For $|\theta| \le \pi/(2\beta)$ we have $|\gamma\theta| \le \gamma\pi/(2\beta) < \pi/2$, so

$$\cos(\gamma\theta) \ge \delta := \cos\Big(\frac{\gamma\pi}{2\beta}\Big) > 0.$$

For $\epsilon > 0$ put $F_\epsilon(z) = F(z)e^{-\epsilon z^\gamma}$. Then $|e^{-\epsilon z^\gamma}| = e^{-\epsilon r^\gamma\cos\gamma\theta} \le e^{-\epsilon\delta r^\gamma} \le 1$ on $\overline{S}$.

Step 3. (Estimates.) On ∂S, $|F_\epsilon| \le |F| \le 1$. In S, $|F_\epsilon(z)| \le C\exp(cr^\alpha - \epsilon\delta r^\gamma) \to 0$ as $r \to \infty$, because $\gamma > \alpha$. Choose R_0 such that $|F_\epsilon| \le 1$ on $\overline{S} \cap \{|z| \ge R_0\}$.

Step 4. (Maximum modulus.) For $R \ge R_0$, $S_R = S \cap \{|z| < R\}$ is bounded with $|F_\epsilon| \le 1$ on its boundary (two segments of ∂S and an arc), so $|F_\epsilon| \le 1$ on S_R (Corollary 4.6), hence on S.

Step 5. For fixed $z \in S$, $|F(z)| \le e^{\epsilon|z^\gamma|} = e^{\epsilon|z|^\gamma}$. Letting $\epsilon \to 0$ gives $|F(z)| \le 1$. ■

Chapter 5. Entire Functions

Overview

Entire functions are holomorphic in the whole complex plane. The chapter is organized around three questions.

1. *Where can they vanish?* On any sequence without finite limit point (Weierstrass canonical factors, after Euler's product for $\sin \pi z$).
2. *How do they grow?* Faster growth allows more zeros; *Jensen's formula* links the zeros in a disc to the average of $\log|f|$ on its boundary.
3. *Are they determined by their zeros?* For finite order, up to a simple factor (*Hadamard factorization*).

1 Jensen's formula

Motivation. Jensen's formula lets growth control the number of zeros in a disc.

D_R and C_R denote the open disc and circle of radius R centered at 0. We exclude the function that vanishes identically.

Theorem 1.1 (Jensen's formula)

Let Ω be open with $\overline{D_R} \subset \Omega$, f holomorphic in Ω, $f(0) \neq 0$, $f \neq 0$ on C_R. If $z_1, \dots, z_N$ are the zeros of f in D_R (with multiplicities), then

$$\log|f(0)| = \sum_{k=1}^{N} \log\left(\frac{|z_k|}{R}\right) + \frac{1}{2\pi}\int_0^{2\pi} \log|f(Re^{i\theta})|\, d\theta. \tag{1}$$

Proof.

Step 1. If f_1, f_2 satisfy the hypotheses and (1), so does $f_1 f_2$, since $\log xy = \log x + \log y$ for $x, y > 0$ and the zeros of $f_1 f_2$ are the union of those of f_1 and f_2.

Step 2. $g(z) = f(z)/((z - z_1)\cdots(z - z_N))$ is bounded near each z_j, hence extends holomorphically to Ω, nowhere zero on $\overline{D_R}$. By Step 1 it suffices to prove (1) for such g and for $z - z_j$.

Step 3. For such g: in a slightly larger disc $g = e^h$ (Theorem 6.2), so $\log|g| = \mathrm{Re}(h)$, and the mean value property (Corollary 7.3) gives $\log|g(0)| = \frac{1}{2\pi}\int_0^{2\pi} \log|g(Re^{i\theta})|\, d\theta$.

Step 4. For $f(z) = z - w$, $w \in D_R$: we need $\log|w| = \log\frac{|w|}{R} + \frac{1}{2\pi}\int_0^{2\pi} \log|Re^{i\theta} - w|\, d\theta$. As $\log|Re^{i\theta} - w| = \log R + \log|e^{i\theta} - w/R|$, it suffices that $\int_0^{2\pi} \log|e^{i\theta} - a|\, d\theta = 0$ for $|a| < 1$, i.e. $(\theta \mapsto -\theta)$ $\int_0^{2\pi} \log|1 - ae^{i\theta}|\, d\theta = 0$. $F(z) = 1 - az$ has no zeros on the closed unit disc, so $\log|F| = \mathrm{Re}(G)$, $F = e^G$, is harmonic near it, and $\log|F(0)| = 0$; apply the mean value property. ∎

For f holomorphic on $\overline{D_R}$, let $n(r) = n_f(r)$ be the number of zeros of f (with multiplicities) in D_r, $0 < r < R$; $n(r)$ is non-decreasing. If $f(0) \neq 0$ and f does not vanish on C_R, then

$$\int_0^R n(r)\frac{dr}{r} = \frac{1}{2\pi}\int_0^{2\pi} \log|f(Re^{i\theta})|\, d\theta - \log|f(0)|, \tag{2}$$

which follows from Jensen's formula and:

Lemma 1.2

If $z_1, \dots, z_N$ are the zeros of f inside the disc D_R, then

$$\int_0^R n(r)\frac{dr}{r} = \sum_{k=1}^{N} \log\left|\frac{R}{z_k}\right|.$$

Proof. $\sum_{k=1}^N \log\left|\frac{R}{z_k}\right| = \sum_{k=1}^N \int_{|z_k|}^R \frac{dr}{r}$. With $\eta_k(r) = 1$ if $r > |z_k|$ and 0 otherwise, $\sum_k \eta_k(r) = n(r)$, so $\sum_k \int_{|z_k|}^R \frac{dr}{r} = \sum_k \int_0^R \eta_k(r)\frac{dr}{r} = \int_0^R n(r)\frac{dr}{r}$. ■

2 Functions of finite order

Motivation. With growth of order ρ, Jensen's formula bounds the number of zeros.

Definition (order of growth)

An entire function f has an **order of growth** $\le \rho$ if $|f(z)| \le Ae^{B|z|^\rho}$ for all $z \in \mathbb{C}$, for some constants $A, B > 0$. The **order of growth** of f is $\rho_f = \inf \rho$ over all such $\rho > 0$.

For example, e^{z^2} has order of growth 2.

Theorem 2.1

If f is an entire function that has an order of growth $\le \rho$, then:
(i) $n(r) \le Cr^\rho$ for some $C > 0$ and all sufficiently large r.
(ii) If $z_1, z_2, \dots$ denote the zeros of f, with $z_k \neq 0$, then for all $s > \rho$ we have $\sum_{k=1}^{\infty} \frac{1}{|z_k|^s} < \infty$.

Proof. Assume $f(0) \neq 0$ (else use $F = f/z^\ell$, ℓ the order of the zero at 0: $n_f - n_F$ is constant and F has order $\le \rho$). By (2) with $R = 2r$,

$$\int_r^{2r} n(x)\frac{dx}{x} \le \int_0^{2r} n(x)\frac{dx}{x} = \frac{1}{2\pi}\int_0^{2\pi} \log|f(2re^{i\theta})|\,d\theta - \log|f(0)|.$$

Since n is increasing, the left side is $\ge n(r)\int_r^{2r}\frac{dx}{x} = n(r)\log 2$; and by the growth condition $\int_0^{2\pi}\log|f(2re^{i\theta})|\,d\theta \le \int_0^{2\pi}\log|Ae^{B(2r)^\rho}|\,d\theta \le C'r^\rho$ for large r. Hence $n(r) \le Cr^\rho$. For (ii),

$$\sum_{|z_k|\ge 1}|z_k|^{-s} = \sum_{j=0}^{\infty}\sum_{2^j\le|z_k|<2^{j+1}}|z_k|^{-s} \le \sum_{j=0}^{\infty} 2^{-js}n(2^{j+1}) \le c\sum_{j=0}^{\infty}2^{-js}2^{(j+1)\rho} \le c'\sum_{j=0}^{\infty}(2^{\rho-s})^j < \infty,$$

since $s > \rho$. ■

The condition $s > \rho$ cannot be improved:

Example 1

$f(z) = \sin \pi z = \frac{e^{i\pi z} - e^{-i\pi z}}{2i}$ satisfies $|f(z)| \le e^{\pi|z|}$, and taking $z = ix$ shows the order of growth is exactly 1. f vanishes to order 1 at each $n \in \mathbb{Z}$, and $\sum_{n\neq 0} 1/|n|^s < \infty$ precisely when $s > 1$.

Example 2

$f(z) = \cos z^{1/2} = \sum_{n=0}^{\infty}(-1)^n \frac{z^n}{(2n)!}$ is entire with $|f(z)| \le e^{|z|^{1/2}}$, of order $1/2$. It vanishes at $z_n = ((n+1/2)\pi)^2$, and $\sum_n 1/|z_n|^s < \infty$ exactly when $s > 1/2$.

Is there an entire f with zeros exactly at given $z_1, z_2, \ldots$? Necessarily $|z_k| \to \infty$ (Theorem 4.8); Weierstrass showed this suffices, inserting convergence factors into $(z-z_1)(z-z_2)\cdots$ without new zeros.

3 Infinite products

Motivation. Entire functions with infinitely many zeros need infinite products in place of $\prod(z-a_k)$.

3.1 Generalities

For complex a_n, the product $\prod_{n=1}^{\infty}(1+a_n)$ **converges** if $\lim_{N\to\infty}\prod_{n=1}^{N}(1+a_n)$ exists.

Proposition 3.1

If $\sum|a_n| < \infty$, then the product $\prod_{n=1}^{\infty}(1+a_n)$ converges. Moreover, the product converges to 0 if and only if one of its factors is 0.

Proof. Discarding finitely many terms, $|a_n| < 1/2$. With $b_n = \log(1+a_n)$ (power series (6) of Chapter 3), $\prod_{n=1}^{N}(1+a_n) = e^{B_N}$, $B_N = \sum_{n=1}^{N} b_n$. As $|\log(1+z)| \le 2|z|$ for $|z| < 1/2$, $|b_n| \le 2|a_n|$, so $B_N \to B$ and $e^{B_N} \to e^B$, which is $\neq 0$; with all $1+a_n \neq 0$ the full product is then non-zero. ■

Proposition 3.2

Suppose $\{F_n\}$ is a sequence of holomorphic functions on the open set Ω. If there exist constants $c_n > 0$ such that $\sum c_n < \infty$ and $|F_n(z) - 1| \le c_n$ for all $z \in \Omega$, then:
(i) The product $\prod_{n=1}^{\infty} F_n(z)$ converges uniformly in Ω to a holomorphic function $F(z)$.
(ii) If $F_n(z)$ does not vanish for any n, then

$$\frac{F'(z)}{F(z)} = \sum_{n=1}^{\infty} \frac{F_n'(z)}{F_n(z)}.$$

Proof. (i) Argue as in Proposition 3.1 with $|a_n(z)| \le c_n$; the estimates are uniform in z, and the uniform limit is holomorphic (Theorem 5.2).

(ii) Let $K \subset \Omega$ be compact and $G_N = \prod_{n=1}^{N} F_n$. Since $G_N \to F$ uniformly, Theorem 5.3 gives $G_N' \to F'$ uniformly on K. G_N is uniformly bounded below on K, so $G_N'/G_N \to F'/F$ uniformly on K; and $G_N'/G_N = \sum_{n=1}^{N} F_n'/F_n$ (Section 4). ■

3.2 Example: the product formula for the sine function

$$\sin \pi z = \pi z \prod_{n=1}^{\infty}\left(1 - \frac{z^2}{n^2}\right) \tag{3}$$

for all $z \in \mathbb{C}$; it is derived from the sum formula for the cotangent

$$\pi \cot \pi z = \sum_{n=-\infty}^{\infty} \frac{1}{z+n} = \lim_{N\to\infty} \sum_{|n|\le N} \frac{1}{z+n} = \frac{1}{z} + \sum_{n=1}^{\infty} \frac{2z}{z^2 - n^2}, \tag{4}$$

valid for $z \notin \mathbb{Z}$. The halves of $\sum 1/(z+n)$ with $n > 0$ and $n < 0$ diverge separately; only the symmetric interpretation converges.

Proof of (4). $F(z) = \pi \cot \pi z$ has the properties:

(i) $F(z+1) = F(z)$ for $z \notin \mathbb{Z}$;

(ii) $F(z) = \frac{1}{z} + F_0(z)$ with F_0 analytic near 0;

(iii) F has simple poles at the integers and no other singularities. The function $\sum_{n=-\infty}^{\infty} \frac{1}{z+n} = \lim_N \sum_{|n|\le N} \frac{1}{z+n}$ has the same three properties: for (i),

$$\sum_{|n|\le N} \frac{1}{z+1+n} = \frac{1}{z+1+N} - \frac{1}{z-N} + \sum_{|n|\le N} \frac{1}{z+n},$$

and let $N \to \infty$; parts (ii) and (iii) are evident from $\frac{1}{z} + \sum_{n\ge 1} \frac{2z}{z^2-n^2}$. So $\Delta(z) = F(z) - \sum \frac{1}{z+n}$ is 1-periodic with a removable singularity at 0 by (ii), hence at all integers, and is entire. It suffices to bound Δ in the strip $|\mathrm{Re}(z)| \le 1/2$; it is bounded for $|\mathrm{Im}(z)| \le 1$. For $z = x + iy$, $y > 1$,

$$\pi \cot \pi z = \pi i \frac{e^{i\pi z} + e^{-i\pi z}}{e^{i\pi z} - e^{-i\pi z}} = \pi i \frac{e^{-2\pi y} + e^{-2\pi i x}}{e^{-2\pi y} - e^{-2\pi i x}}$$

is bounded, and

$$\Big|\frac{1}{z} + \sum_{n=1}^{\infty} \frac{2z}{z^2-n^2}\Big| = \Big|\frac{1}{x+iy} + \sum_{n=1}^{\infty} \frac{2(x+iy)}{x^2-y^2-n^2+2ixy}\Big| \le C + C\sum_{n=1}^{\infty} \frac{y}{y^2+n^2}.$$

The last sum is majorized by $\int_0^{\infty} \frac{y}{y^2+x^2}\,dx$ (decreasing integrand), independent of y. Similarly for $y < -1$. So Δ is bounded, constant by Liouville, and 0 as it is odd. ∎

Proof of (3). Let $G(z) = \sin \pi z$ and $P(z) = \pi z \prod_{n=1}^{\infty}(1 - z^2/n^2)$. Since $\sum 1/n^2 < \infty$, Proposition 3.2 shows P converges and, away from the integers, $\frac{P'(z)}{P(z)} = \frac{1}{z} + \sum_{n=1}^{\infty} \frac{2z}{z^2-n^2}$. Since $G'/G = \pi \cot \pi z$, (4) gives

$$\Big(\frac{P}{G}\Big)' = \frac{P}{G}\Big[\frac{P'}{P} - \frac{G'}{G}\Big] = 0,$$

so $P = cG$. Dividing by z and letting $z \to 0$, $c = 1$. ∎

4 Weierstrass infinite products

Motivation. Weierstrass's canonical factors make $\prod(1 - z/a_n)$ converge, so any zero sequence tending to infinity is attained.

Theorem 4.1 (Weierstrass)

For any sequence $\{a_n\}$ with $|a_n| \to \infty$ there is an entire f vanishing exactly at the a_n. Any other such function is $f(z)e^{g(z)}$ with g entire.

Repetitions are allowed in $\{a_n\}$, so this gives entire functions with prescribed zeros *and multiplicities* (the multiplicity of a zero a is the m with $f(z) = (z-a)^m g(z)$, $g(a) \neq 0$).

Proof. If f_1, f_2 both vanish exactly at the a_n, f_1/f_2 is entire and nowhere zero, so $f_1/f_2 = e^g$ (Theorem 6.2): the last statement.

To construct f: the naive product $\prod(1 - z/a_n)$ converges only for suitable $\{a_n\}$, so we insert exponential factors. Define the **canonical factors** of degree k:

$$E_0(z) = 1 - z, \qquad E_k(z) = (1-z)e^{z + z^2/2 + \cdots + z^k/k} \quad (k \ge 1).$$

Suppose f is to have a zero of order m at the origin and $a_1, a_2, \ldots$ are non-zero. The **Weierstrass product** is

$$f(z) = z^m \prod_{n=1}^{\infty} E_n(z/a_n).$$

Fix $|z| < R$. The finitely many factors with $|a_n| \le 2R$ vanish exactly at the a_n with $|a_n| < R$. For $|a_n| > 2R$, $|z/a_n| \le 1/2$, and by Lemma 4.2 (c independent of n)

$$|1 - E_n(z/a_n)| \le c\left|\frac{z}{a_n}\right|^{n+1} \le \frac{c}{2^{n+1}}.$$

By Section 3, $\prod_{|a_n|>2R} E_n(z/a_n)$ is holomorphic and non-vanishing in $|z| < R$. As R is arbitrary, f has the required properties. ■

Lemma 4.2

If $|z| \le 1/2$, then $|1 - E_k(z)| \le c|z|^{k+1}$ for some $c > 0$.

Proof. For $|z| \le 1/2$, $1 - z = e^{\log(1-z)}$ (power series), so $E_k(z) = e^{\log(1-z)+z+z^2/2+\cdots+z^k/k} = e^w$ with $w = -\sum_{n=k+1}^{\infty} z^n/n$. Since $|z| \le 1/2$,

$$|w| \le |z|^{k+1} \sum_{n=k+1}^{\infty} |z|^{n-k-1}/n \le |z|^{k+1} \sum_{j=0}^{\infty} 2^{-j} \le 2|z|^{k+1}.$$

In particular $|w| \le 1$, so $|1 - E_k(z)| = |1 - e^w| \le c'|w| \le c|z|^{k+1}$. ■

Remark

The constant c is independent of k: one may take $c' = e$, $c = 2e$.

5 Hadamard's factorization theorem

Weierstrass writes $f = e^{g(z)} z^m \prod E_n(z/a_n)$. For finite order, Hadamard takes the canonical factors of constant degree and g a polynomial. Recall: order $\le \rho$ gives $n(r) \le Cr^\rho$ for large r and $\sum |a_n|^{-s} < \infty$ for $s > \rho$ (Theorem 2.1).

Theorem 5.1 (Hadamard)

Suppose f is entire and has growth order ρ_0. Let k be the integer so that $k \le \rho_0 < k + 1$. If $a_1, a_2, \ldots$ denote the (non-zero) zeros of f, then

$$f(z) = e^{P(z)} z^m \prod_{n=1}^{\infty} E_k(z/a_n),$$

where P is a polynomial of degree $\le k$, and m is the order of the zero of f at $z = 0$.

Main lemmas.

Lemma 5.2

The canonical products satisfy

$$|E_k(z)| \ge e^{-c|z|^{k+1}} \text{ if } |z| \le 1/2 \quad \text{and} \quad |E_k(z)| \ge |1 - z|e^{-c'|z|^k} \text{ if } |z| \ge 1/2.$$

Proof. If $|z| \le 1/2$, $E_k(z) = e^{\log(1-z)+\sum_{n=1}^{k} z^n/n} = e^{-\sum_{n=k+1}^{\infty} z^n/n} = e^w$, and $|e^w| \ge e^{-|w|}$ with $|w| \le c|z|^{k+1}$. If $|z| \ge 1/2$, $|E_k(z)| = |1-z||e^{z+z^2/2+\cdots+z^k/k}|$, and $|e^{z+\cdots+z^k/k}| \ge e^{-|z+\cdots+z^k/k|} \ge e^{-c'|z|^k}$ (as $|z| \ge 1/2$). ∎

The key is a lower bound for the product of canonical factors away from the zeros.

Lemma 5.3

For any s with $\rho_0 < s < k+1$, we have

$$\Big|\prod_{n=1}^{\infty} E_k(z/a_n)\Big| \ge e^{-c|z|^s},$$

except possibly when z belongs to the union of the discs centered at a_n of radius $|a_n|^{-k-1}$, for $n = 1, 2, 3, \ldots$.

Proof. Write $\prod_n E_k(z/a_n) = \prod_{|a_n|\le 2|z|} E_k(z/a_n) \prod_{|a_n|>2|z|} E_k(z/a_n)$. For the second product, with no restriction on z, Lemma 5.2 gives

$$\Big|\prod_{|a_n|>2|z|} E_k(z/a_n)\Big| \ge \prod_{|a_n|>2|z|} e^{-c|z/a_n|^{k+1}} = e^{-c|z|^{k+1}\sum_{|a_n|>2|z|}|a_n|^{-k-1}},$$

and since $|a_n| > 2|z|$ and $s < k+1$, $|a_n|^{-k-1} = |a_n|^{-s}|a_n|^{s-k-1} \le C|a_n|^{-s}|z|^{s-k-1}$; as $\sum|a_n|^{-s} < \infty$, this product is $\ge e^{-c|z|^s}$. For the first product, by the second part of Lemma 5.2,

$$\Big|\prod_{|a_n|\le 2|z|} E_k(z/a_n)\Big| \ge \prod_{|a_n|\le 2|z|} \Big|1-\frac{z}{a_n}\Big| \prod_{|a_n|\le 2|z|} e^{-c'|z/a_n|^k}. \tag{5}$$

Here $\prod_{|a_n|\le 2|z|} e^{-c'|z/a_n|^k} = e^{-c'|z|^k\sum_{|a_n|\le 2|z|}|a_n|^{-k}}$ and $|a_n|^{-k} = |a_n|^{-s}|a_n|^{s-k} \le C|a_n|^{-s}|z|^{s-k}$, so this is $\ge e^{-c|z|^s}$. For the first factor of (5): if z is outside the disc of radius $|a_n|^{-k-1}$ about a_n, $|a_n - z| \ge |a_n|^{-k-1}$, so

$$\prod_{|a_n|\le 2|z|} \Big|1-\frac{z}{a_n}\Big| = \prod_{|a_n|\le 2|z|} \Big|\frac{a_n-z}{a_n}\Big| \ge \prod_{|a_n|\le 2|z|} |a_n|^{-k-1}|a_n|^{-1} = \prod_{|a_n|\le 2|z|} |a_n|^{-k-2}.$$

Finally $\log\prod_{|a_n|\le 2|z|}|a_n|^{k+2} = (k+2)\sum_{|a_n|\le 2|z|}\log|a_n| \le (k+2)n(2|z|)\log 2|z| \le c|z|^s \log 2|z| \le c'|z|^{s'}$ for any $s' > s$, using $n(2|z|) \le c|z|^s$ (Theorem 2.1). Since $s > \rho_0$ was arbitrary, starting with s close enough to ρ_0 gives the lemma (with s' in place of s). ∎

Corollary 5.4

There exists a sequence of radii $r_1, r_2, \ldots$ with $r_m \to \infty$, such that

$$\Big|\prod_{n=1}^{\infty} E_k(z/a_n)\Big| \ge e^{-c|z|^s} \qquad \text{for } |z| = r_m.$$

Proof. Choose N with $\sum_{n\ge N}|a_n|^{-k-1} < 1/10$. For large integers L some $r \in [L, L+1]$ gives a circle missing the forbidden discs of Lemma 5.3: otherwise the intervals $I_n = [|a_n| - |a_n|^{-k-1}, |a_n| + |a_n|^{-k-1}]$ would cover $[L, L+1]$, forcing $2\sum_{n\ge N}|a_n|^{-k-1} \ge 1$. Apply Lemma 5.3 with $|z| = r$. ∎

Proof of Hadamard's theorem. Let $E(z) = z^m \prod_{n=1}^{\infty} E_k(z/a_n)$, entire as in Theorem 4.1: $|1 - E_k(z/a_n)| \le c|z/a_n|^{k+1}$ for large n (Lemma 4.2) and $\sum|a_n|^{-k-1} < \infty$ ($\rho_0 < s < k+1$). f/E is entire and nowhere zero, so $f/E = e^g$, g entire. By the order of f and Corollary 5.4,

$$e^{\mathrm{Re}(g(z))} = \Big|\frac{f(z)}{E(z)}\Big| \le c'e^{c|z|^s} \qquad \text{for } |z| = r_m,$$

so $\operatorname{Re}(g(z)) \le C|z|^s$ for $|z| = r_m$. Lemma 5.5 gives that g is a polynomial of degree $\le s$, hence $\le k$ (as $s < k+1$). ■

Lemma 5.5

Suppose g is entire and $u = \operatorname{Re}(g)$ satisfies $u(z) \le Cr^s$ whenever $|z| = r$, for a sequence of positive real numbers r that tends to infinity. Then g is a polynomial of degree $\le s$.

Proof. Write $g(z) = \sum_{n=0}^{\infty} a_n z^n$. By Theorem 7.1,

$$\frac{1}{2\pi}\int_0^{2\pi} g(re^{i\theta})e^{-in\theta}\,d\theta = \begin{cases} a_n r^n & n \ge 0, \\ 0 & n < 0. \end{cases} \tag{6}$$

Taking complex conjugates, $\frac{1}{2\pi}\int_0^{2\pi} \overline{g(re^{i\theta})}e^{-in\theta}\,d\theta = 0$ for $n > 0$ (7). Since $2u = g + \overline{g}$, adding (6) and (7),

$$a_n r^n = \frac{1}{\pi}\int_0^{2\pi} u(re^{i\theta})e^{-in\theta}\,d\theta \quad (n>0), \qquad 2\operatorname{Re}(a_0) = \frac{1}{\pi}\int_0^{2\pi} u(re^{i\theta})\,d\theta.$$

[Supplement]Since $\int_0^{2\pi} e^{-in\theta}d\theta = 0$ for $n > 0$, $a_n r^n = -\frac{1}{\pi}\int_0^{2\pi}[Cr^s - u(re^{i\theta})]e^{-in\theta}\,d\theta$, and $Cr^s - u \ge 0$ on the circles in question, so

$$|a_n| r^n \le \frac{1}{\pi}\int_0^{2\pi}[Cr^s - u(re^{i\theta})]\,d\theta = 2Cr^s - 2\operatorname{Re}(a_0).$$

Letting $r \to \infty$ along the sequence, $a_n = 0$ for $n > s$. ■

6 Selected exercises

Motivation. Four exercises from Stein–Shakarchi, Chapter 5 (book numbers kept).

Exercise 5.6 (Wallis's product)

(a) Prove Wallis's product formula

$$\frac{\pi}{2} = \prod_{m=1}^{\infty} \frac{2m\cdot 2m}{(2m-1)(2m+1)} = \frac{2\cdot 2}{1\cdot 3}\cdot\frac{4\cdot 4}{3\cdot 5}\cdots.$$

(b) Deduce that $\lim_{n\to\infty} \dfrac{2^{2n}(n!)^2}{(2n)!\,(2n+1)^{1/2}} = \sqrt{\pi/2}$.

Solution.

(a) Put $z = 1/2$ in the product formula (3) for the sine:

$$1 = \sin\frac{\pi}{2} = \frac{\pi}{2}\prod_{m=1}^{\infty}\Big(1 - \frac{1}{4m^2}\Big).$$

So the partial products $P_N = \prod_{m\le N}(1 - \frac{1}{4m^2})$ converge to $2/\pi \ne 0$. Hence $1/P_N \to \pi/2$, and

$$\frac{1}{P_N} = \prod_{m=1}^{N}\frac{4m^2}{4m^2-1} = \prod_{m=1}^{N}\frac{2m\cdot 2m}{(2m-1)(2m+1)}.$$

(b) Let $W_N = 1/P_N$.

Step 1. We have $\prod_{m\le N} 2m = 2^N N!$. The odd numbers below $2N$ and below $2N+2$ have products

$$\prod_{m\le N}(2m-1) = \frac{(2N)!}{2^N N!}, \qquad \prod_{m\le N}(2m+1) = \frac{(2N+1)!}{2^N N!}.$$

Step 2. Therefore

$$W_N = \frac{(2^N N!)^4}{(2N)!\,(2N+1)!} = \frac{2^{4N}(N!)^4}{((2N)!)^2(2N+1)} = \Big(\frac{2^{2N}(N!)^2}{(2N)!\,(2N+1)^{1/2}}\Big)^2.$$

Step 3. The bracket is positive and its square tends to $\pi/2$ by (a), so it tends to $\sqrt{\pi/2}$. ∎

Exercise 5.3 (Order of the theta function)

Fix τ with $t = \operatorname{Im}\tau > 0$. Show that the Jacobi theta function

$$\Theta(z\mid\tau) = \sum_{n=-\infty}^{\infty} e^{i\pi n^2\tau} e^{2\pi i n z}$$

is an entire function of z of order exactly 2.

Solution.

Step 1. (Upper bound.) The n-th term is $\le e^{-\pi n^2 t + 2\pi|n||z|}$. For $|n| \ge 4|z|/t$ this is $\le e^{-\pi n^2 t/2}$; for the at most $8|z|/t+1$ values $|n| < 4|z|/t$ it is $\le e^{8\pi|z|^2/t}$. Hence

$$|\Theta(z\mid\tau)| \le \Big(\frac{8|z|}{t}+1\Big)e^{8\pi|z|^2/t} + \sum_n e^{-\pi n^2 t/2} \le A e^{B|z|^2}.$$

The series converges uniformly on compact sets by the same bound, so Θ is entire (Theorem 5.2) of order ≤ 2.

Step 2. (Quasi-periodicity.) Replacing n by $n+1$ in the sum,

$$\Theta(z+\tau\mid\tau) = \sum_n e^{i\pi(n^2+2n)\tau} e^{2\pi i n z} = e^{-i\pi\tau} e^{-2\pi i z} \sum_n e^{i\pi(n+1)^2\tau} e^{2\pi i (n+1) z} = e^{-i\pi\tau - 2\pi i z}\,\Theta(z\mid\tau).$$

Step 3. ($\Theta \not\equiv 0$.) On the real line the series converges uniformly, so $\int_0^1 \Theta(x\mid\tau)\,dx = 1$ (only $n=0$ survives). Pick z_0 with $\Theta(z_0\mid\tau)\neq 0$ and write $y_0 = \operatorname{Im} z_0$.

Step 4. (Lower bound.) Let $z_m = z_0 + m\tau$. Since $|e^{-i\pi\tau - 2\pi i z}| = e^{\pi t + 2\pi \operatorname{Im} z}$ and $\operatorname{Im} z_j = y_0 + jt$, Step 2 gives

$$|\Theta(z_m\mid\tau)| = |\Theta(z_0\mid\tau)| \prod_{j=0}^{m-1} e^{\pi t + 2\pi(y_0+jt)} = |\Theta(z_0\mid\tau)|\, e^{\pi t m^2 + 2\pi y_0 m}.$$

If $|\Theta| \le A e^{B|z|^\rho}$ with $\rho < 2$, then, since $|z_m| \le |z_0| + m|\tau|$,

$$\pi t m^2 + 2\pi y_0 m + \log|\Theta(z_0\mid\tau)| \le \log A + B(|z_0| + m|\tau|)^\rho,$$

which fails for large m. Hence the order is exactly 2. ∎

Exercise 5.11 (Picard's little theorem for finite order)

Show that an entire function of finite order that omits two distinct values is constant.

Solution.

Step 1. If f has order $\leq \rho$ and omits $a \neq b$, $g = f - a$ is entire, nowhere zero, with $|g| \leq (A + |a|)e^{B|z|^\rho}$; by Hadamard (Theorem 5.1) $g = e^p$, p a polynomial.

Step 2. Since f omits b, $e^{p(z)} \neq b - a$ for all z. As $b - a \neq 0$, there is $w \in \mathbb{C}$ with $e^w = b - a$.

Step 3. If p were non-constant, some z has $p(z) = w$ (Corollary 4.6), so $e^{p(z)} = b - a$, a contradiction. So $f = a + e^p$ is constant. ■

Exercise 5.13 (Solutions of $e^z = z$)

Show that the equation $e^z - z = 0$ has infinitely many solutions in $\mathbb{C}$.

Solution.

Step 1. $f(z) = e^z - z$ satisfies $|f| \leq e^{|z|} + |z| \leq 2e^{|z|}$, so its order is $\rho_0 \leq 1$, and the integer k of Theorem 5.1 is 0 or 1. Also $f(0) = 1 \neq 0$.

Step 2. Suppose f had only finitely many zeros $a_1, \ldots, a_N$. By Hadamard's theorem, $f = e^P \prod_{n \leq N} E_k(z/a_n)$ with $\deg P \leq k \leq 1$. Each $E_k(z/a_n)$ is $(1 - z/a_n)$ times the exponential of a polynomial of degree $\leq k$. Collecting factors,

$$e^z - z = Q(z)e^{\alpha z + \beta}$$

for a non-zero polynomial Q and constants $\alpha, \beta \in \mathbb{C}$.

Step 3. ($x \to -\infty$.) $|Q(x)e^{\alpha x+\beta}| = |Q(x)|e^{x \operatorname{Re}\alpha + \operatorname{Re}\beta}$ while $|e^x - x| = |x| + o(1)$. $\operatorname{Re}\alpha > 0$ makes the left side $\to 0$, $\operatorname{Re}\alpha < 0$ makes it grow faster than any power; so $\operatorname{Re}\alpha = 0$.

Step 4. (Behaviour as $x \to +\infty$.) Then $|Q(x)e^{\alpha x+\beta}| = |Q(x)|e^{\operatorname{Re}\beta}$ grows at most polynomially. But $e^x - x \geq e^x/2$ for large x, a contradiction. So f has infinitely many zeros. ■

Chapter 6. The Gamma and Zeta Functions

Overview

$1/\Gamma(s)$ is essentially the simplest entire function with zeros exactly at $s = 0, -1, -2, \ldots$. The zeta function enters number theory through Euler's identity

$$\prod_p \frac{1}{1-p^{-s}} = \sum_{n=1}^{\infty} \frac{1}{n^s} = \zeta(s)$$

over all primes. ζ is analytic for $\mathrm{Re}(s) > 1$; Riemann tied the primes to its meromorphic continuation to $\mathbb{C}$. We prove this and the functional equation (symmetry about $\mathrm{Re}(s) = 1/2$, via the theta function), and estimate ζ near $\mathrm{Re}(s) = 1$ for the prime number theorem (Chapter 7). The variable is s.

1 The gamma function

Motivation. Continuation, product formula and $\Gamma(s)\Gamma(1-s) = \pi/\sin \pi s$ are the tools needed for ζ.

For $s > 0$ the **gamma function** is

$$\Gamma(s) = \int_0^{\infty} e^{-t} t^{s-1}\, dt. \tag{1}$$

The integral converges: t^{s-1} is integrable near 0, and e^{-t} decays at infinity.

Proposition 1.1

The gamma function extends to an analytic function in the half-plane $\mathrm{Re}(s) > 0$, and is still given there by the integral formula (1).

Proof. It suffices to work in strips $S_{\delta,M} = \{\delta < \mathrm{Re}(s) < M\}$. With $\sigma = \mathrm{Re}(s)$, $|e^{-t}t^{s-1}| = e^{-t}t^{\sigma-1}$, so $\Gamma(s) = \lim_{\epsilon\to 0} F_\epsilon(s)$, $F_\epsilon(s) = \int_\epsilon^{1/\epsilon} e^{-t}t^{s-1}\, dt$, holomorphic by Theorem 5.4. By Theorem 5.2 it suffices that $F_\epsilon \to \Gamma$ uniformly on the strip. Now

$$|\Gamma(s) - F_\epsilon(s)| \le \int_0^{\epsilon} e^{-t}t^{\sigma-1}\, dt + \int_{1/\epsilon}^{\infty} e^{-t}t^{\sigma-1}\, dt.$$

The first integral is $\le \epsilon^\delta/\delta$ for $0 < \epsilon < 1$, and the second is $\le \int_{1/\epsilon}^{\infty} e^{-t}t^{M-1}\, dt \le C\int_{1/\epsilon}^{\infty} e^{-t/2}\, dt \to 0$, uniformly in the strip. ∎

1.1 Analytic continuation

Though the integral diverges elsewhere, a meromorphic function on $\mathbb{C}$ equals Γ for $\mathrm{Re}(s) > 0$: its **analytic continuation**, still Γ (unique, as the complement of the poles is connected).

Lemma 1.2

If $\mathrm{Re}(s) > 0$, then

$$\Gamma(s+1) = s\Gamma(s). \tag{2}$$

As a consequence $\Gamma(n+1) = n!$ for $n = 0, 1, 2, \ldots$.

Proof. Integrating by parts,

$$\int_\epsilon^{1/\epsilon} \frac{d}{dt}(e^{-t}t^s)\,dt = -\int_\epsilon^{1/\epsilon} e^{-t}t^s\,dt + s\int_\epsilon^{1/\epsilon} e^{-t}t^{s-1}\,dt,$$

and letting $\epsilon \to 0$ the left side vanishes, since $e^{-t}t^s \to 0$ as $t \to 0$ or $t \to \infty$. Also $\Gamma(1) = \int_0^\infty e^{-t}\,dt = 1$; apply (2) successively. ∎

Theorem 1.3

Γ continues to a meromorphic function on $\mathbb{C}$ whose only singularities are simple poles at $s = 0, -1, \ldots$, with $\operatorname{res}_{s=-n}\Gamma = (-1)^n/n!$.

Proof. It suffices to extend Γ to each half-plane $\operatorname{Re}(s) > -m$, $m \geq 1$. For $\operatorname{Re}(s) > -m$ define

$$F_m(s) = \frac{\Gamma(s+m)}{(s+m-1)(s+m-2)\cdots s}.$$

F_m is meromorphic in $\operatorname{Re}(s) > -m$ with simple poles at $s = 0, -1, \ldots, -m+1$ and residues

$$\operatorname{res}_{s=-n} F_m(s) = \frac{\Gamma(-n+m)}{(m-1-n)!(-1)(-2)\cdots(-n)} = \frac{(m-n-1)!}{(m-1-n)!(-1)(-2)\cdots(-n)} = \frac{(-1)^n}{n!}.$$

By Lemma 1.2, $F_m = \Gamma$ for $\operatorname{Re}(s) > 0$, so by uniqueness $F_m = F_k$ $(1 \leq k \leq m)$ on the domain of F_k; this gives the continuation. ∎

Remark

By analytic continuation, $\Gamma(s+1) = s\Gamma(s)$ holds whenever $s \neq 0, -1, -2, \ldots$. At $s = -n$, $n \geq 1$, both sides are infinite and $\operatorname{res}_{s=-n}\Gamma(s+1) = -n\operatorname{res}_{s=-n}\Gamma(s)$. At $s = 0$, $\Gamma(1) = \lim_{s\to 0} s\Gamma(s)$.

An alternate proof. For $\operatorname{Re}(s) > 0$ split $\Gamma(s) = \int_0^1 e^{-t}t^{s-1}\,dt + \int_1^\infty e^{-t}t^{s-1}\,dt$. The second integral is entire; expanding e^{-t} and integrating termwise, $\int_0^1 e^{-t}t^{s-1}\,dt = \sum_{n=0}^\infty \frac{(-1)^n}{n!(n+s)}$, so

$$\Gamma(s) = \sum_{n=0}^\infty \frac{(-1)^n}{n!(n+s)} + \int_1^\infty e^{-t}t^{s-1}\,dt \qquad \text{for } \operatorname{Re}(s) > 0. \tag{3}$$

The series is meromorphic on $\mathbb{C}$ with poles at $-n$, residues $(-1)^n/n!$: for $|s| < R$ split it at $N > 2R$; the finite part has the right poles, and for $n > N$, $|\frac{(-1)^n}{n!(n+s)}| \leq \frac{1}{n!R}$, so the tail is holomorphic. Hence (3) holds on $\mathbb{C}$.

1.2 Further properties of Γ

Theorem 1.4

For all $s \in \mathbb{C}$,

$$\Gamma(s)\Gamma(1-s) = \frac{\pi}{\sin \pi s}. \tag{4}$$

$\Gamma(s)\Gamma(1-s)$ and $\pi/\sin \pi s$ are meromorphic with simple poles at the integers, so by analytic continuation it suffices to prove (4) for $0 < s < 1$.

Lemma 1.5

For $0 < a < 1$, $\displaystyle\int_0^\infty \frac{v^{a-1}}{1+v}\,dv = \frac{\pi}{\sin \pi a}$.

Proof. With $v = e^x$, $\int_0^\infty \frac{v^{a-1}}{1+v}\,dv = \int_{-\infty}^\infty \frac{e^{ax}}{1+e^x}\,dx$, which equals $\pi/\sin \pi a$ by Example 2. ∎

Proof of Theorem 1.4. For $0 < s < 1$ and $t > 0$, the change of variables $u = vt$ gives $\Gamma(1-s) = \int_0^\infty e^{-u}u^{-s}\,du = t\int_0^\infty e^{-vt}(vt)^{-s}\,dv$. Hence

$$\begin{aligned}\Gamma(1-s)\Gamma(s) &= \int_0^\infty e^{-t}t^{s-1}\Gamma(1-s)\,dt = \int_0^\infty e^{-t}t^{s-1}\Big(t\int_0^\infty e^{-vt}(vt)^{-s}\,dv\Big)dt \\ &= \int_0^\infty\int_0^\infty e^{-t[1+v]}v^{-s}\,dt\,dv = \int_0^\infty \frac{v^{-s}}{1+v}\,dv = \frac{\pi}{\sin\pi(1-s)} = \frac{\pi}{\sin \pi s}.\end{aligned}$$

∎

With $s = 1/2$, since $\Gamma(s) > 0$ for $s > 0$, $\Gamma(1/2) = \sqrt{\pi}$.

Theorem 1.6

The function Γ has the following properties:
(i) $1/\Gamma(s)$ is an entire function of s with simple zeros at $s = 0, -1, -2, \ldots$ and it vanishes nowhere else.
(ii) $1/\Gamma(s)$ has growth $\left|\frac{1}{\Gamma(s)}\right| \le c_1 e^{c_2|s|\log|s|}$. Therefore, $1/\Gamma$ is of order 1 in the sense that for every $\epsilon > 0$, there exists a bound $c(\epsilon)$ so that $\left|\frac{1}{\Gamma(s)}\right| \le c(\epsilon)e^{c_2|s|^{1+\epsilon}}$.

Proof. By Theorem 1.4,

$$\frac{1}{\Gamma(s)} = \Gamma(1-s)\frac{\sin \pi s}{\pi}, \tag{5}$$

so the simple poles of $\Gamma(1-s)$ at $s = 1, 2, \ldots$ are cancelled by zeros of $\sin \pi s$, and $1/\Gamma$ is entire with simple zeros at $0, -1, -2, \ldots$.

For the estimate, first $\int_1^\infty e^{-t}t^\sigma\,dt \le e^{(\sigma+1)\log(\sigma+1)}$ for $\sigma > 0$: choose n with $\sigma \le n \le \sigma + 1$; then $\int_1^\infty e^{-t}t^\sigma\,dt \le \int_0^\infty e^{-t}t^n\,dt = n! \le n^n = e^{n\log n} \le e^{(\sigma+1)\log(\sigma+1)}$. Since (3) holds on $\mathbb{C}$, by (5)

$$\frac{1}{\Gamma(s)} = \Big(\sum_{n=0}^\infty \frac{(-1)^n}{n!(n+1-s)}\Big)\frac{\sin \pi s}{\pi} + \Big(\int_1^\infty e^{-t}t^{-s}\,dt\Big)\frac{\sin \pi s}{\pi}.$$

$\left|\int_1^\infty e^{-t}t^{-s}\,dt\right| \le \int_1^\infty e^{-t}t^{|\sigma|}\,dt \le e^{(|s|+1)\log(|s|+1)}$ and $|\sin \pi s| \le e^{\pi|s|}$, so the second term is $\le c_1 e^{c_2|s|\log|s|}$. For the first term: if $|\mathrm{Im}(s)| > 1$ it is $\le ce^{\pi|s|}$. If $|\mathrm{Im}(s)| \le 1$, let k be the integer with $k - 1/2 \le \mathrm{Re}(s) < k + 1/2$. If $k \ge 1$,

$$\sum_{n=0}^\infty \frac{(-1)^n}{n!(n+1-s)}\frac{\sin \pi s}{\pi} = (-1)^{k-1}\frac{\sin \pi s}{\pi(k-1)!(k-s)} + \sum_{n\ne k-1}\frac{(-1)^n}{n!(n+1-s)}\frac{\sin \pi s}{\pi};$$

the first term is bounded since $\sin \pi s$ vanishes at $s = k$, and the second is majorized by $c\sum 1/n!$. If $k \le 0$ then $\mathrm{Re}(s) < 1/2$ and the series is again bounded by $c\sum 1/n!$. ∎

Theorem 1.7

For all $s \in \mathbb{C}$,

$$\frac{1}{\Gamma(s)} = e^{\gamma s} s \prod_{n=1}^{\infty} \left(1 + \frac{s}{n}\right) e^{-s/n}.$$

Here γ is **Euler's constant**, $\gamma = \lim_{N\to\infty} \left(\sum_{n=1}^{N} \frac{1}{n} - \log N\right)$. The limit exists: $\sum_{n=1}^{N} \frac{1}{n} - \log N = \sum_{n=1}^{N-1} \int_n^{n+1} \left[\frac{1}{n} - \frac{1}{x}\right] dx + \frac{1}{N}$, and by the mean value theorem $0 \le \frac{1}{n} - \frac{1}{x} \le \frac{1}{n^2}$ for $n \le x \le n+1$; so the sum is $\sum_{n=1}^{N-1} a_n + \frac{1}{N}$ with $0 \le a_n \le 1/n^2$, and $\sum a_n$ converges.

Proof. By Hadamard's theorem, since $1/\Gamma$ is entire of growth order 1 with simple zeros at $0, -1, -2, \ldots$,

$$\frac{1}{\Gamma(s)} = e^{As+B} s \prod_{n=1}^{\infty} \left(1 + \frac{s}{n}\right) e^{-s/n}.$$

Since $s\Gamma(s) \to 1$ as $s \to 0$, $B = 0$ (or an integer multiple of $2\pi i$, which gives the same result). Putting $s = 1$ and using $\Gamma(1) = 1$,

$$e^{-A} = \prod_{n=1}^{\infty} \left(1 + \frac{1}{n}\right) e^{-1/n} = \lim_{N\to\infty} e^{\sum_{n=1}^{N} [\log(1+1/n) - 1/n]} = \lim_{N\to\infty} e^{-(\sum_{n=1}^{N} 1/n) + \log N + \log(1+1/N)} = e^{-\gamma}.$$

So $A = \gamma + 2\pi i k$; since $\Gamma(s)$ is real for real s, $k = 0$. ∎

So $1/\Gamma$ is characterized (up to two constants) as entire with (i) simple zeros exactly at $s = 0, -1, -2, \ldots$ and (ii) order 1. $\sin \pi s$ (zeros at all integers) satisfies the stricter $O(e^{c|s|})$, which fails for $1/\Gamma$.

2 The zeta function

Motivation. To use complex analysis on ζ, continue it to $\mathbb{C} - \{1\}$ and relate $\zeta(s)$ to $\zeta(1-s)$.

The **Riemann zeta function** is initially defined for real $s > 1$ by $\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}$.

2.1 Functional equation and analytic continuation

Proposition 2.1

The series defining $\zeta(s)$ converges for $\mathrm{Re}(s) > 1$, and the function ζ is holomorphic in this half-plane.

Proof. For $s = \sigma + it$, $|n^{-s}| = |e^{-s \log n}| = e^{-\sigma \log n} = n^{-\sigma}$. If $\sigma \ge 1 + \delta > 1$ the series is uniformly bounded by $\sum 1/n^{1+\delta} < \infty$; so it converges uniformly on each half-plane $\mathrm{Re}(s) \ge 1 + \delta$ and defines a holomorphic function in $\mathrm{Re}(s) > 1$. ∎

The continuation is subtler than for Γ; it relates ζ to the **theta function** $\vartheta(t) = \sum_{n=-\infty}^{\infty} e^{-\pi n^2 t}$, $t > 0$, which satisfies (by Poisson summation, (4) of Chapter 4) $\vartheta(t) = t^{-1/2} \vartheta(1/t)$. We need

$$\vartheta(t) \le C t^{-1/2} \quad \text{as } t \to 0, \qquad |\vartheta(t) - 1| \le C e^{-\pi t} \quad \text{for } t \ge 1 :$$

the first follows from the functional equation, the second from $\sum_{n\ge1} e^{-\pi n^2 t} \le \sum_{n\ge1} e^{-\pi n t} \le C e^{-\pi t}$ for $t \ge 1$.

Theorem 2.2

If $\mathrm{Re}(s) > 1$, then

$$\pi^{-s/2}\Gamma(s/2)\zeta(s) = \frac{1}{2}\int_0^\infty u^{(s/2)-1}[\vartheta(u) - 1]\,du.$$

Proof. The key is

$$\int_0^\infty e^{-\pi n^2 u} u^{(s/2)-1}\,du = \pi^{-s/2}\Gamma(s/2)n^{-s}, \qquad n \geq 1 : \tag{6}$$

with $u = t/\pi n^2$ the left side becomes $\left(\int_0^\infty e^{-t}t^{(s/2)-1}\,dt\right)(\pi n^2)^{-s/2}$. Since $\frac{\vartheta(u)-1}{2} = \sum_{n=1}^\infty e^{-\pi n^2 u}$, and the estimates on ϑ justify interchanging sum and integral,

$$\frac{1}{2}\int_0^\infty u^{(s/2)-1}[\vartheta(u) - 1]\,du = \sum_{n=1}^\infty \int_0^\infty u^{(s/2)-1}e^{-\pi n^2 u}\,du = \pi^{-s/2}\Gamma(s/2)\sum_{n=1}^\infty n^{-s} = \pi^{-s/2}\Gamma(s/2)\zeta(s).$$

■

The **xi function**, a more symmetric modification of ζ, is defined for $\mathrm{Re}(s) > 1$ by

$$\xi(s) = \pi^{-s/2}\Gamma(s/2)\zeta(s). \tag{7}$$

Theorem 2.3

ξ is holomorphic for $\mathrm{Re}(s) > 1$ and continues meromorphically to $\mathbb{C}$ with simple poles at $s = 0, 1$, and $\xi(s) = \xi(1 - s)$.

Proof. Heuristically: multiply $\sum_n e^{-\pi n^2 u} = u^{-1/2}\sum_n e^{-\pi n^2/u}$ by $u^{(s/2)-1}$ and integrate, using (6) and the change $u \mapsto 1/u$ (discarding the $n = 0$ terms). Precisely: let $\psi(u) = [\vartheta(u) - 1]/2$. The functional equation of ϑ gives

$$\psi(u) = u^{-1/2}\psi(1/u) + \frac{1}{2u^{1/2}} - \frac{1}{2}.$$

By Theorem 2.2, for $\mathrm{Re}(s) > 1$,

$$\begin{aligned}
\pi^{-s/2}\Gamma(s/2)\zeta(s) &= \int_0^1 u^{(s/2)-1}\psi(u)\,du + \int_1^\infty u^{(s/2)-1}\psi(u)\,du \\
&= \int_0^1 u^{(s/2)-1}\Big[u^{-1/2}\psi(1/u) + \frac{1}{2u^{1/2}} - \frac{1}{2}\Big]du + \int_1^\infty u^{(s/2)-1}\psi(u)\,du \\
&= \frac{1}{s-1} - \frac{1}{s} + \int_1^\infty (u^{(-s/2)-1/2} + u^{(s/2)-1})\psi(u)\,du
\end{aligned}$$

(substituting $u \mapsto 1/u$ in the first integral). Therefore

$$\xi(s) = \frac{1}{s-1} - \frac{1}{s} + \int_1^\infty (u^{(-s/2)-1/2} + u^{(s/2)-1})\psi(u)\,du.$$

As ψ decays exponentially, the integral is entire, giving the continuation with simple poles at $0, 1$; the integral and $\frac{1}{s-1} - \frac{1}{s}$ are invariant under $s \mapsto 1 - s$. ■

Theorem 2.4

The zeta function has a meromorphic continuation into the entire complex plane, whose only singularity is a simple pole at $s = 1$.

Proof. By (7), $\zeta(s) = \pi^{s/2}\xi(s)/\Gamma(s/2)$; the simple zero of $1/\Gamma(s/2)$ at 0 cancels the pole of ξ, so ζ has only a simple pole at $s = 1$. ■

A more elementary approach gives the continuation to $\mathrm{Re}(s) > 0$ and is useful for the growth of ζ near $\mathrm{Re}(s) = 1$: compare $\sum n^{-s}$ with $\int_1^\infty x^{-s}\,dx$.

Proposition 2.5

There is a sequence of entire functions $\{\delta_n(s)\}_{n=1}^{\infty}$ that satisfy the estimate $|\delta_n(s)| \le |s|/n^{\sigma+1}$, where $s = \sigma + it$, and such that

$$\sum_{1\le n<N} \frac{1}{n^s} - \int_1^N \frac{dx}{x^s} = \sum_{1\le n<N} \delta_n(s), \tag{8}$$

whenever N is an integer > 1.

Proof. Set

$$\delta_n(s) = \int_n^{n+1} \Big[\frac{1}{n^s} - \frac{1}{x^s}\Big] dx. \tag{9}$$

By the mean-value theorem applied to x^{-s}, $\big|\frac{1}{n^s} - \frac{1}{x^s}\big| \le \frac{|s|}{n^{\sigma+1}}$ for $n \le x \le n+1$, so $|\delta_n(s)| \le |s|/n^{\sigma+1}$; and $\int_1^N \frac{dx}{x^s} = \sum_{1\le n<N} \int_n^{n+1} \frac{dx}{x^s}$. ∎

Corollary 2.6

For $\mathrm{Re}(s) > 0$ we have

$$\zeta(s) - \frac{1}{s-1} = H(s),$$

where $H(s) = \sum_{n=1}^{\infty} \delta_n(s)$ is holomorphic in the half-plane $\mathrm{Re}(s) > 0$.

Proof. For $\mathrm{Re}(s) > 1$ let $N \to \infty$ in (8): $\int_1^N x^{-s} dx \to \frac{1}{s-1}$, and $|\delta_n(s)| \le |s|/n^{\sigma+1}$ makes $\sum \delta_n(s)$ converge uniformly on compacta of $\mathrm{Re}(s) \ge \delta > 0$. So the identity holds for $\mathrm{Re}(s) > 1$, and H is holomorphic in $\mathrm{Re}(s) > 0$, extending ζ there. ∎

For $\mathrm{Re}(s) > 1$, $|\zeta(s)| \le \sum n^{-\sigma}$, so ζ is bounded in $\mathrm{Re}(s) \ge 1 + \delta$. The growth near $\mathrm{Re}(s) = 1$ is mild (the following estimates are crude but suffice for Chapter 7).

Proposition 2.7

Suppose $s = \sigma + it$ with $\sigma, t \in \mathbb{R}$. Then for each σ_0, $0 \le \sigma_0 \le 1$, and every $\epsilon > 0$, there exists a constant c_ϵ so that (i) $|\zeta(s)| \le c_\epsilon |t|^{1-\sigma_0+\epsilon}$, if $\sigma_0 \le \sigma$ and $|t| \ge 1$.
(ii) $|\zeta'(s)| \le c_\epsilon |t|^{\epsilon}$, if $1 \le \sigma$ and $|t| \ge 1$.

In particular $\zeta(1 + it) = O(|t|^{\epsilon})$ as $|t| \to \infty$, and likewise for ζ'.

Proof. Besides $|\delta_n(s)| \le |s|/n^{\sigma+1}$ we have $|\delta_n(s)| \le 2/n^{\sigma}$, from (9) and $|n^{-s}| = n^{-\sigma}$, $|x^{-s}| \le n^{-\sigma}$ for $x \ge n$. Combining via $A = A^{\delta} A^{1-\delta}$ $(0 \le \delta \le 1)$,

$$|\delta_n(s)| \le \Big(\frac{|s|}{n^{\sigma_0+1}}\Big)^{\delta} \Big(\frac{2}{n^{\sigma_0}}\Big)^{1-\delta} \le \frac{2|s|^{\delta}}{n^{\sigma_0+\delta}} \qquad (\sigma \ge \sigma_0).$$

Take $\delta = 1 - \sigma_0 + \epsilon$ and use Corollary 2.6: for $\sigma \ge \sigma_0$,

$$|\zeta(s)| \le \Big|\frac{1}{s-1}\Big| + 2|s|^{1-\sigma_0+\epsilon} \sum_{n=1}^{\infty} \frac{1}{n^{1+\epsilon}},$$

which gives (i). For (ii), by the Cauchy integral formula on the circle of radius r about s,

$$\zeta'(s) = \frac{1}{2\pi r} \int_0^{2\pi} \zeta(s + re^{i\theta}) e^{-i\theta}\, d\theta.$$

Take $r = \epsilon$; this circle lies in $\mathrm{Re} \ge 1 - \epsilon$, and (ii) follows from (i) (with $\sigma_0 = 1 - \epsilon$ and 2ϵ in place of ϵ). ∎

3 Selected exercises

Motivation. Four exercises from Stein–Shakarchi, Chapter 6 (book numbers kept).

Exercise 6.1 (Gauss's limit formula)

Prove that

$$\Gamma(s) = \lim_{n\to\infty} \frac{n^s\, n!}{s(s+1)\cdots(s+n)} \qquad \text{whenever } s \neq 0, -1, -2, \dots.$$

Solution.

Step 1. Let $H_n = \sum_{k=1}^{n} 1/k$. The partial products of Theorem 1.7 are

$$P_n(s) = e^{\gamma s} s \prod_{k=1}^{n} \Big(1 + \frac{s}{k}\Big) e^{-s/k},$$

and $P_n(s) \to 1/\Gamma(s)$ for every s.

Step 2. Since $s\prod_{k\le n}(1+s/k) = s(s+1)\cdots(s+n)/n!$ and $n^{-s} = e^{-s\log n}$,

$$P_n(s) = \frac{s(s+1)\cdots(s+n)}{n!\, n^s}\, e^{-s(H_n - \log n - \gamma)}.$$

Step 3. $H_n - \log n - \gamma \to 0$, so $s(s+1)\cdots(s+n)/(n!\,n^s) \to 1/\Gamma(s)$, non-zero for $s \neq 0, -1, \dots$ (Theorem 1.6); take reciprocals. ∎

Exercise 6.3 (Duplication formula)

Using Exercise 5.6(b), prove that $\Gamma(s)\Gamma(s+1/2) = 2^{1-2s}\sqrt{\pi}\,\Gamma(2s)$ for all s with $2s \neq 0, -1, -2, \dots$.

Solution. Fix s with $2s \notin \{0, -1, -2, \dots\}$, so none of s, $s+1/2$, $2s$ is a pole of Γ.

Step 1. By Exercise 6.1 applied to s and $s+\frac{1}{2}$ (with the same n),

$$\Gamma(s)\Gamma(s+\tfrac{1}{2}) = \lim_{n\to\infty} \frac{n^{2s+1/2}(n!)^2}{\prod_{k=0}^{n}(s+k)(s+k+\frac{1}{2})}.$$

Step 2. Since $(s+k)(s+k+\frac{1}{2}) = \frac{1}{4}(2s+2k)(2s+2k+1)$,

$$\prod_{k=0}^{n}(s+k)(s+k+\tfrac{1}{2}) = 2^{-2n-2}\prod_{j=0}^{2n+1}(2s+j).$$

Step 3. Exercise 6.1 applied to $2s$, along the subsequence $2n+1$, gives

$$\Gamma(2s) = \lim_{n\to\infty} \frac{(2n+1)^{2s}(2n+1)!}{\prod_{j=0}^{2n+1}(2s+j)}.$$

Step 4. Dividing the limits in Steps 1 and 3 (the second is non-zero), and using Step 2,

$$\frac{\Gamma(s)\Gamma(s+\frac{1}{2})}{\Gamma(2s)} = \lim_{n\to\infty} \Big(\frac{n}{2n+1}\Big)^{2s} \cdot \frac{2^{2n+2}\, n^{1/2}(n!)^2}{(2n+1)!}.$$

Step 5. We evaluate the two factors. First, $(n/(2n+1))^{2s} = e^{2s\log(n/(2n+1))} \to e^{-2s\log 2} = 2^{-2s}$. Second, since $(2n+1)! = (2n)!\,(2n+1)$,

$$\frac{2^{2n+2}n^{1/2}(n!)^2}{(2n+1)!} = 4\cdot\frac{2^{2n}(n!)^2}{(2n)!\,(2n+1)^{1/2}}\cdot\Big(\frac{n}{2n+1}\Big)^{1/2} \to 4\sqrt{\frac{\pi}{2}}\cdot\frac{1}{\sqrt{2}} = 2\sqrt{\pi},$$

by Exercise 5.6(b).

Step 6. Hence $\Gamma(s)\Gamma(s+\frac12) = 2^{-2s}\cdot 2\sqrt{\pi}\,\Gamma(2s) = 2^{1-2s}\sqrt{\pi}\,\Gamma(2s)$. ∎

Exercise 6.5 (Γ on the line $\mathrm{Re}(s) = 1/2$)

Show that $|\Gamma(\frac12 + it)|^2 = \dfrac{2\pi}{e^{\pi t} + e^{-\pi t}}$ for $t \in \mathbb{R}$.

Solution.

Step 1. For $\mathrm{Re}(s) > 0$, conjugating the integral (1) gives $\overline{\Gamma(s)} = \int_0^\infty e^{-u}u^{\bar s - 1}\,du = \Gamma(\bar s)$. Both $s \mapsto \overline{\Gamma(\bar s)}$ and Γ are meromorphic on $\mathbb{C}$ and agree on $\mathrm{Re}(s) > 0$, so $\overline{\Gamma(s)} = \Gamma(\bar s)$ for all s off the poles.

Step 2. For $s = \frac12 + it$ we have $\bar s = \frac12 - it = 1 - s$. By Theorem 1.4,

$$|\Gamma(s)|^2 = \Gamma(s)\Gamma(\bar s) = \Gamma(s)\Gamma(1-s) = \frac{\pi}{\sin(\frac{\pi}{2} + i\pi t)} = \frac{\pi}{\cos(i\pi t)} = \frac{\pi}{\cosh \pi t} = \frac{2\pi}{e^{\pi t} + e^{-\pi t}}.$$

∎

Exercise 6.7 (The Beta function)

For $\mathrm{Re}(\alpha) > 0$ and $\mathrm{Re}(\beta) > 0$ let $B(\alpha,\beta) = \int_0^1 (1-t)^{\alpha-1}t^{\beta-1}\,dt$.

(a) Prove that $B(\alpha,\beta) = \dfrac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$.

(b) Show that $B(\alpha,\beta) = \displaystyle\int_0^\infty \frac{u^{\alpha-1}}{(1+u)^{\alpha+\beta}}\,du$.

Solution. All powers of positive reals are principal, so $(xy)^w = x^w y^w$ and $|x^w| = x^{\mathrm{Re}\,w}$ for $x, y > 0$. In particular $|(1-t)^{\alpha-1}t^{\beta-1}| = (1-t)^{\mathrm{Re}\,\alpha-1}t^{\mathrm{Re}\,\beta-1}$ is integrable on $(0,1)$, so B is well defined.

(a)

Step 1. The double integral

$$\Gamma(\alpha)\Gamma(\beta) = \int_0^\infty\int_0^\infty t^{\alpha-1}s^{\beta-1}e^{-t-s}\,dt\,ds$$

converges absolutely, since the modulus of the integrand is $t^{\mathrm{Re}\,\alpha-1}s^{\mathrm{Re}\,\beta-1}e^{-t-s}$. Hence Fubini's theorem and changes of variables are allowed.

Step 2. The map $(u,r) \mapsto (t,s) = (u(1-r), ur)$ is a C^1 bijection from $(0,\infty)\times(0,1)$ onto $(0,\infty)^2$, with inverse $u = t+s$, $r = s/(t+s)$. Its Jacobian determinant is $(1-r)u + ru = u$.

Step 3. Therefore

$$\Gamma(\alpha)\Gamma(\beta) = \int_0^\infty\int_0^1 u^{\alpha+\beta-1}(1-r)^{\alpha-1}r^{\beta-1}e^{-u}\,dr\,du = \Gamma(\alpha+\beta)\,B(\alpha,\beta).$$

Since Γ never vanishes (Theorem 1.6(i)), we may divide by $\Gamma(\alpha+\beta)$.

(b) Substitute $t = 1/(1+u)$, $u \in (0,\infty)$. Then $1-t = u/(1+u)$ and $dt = -du/(1+u)^2$, and t decreases from 1 to 0. Hence

$$B(\alpha,\beta) = \int_0^\infty \frac{u^{\alpha-1}}{(1+u)^{\alpha-1}} \frac{1}{(1+u)^{\beta-1}} \frac{du}{(1+u)^2} = \int_0^\infty \frac{u^{\alpha-1}}{(1+u)^{\alpha+\beta}}\, du.$$

■

Chapter 7. The Zeta Function and Prime Number Theorem

Overview

Euler's product links ζ to the primes (so $\sum_p 1/p$ diverges). Let

$$\pi(x) = \text{number of primes less than or equal to } x.$$

Legendre and Gauss conjectured

$$\pi(x) \sim \frac{x}{\log x} \qquad \text{as } x \to \infty, \tag{1}$$

where $f \sim g$ means $f/g \to 1$. Tchebychev proved $A\frac{x}{\log x} \le \pi(x) \le B\frac{x}{\log x}$ for large x (2); Hadamard and de la Vallée Poussin (1896) proved (1), the *prime number theorem*, by complex analysis. The heart is $\zeta(s) \neq 0$ on $\mathrm{Re}(s) = 1$ (in fact equivalent).

1 Zeros of the zeta function

Motivation. We prove $\zeta \neq 0$ on $\mathrm{Re}\, s = 1$ and the bounds on $1/\zeta$ needed for the contour integration.

Euler's identity. For $\mathrm{Re}(s) > 1$,

$$\zeta(s) = \prod_p \frac{1}{1 - p^{-s}}.$$

Proof. Each factor is a geometric series $1 + p^{-s} + p^{-2s} + \cdots$. Let $s > 1$ and $M > N$. By unique factorization each $n \le N$ is a product of primes $\le N$, each repeated fewer than M times, so

$$\sum_{n=1}^{N} \frac{1}{n^s} \le \prod_{p \le N} \left(1 + \frac{1}{p^s} + \frac{1}{p^{2s}} + \cdots + \frac{1}{p^{Ms}}\right) \le \prod_{p \le N} \frac{1}{1 - p^{-s}} \le \prod_p \frac{1}{1 - p^{-s}},$$

and $N \to \infty$ gives $\sum n^{-s} \le \prod_p (1 - p^{-s})^{-1}$. Conversely $\prod_{p \le N}(1 + p^{-s} + \cdots + p^{-Ms}) \le \sum n^{-s}$; let $M \to \infty$, then $N \to \infty$. (Complex s with $\mathrm{Re}(s) > 1$: analytic continuation, or the same argument with absolute convergence.) ∎

By Proposition 3.1, $\zeta(s) \neq 0$ for $\mathrm{Re}(s) > 1$. The functional equation $\xi(s) = \xi(1 - s)$ reads $\pi^{-s/2}\Gamma(s/2)\zeta(s) = \pi^{-(1-s)/2}\Gamma((1-s)/2)\zeta(1-s)$, so

$$\zeta(s) = \pi^{s-1/2}\frac{\Gamma((1-s)/2)}{\Gamma(s/2)}\zeta(1-s).$$

For $\mathrm{Re}(s) < 0$:

(i) $\zeta(1 - s)$ has no zeros, as $\mathrm{Re}(1 - s) > 1$;

(ii) $\Gamma((1 - s)/2)$ is zero free;

(iii) $1/\Gamma(s/2)$ has zeros at $s = -2, -4, -6, \ldots$. Hence:

Theorem 1.1

The only zeros of ζ outside the strip $0 \le \mathrm{Re}(s) \le 1$ are at the negative even integers, $-2, -4, -6, \ldots$.

$0 \le \mathrm{Re}(s) \le 1$ is the **critical strip**; zeros outside it are **trivial**. The **Riemann hypothesis**, still open: *the zeros in the critical strip lie on* $\mathrm{Re}(s) = 1/2$. We prove:

Theorem 1.2

The zeta function has no zeros on the line $\mathrm{Re}(s) = 1$.

(Near $s = 1$ there are no zeros because of the pole; the point is $\zeta(1 + it) \neq 0$ for all t.) By the functional equation there are none on $\mathrm{Re}(s) = 0$ either.

Lemma 1.3

If $\mathrm{Re}(s) > 1$, then

$$\log \zeta(s) = \sum_{p,m} \frac{p^{-ms}}{m} = \sum_{n=1}^{\infty} c_n n^{-s}$$

for some $c_n \geq 0$.

Proof. For $s > 1$, take the logarithm of Euler's product and use $\log \frac{1}{1-x} = \sum_{m=1}^{\infty} \frac{x^m}{m}$ $(0 \leq x < 1)$:

$$\log \zeta(s) = \sum_{p} \log\Big(\frac{1}{1 - p^{-s}}\Big) = \sum_{p,m} \frac{p^{-ms}}{m}.$$

The double sum converges absolutely (Note at the end of the chapter). The formula holds for $\mathrm{Re}(s) > 1$ by analytic continuation, $\log \zeta$ being defined on the simply connected half-plane where $\zeta \neq 0$ (Theorem 6.2). And $c_n = 1/m$ if $n = p^m$, 0 otherwise. ■

Lemma 1.4

If $\theta \in \mathbb{R}$, then $3 + 4\cos\theta + \cos 2\theta \geq 0$.

This follows from $3 + 4\cos\theta + \cos 2\theta = 2(1 + \cos\theta)^2$.

Corollary 1.5

If $\sigma > 1$ and t is real, then $\log|\zeta^3(\sigma)\zeta^4(\sigma + it)\zeta(\sigma + 2it)| \geq 0$.

Proof. With $s = \sigma + it$, $\mathrm{Re}(n^{-s}) = \mathrm{Re}(e^{-(\sigma+it)\log n}) = n^{-\sigma}\cos(t \log n)$. Hence, with $\theta_n = t \log n$,

$$\begin{aligned}\log|\zeta^3(\sigma)\zeta^4(\sigma + it)\zeta(\sigma + 2it)| &= 3\,\mathrm{Re}[\log \zeta(\sigma)] + 4\,\mathrm{Re}[\log \zeta(\sigma + it)] + \mathrm{Re}[\log \zeta(\sigma + 2it)] \\ &= \sum c_n n^{-\sigma}(3 + 4\cos\theta_n + \cos 2\theta_n) \geq 0,\end{aligned}$$

by Lemma 1.4 and $c_n \geq 0$. ■

Proof of Theorem 1.2. Suppose $\zeta(1 + it_0) = 0$, $t_0 \neq 0$. Then $|\zeta(\sigma + it_0)|^4 \leq C(\sigma - 1)^4$, the simple pole gives $|\zeta(\sigma)|^3 \leq C'(\sigma - 1)^{-3}$, and $|\zeta(\sigma + 2it_0)|$ stays bounded as $\sigma \to 1$. So $|\zeta^3(\sigma)\zeta^4(\sigma + it_0)\zeta(\sigma + 2it_0)| \to 0$ as $\sigma \to 1^+$, contradicting Corollary 1.5 (the product has modulus ≥ 1). ■

1.1 Estimates for $1/\zeta(s)$

The proof uses ζ'/ζ near $\mathrm{Re}(s) = 1$, so we need bounds on ζ' (Proposition 2.7) and $1/\zeta$; this is a quantitative Theorem 1.2.

Proposition 1.6

For every $\epsilon > 0$, we have $1/|\zeta(s)| \leq c_\epsilon |t|^\epsilon$ when $s = \sigma + it$, $\sigma \geq 1$, and $|t| \geq 1$.

Proof. From Corollary 1.5, $|\zeta^3(\sigma)\zeta^4(\sigma+it)\zeta(\sigma+2it)| \geq 1$ for $\sigma \geq 1$. With the estimate for ζ in Proposition 2.7 and $|\zeta(\sigma)| \leq c/(\sigma-1)$,

$$|\zeta^4(\sigma+it)| \geq c|\zeta^{-3}(\sigma)||t|^{-\epsilon} \geq c'(\sigma-1)^3|t|^{-\epsilon},$$

so

$$|\zeta(\sigma+it)| \geq c'(\sigma-1)^{3/4}|t|^{-\epsilon/4} \qquad (\sigma \geq 1,\ |t| \geq 1). \tag{3}$$

Two cases, A to be chosen. If $\sigma - 1 \geq A|t|^{-5\epsilon}$, (3) gives $|\zeta(\sigma+it)| \geq A'|t|^{-4\epsilon}$ (rename 4ϵ as ϵ). If $\sigma - 1 < A|t|^{-5\epsilon}$, take $\sigma' > \sigma$ with $\sigma' - 1 = A|t|^{-5\epsilon}$; then $|\zeta(\sigma+it)| \geq |\zeta(\sigma'+it)| - |\zeta(\sigma'+it) - \zeta(\sigma+it)|$, and by the mean value theorem and Proposition 2.7, $|\zeta(\sigma'+it) - \zeta(\sigma+it)| \leq c''|\sigma'-\sigma||t|^{\epsilon} \leq c''(\sigma'-1)|t|^{\epsilon}$. With (3) at σ',

$$|\zeta(\sigma+it)| \geq c'(\sigma'-1)^{3/4}|t|^{-\epsilon/4} - c''(\sigma'-1)|t|^{\epsilon}.$$

Choose $A = (c'/(2c''))^4$; since $\sigma' - 1 = A|t|^{-5\epsilon}$, this makes $c'(\sigma'-1)^{3/4}|t|^{-\epsilon/4} = 2c''(\sigma'-1)|t|^{\epsilon}$, hence $|\zeta(\sigma+it)| \geq c''(\sigma'-1)|t|^{\epsilon} = A''|t|^{-4\epsilon}$. Replace 4ϵ by ϵ. ∎

2 Reduction to the functions ψ and ψ_1

Tchebychev introduced a function easier to handle than $\pi(x)$ but carrying the same information:

$$\psi(x) = \sum_{p^m \leq x} \log p,$$

summed over integers of the form $p^m \leq x$ (p prime, $m \geq 1$). Equivalently, with $\Lambda(n) = \log p$ if $n = p^m$ for some prime p and $m \geq 1$, and $\Lambda(n) = 0$ otherwise,

$$\psi(x) = \sum_{1 \leq n \leq x} \Lambda(n), \qquad \psi(x) = \sum_{p \leq x} \left[\frac{\log x}{\log p}\right] \log p,$$

$[u]$ the greatest integer $\leq u$ (since $p^m \leq x$ iff $m \leq \log x/\log p$).

Proposition 2.1

If $\psi(x) \sim x$ as $x \to \infty$, then $\pi(x) \sim x/\log x$ as $x \to \infty$.

Proof. We show $1 \leq \liminf \pi(x)\frac{\log x}{x}$ and $\limsup \pi(x)\frac{\log x}{x} \leq 1$ (4). Crudely, $\psi(x) = \sum_{p\leq x}\left[\frac{\log x}{\log p}\right]\log p \leq \sum_{p\leq x}\frac{\log x}{\log p}\log p = \pi(x)\log x$, so $\frac{\psi(x)}{x} \leq \frac{\pi(x)\log x}{x}$, and $\psi(x) \sim x$ gives the first inequality. For the second, fix $0 < \alpha < 1$:

$$\psi(x) \geq \sum_{p\leq x} \log p \geq \sum_{x^\alpha < p \leq x} \log p \geq (\pi(x) - \pi(x^\alpha))\alpha \log x,$$

so $\psi(x) + \alpha\pi(x^\alpha)\log x \geq \alpha\pi(x)\log x$. Dividing by x, and using $\pi(x^\alpha) \leq x^\alpha$, $\alpha < 1$, $\psi(x) \sim x$, we get $1 \geq \alpha \limsup \pi(x)\frac{\log x}{x}$. As $\alpha < 1$ is arbitrary, the proof is complete. ∎

(The converse also holds.) It is more convenient to work with

$$\psi_1(x) = \int_1^x \psi(u)\,du.$$

Proposition 2.2

If $\psi_1(x) \sim x^2/2$ as $x \to \infty$, then $\psi(x) \sim x$ as $x \to \infty$, and therefore $\pi(x) \sim x/\log x$ as $x \to \infty$.

Proof. By Proposition 2.1 it suffices to show $\psi(x) \sim x$. Since ψ is increasing, for $\alpha < 1 < \beta$,

$$\frac{1}{(1-\alpha)x}\int_{\alpha x}^{x} \psi(u)\,du \le \psi(x) \le \frac{1}{(\beta-1)x}\int_{x}^{\beta x} \psi(u)\,du.$$

So $\psi(x) \le \frac{1}{(\beta-1)x}[\psi_1(\beta x) - \psi_1(x)]$, i.e. $\frac{\psi(x)}{x} \le \frac{1}{\beta-1}\Big[\frac{\psi_1(\beta x)}{(\beta x)^2}\beta^2 - \frac{\psi_1(x)}{x^2}\Big]$, whence

$$\limsup_{x\to\infty}\frac{\psi(x)}{x} \le \frac{1}{\beta-1}\Big[\frac{1}{2}\beta^2 - \frac{1}{2}\Big] = \frac{1}{2}(\beta+1).$$

As $\beta > 1$ is arbitrary, $\limsup \psi(x)/x \le 1$; a similar argument with $\alpha < 1$ gives $\liminf \psi(x)/x \ge 1$. ∎

To relate ψ_1 and ζ: differentiating Lemma 1.3, for $\mathrm{Re}(s) > 1$, $\frac{\zeta'(s)}{\zeta(s)} = -\sum_{m,p}(\log p)p^{-ms} = -\sum_{n=1}^{\infty}\frac{\Lambda(n)}{n^s}$, i.e.

$$-\frac{\zeta'(s)}{\zeta(s)} = \sum_{n=1}^{\infty}\frac{\Lambda(n)}{n^s}. \tag{5}$$

Proposition 2.3

For all $c > 1$,

$$\psi_1(x) = \frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty}\frac{x^{s+1}}{s(s+1)}\Big(-\frac{\zeta'(s)}{\zeta(s)}\Big)\,ds. \tag{6}$$

Lemma 2.4

If $c > 0$, then

$$\frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty}\frac{a^s}{s(s+1)}\,ds = \begin{cases} 0 & \text{if } 0 < a \le 1,\\ 1 - 1/a & \text{if } 1 \le a.\end{cases}$$

Here, the integral is over the vertical line $\mathrm{Re}(s) = c$.

Proof. $|a^s| = a^c$ on the line, so the integral converges. Let $a = e^\beta \ge 1$ and $f(s) = \frac{e^{\beta s}}{s(s+1)}$, with $\mathrm{res}_0 f = 1$, $\mathrm{res}_{-1} f = -1/a$. Let $\Gamma(T)$ be the segment $S(T)$ from $c - iT$ to $c + iT$ plus the half-circle $C(T)$ of radius T about c to its left, positively oriented. For large T, $\frac{1}{2\pi i}\int_{\Gamma(T)} f = 1 - 1/a$, and on $C(T)$, $|s(s+1)| \ge \frac{1}{2}T^2$, $|e^{\beta s}| \le e^{\beta c}$ ($\sigma \le c$), so $\big|\int_{C(T)} f\big| \le \frac{C}{T^2}2\pi T \to 0$. For $0 < a \le 1$ use the right half-circle (no poles; $\beta \le 0$, $\sigma \ge c$). ∎

Proof of Proposition 2.3. $\psi(u) = \sum_{n=1}^{\infty}\Lambda(n)f_n(u)$ with $f_n(u) = 1$ if $n \le u$ and 0 otherwise. Hence

$$\psi_1(x) = \int_0^x \psi(u)\,du = \sum_{n=1}^{\infty}\Lambda(n)\int_0^x f_n(u)\,du = \sum_{n\le x}\Lambda(n)(x-n).$$

By (5) and Lemma 2.4 with $a = x/n$,

$$\frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty}\frac{x^{s+1}}{s(s+1)}\Big(-\frac{\zeta'(s)}{\zeta(s)}\Big)ds = x\sum_{n=1}^{\infty}\Lambda(n)\frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty}\frac{(x/n)^s}{s(s+1)}\,ds = x\sum_{n\le x}\Lambda(n)\Big(1-\frac{n}{x}\Big) = \psi_1(x).$$

∎

2.1 Proof of the asymptotics for ψ_1

We show $\psi_1(x) \sim x^2/2$, completing the proof, from (6), $\zeta(1+it) \neq 0$ and Propositions 2.7, 1.6. Move the line from $\mathrm{Re}(s) = c$ to $\mathrm{Re}(s) = 1$, so x^{s+1} has size x^2; the pole at $s = 1$ gives the main term $x^2/2$, and the rest is smaller.

Fix $c = 2$, $x \geq 2$, and let

$$F(s) = \frac{x^{s+1}}{s(s+1)}\Big(-\frac{\zeta'(s)}{\zeta(s)}\Big).$$

Deform $\mathrm{Re}(s) = c$ to $\gamma(T)$ ($T \geq 3$): $\mathrm{Re}(s) = 1$ for $|t| \geq T$, and around $s = 1$ along $\mathrm{Re}(s) = c$, $|t| \leq T$, joined by horizontal segments at $\pm T$. Then

$$\frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty} F(s)\,ds = \frac{1}{2\pi i}\int_{\gamma(T)} F(s)\,ds. \tag{7}$$

Indeed, by Propositions 2.7 and 1.6, $|\zeta'(s)/\zeta(s)| \leq A_\eta |t|^\eta$ for $\sigma \geq 1$, $|t| \geq 1$, so $|F(s)| \leq A'|t|^{-2+\eta}$ in the two infinite regions between the line and $\gamma(T)$, and Cauchy's theorem gives (7).

Next pass to $\gamma(T,\delta)$: γ_1: $\mathrm{Re}(s) = 1$, $t \leq -T$; γ_2: $t = -T$ from 1 to $1-\delta$; γ_3: $\mathrm{Re}(s) = 1-\delta$, $|t| \leq T$; γ_4: $t = T$ from $1-\delta$ to 1; γ_5: $\mathrm{Re}(s) = 1$, $t \geq T$. For fixed T choose δ so that ζ has no zeros in $\{1-\delta \leq \sigma \leq 1,\ |t| \leq T\}$. By Corollary 2.6, $\zeta(s) = \frac{1}{s-1} + H(s)$, so $-\zeta'/\zeta = \frac{1}{s-1} + h(s)$ with h holomorphic near 1, and $\mathrm{res}_{s=1} F = x^2/2$. Thus

$$\frac{1}{2\pi i}\int_{\gamma(T)} F(s)\,ds = \frac{x^2}{2} + \frac{1}{2\pi i}\int_{\gamma(T,\delta)} F(s)\,ds.$$

Let $\epsilon > 0$ be given. γ_1, γ_5. For $\sigma = 1$, $|x^{1+s}| = x^2$, and by Proposition 1.6 (with Proposition 2.7) $|\zeta'/\zeta| \leq A|t|^{1/2}$, so $\big|\int_{\gamma_5} F\big| \leq Cx^2 \int_T^\infty \frac{|t|^{1/2}}{t^2}\,dt$; the integral converges, so we can choose T with $\big|\int_{\gamma_1} F\big|, \big|\int_{\gamma_5} F\big| \leq \epsilon x^2$. γ_3. With T fixed, $|x^{1+s}| = x^{2-\delta}$ on γ_3, so $\big|\int_{\gamma_3} F\big| \leq C_T x^{2-\delta}$. γ_2, γ_4. $\big|\int_{\gamma_2} F\big| \leq C'_T \int_{1-\delta}^1 x^{1+\sigma}\,d\sigma \leq C'_T \frac{x^2}{\log x}$.

Hence there are constants C_T, C'_T with

$$\Big|\psi_1(x) - \frac{x^2}{2}\Big| \leq \epsilon x^2 + C_T x^{2-\delta} + C'_T \frac{x^2}{\log x}, \qquad \Big|\frac{2\psi_1(x)}{x^2} - 1\Big| \leq 2\epsilon + 2C_T x^{-\delta} + \frac{2C'_T}{\log x},$$

and for all large x, $\big|\frac{2\psi_1(x)}{x^2} - 1\big| \leq 4\epsilon$. So $\psi_1(x) \sim x^2/2$, and the prime number theorem is proved.

Note on interchanging double sums

Proposition

Let $\{a_{k\ell}\}_{1 \leq k,\ell < \infty}$ be complex numbers with

$$\sum_{k=1}^\infty \Big(\sum_{\ell=1}^\infty |a_{k\ell}|\Big) < \infty. \tag{8}$$

Then:
(i) The double sum $A = \sum_{k=1}^\infty \sum_{\ell=1}^\infty a_{k\ell}$ converges, and the order of summation may be interchanged: $A = \sum_{\ell=1}^\infty \sum_{k=1}^\infty a_{k\ell}$.
(ii) Given $\epsilon > 0$, there is N so that for all $K, L > N$, $|A - \sum_{k=1}^K \sum_{\ell=1}^L a_{k\ell}| < \epsilon$.
(iii) If $m \mapsto (k(m), \ell(m))$ is a bijection from $\mathbb{N}$ to $\mathbb{N} \times \mathbb{N}$ and $c_m = a_{k(m)\ell(m)}$, then $A = \sum_{m=1}^\infty c_m$.

Proof. (i) Let $b_k = \sum_\ell |a_{k\ell}|$ and $S = \{x_0, x_1, \dots\}$, $x_n \to x_0$. Define $f_k(x_0) = \sum_{\ell \geq 1} a_{k\ell}$, $f_k(x_n) = \sum_{\ell=1}^{n} a_{k\ell}$, $g = \sum_k f_k$. Each f_k is continuous at x_0 and $|f_k| \leq b_k$, $\sum b_k < \infty$, so g is continuous at x_0. Hence

$$\sum_k \sum_\ell a_{k\ell} = g(x_0) = \lim_n g(x_n) = \lim_n \sum_k \sum_{\ell=1}^{n} a_{k\ell} = \lim_n \sum_{\ell=1}^{n} \sum_k a_{k\ell} = \sum_\ell \sum_k a_{k\ell}.$$

(ii) $|A - \sum_{k\leq K}\sum_{\ell\leq L} a_{k\ell}| \leq \sum_{k\leq K}\sum_{\ell>L}|a_{k\ell}| + \sum_{k>K}\sum_\ell |a_{k\ell}|$. Choose N with $\sum_{k>N} b_k < \epsilon/2$; the first sum is $\leq \sum_{k\leq N}\sum_{\ell>L}|a_{k\ell}| + \sum_{k>N} b_k$, and for L large the finitely many tails $\sum_{\ell>L}|a_{k\ell}|$, $k \leq N$, are small. [Supplement]

(iii) [Supplement]Given ϵ, take N as in (ii) (for the absolute values as well), and M so large that $\{c_1, \dots, c_M\}$ contains all $a_{k\ell}$ with $k, \ell \leq N$. For $M' \geq M$, $\sum_{m\leq M'} c_m - \sum_{k,\ell\leq N} a_{k\ell}$ is a sum of terms $a_{k\ell}$ with $k > N$ or $\ell > N$, of total absolute value $< \epsilon$; so $|\sum_{m\leq M'} c_m - A| < 2\epsilon$. ∎

3 Selected exercises

Motivation. Four exercises from Stein–Shakarchi, Chapter 7 (book numbers kept).

Exercise 7.3 (The Möbius function)

Let $\mu(1) = 1$, $\mu(n) = (-1)^k$ if $n = p_1 \cdots p_k$ with distinct primes p_j, and $\mu(n) = 0$ otherwise.

(a) Prove that $\dfrac{1}{\zeta(s)} = \displaystyle\sum_{n=1}^{\infty} \frac{\mu(n)}{n^s}$ for $\mathrm{Re}(s) > 1$.

(b) Show that $\sum_{k|n} \mu(k)$ equals 1 if $n = 1$ and 0 otherwise.

Solution.

(a) Let $\mathrm{Re}(s) = \sigma > 1$.

Step 1. By Euler's identity the partial products $\prod_{p\leq N}(1 - p^{-s})^{-1}$ tend to $\zeta(s) \neq 0$, so their reciprocals tend to $1/\zeta(s)$:

$$\frac{1}{\zeta(s)} = \lim_{N\to\infty} \prod_{p\leq N} (1 - p^{-s}).$$

Step 2. Each term of the expanded product corresponds to a set S of primes $\leq N$ and equals $\prod_{p\in S}(-p^{-s}) = \mu(n)n^{-s}$, $n = \prod_{p\in S} p$; each square-free n with prime factors $\leq N$ arises once, and $\mu(n) = 0$ otherwise. Hence

$$\prod_{p\leq N} (1 - p^{-s}) = \sum_{n\in\mathcal{A}_N} \frac{\mu(n)}{n^s}, \qquad \mathcal{A}_N = \{n : \text{all prime factors of } n \text{ are} \leq N\}.$$

Step 3. Every $n \leq N$ lies in $\mathcal{A}_N$ and $|\mu(n)| \leq 1$. So

$$\Big|\sum_{n=1}^{\infty} \frac{\mu(n)}{n^s} - \sum_{n\in\mathcal{A}_N} \frac{\mu(n)}{n^s}\Big| \leq \sum_{n>N} n^{-\sigma} \to 0.$$

Combining with Step 1 gives (a).

(b) For $n = 1$ the sum is $\mu(1) = 1$. Let $n = p_1^{a_1} \cdots p_r^{a_r}$ with $r \geq 1$ and all $a_j \geq 1$. The divisors k with $\mu(k) \neq 0$ are the products $\prod_{j\in S} p_j$ over subsets $S \subseteq \{1, \dots, r\}$, and for these $\mu(k) = (-1)^{|S|}$. Hence

$$\sum_{k|n} \mu(k) = \sum_{S\subseteq\{1,\dots,r\}} (-1)^{|S|} = \sum_{j=0}^{r} \binom{r}{j} (-1)^j = (1-1)^r = 0.$$

∎

Exercise 7.5 (The alternating zeta function)

Let $\tilde{\zeta}(s) = \sum_{n=1}^{\infty}(-1)^{n+1}n^{-s} = 1 - 2^{-s} + 3^{-s} - \cdots$.
(a) Prove that the series converges for $\mathrm{Re}(s) > 0$ and defines a holomorphic function there.
(b) Show that $\tilde{\zeta}(s) = (1 - 2^{1-s})\zeta(s)$ for $s > 1$.
(c) Conclude that ζ has no zeros on the segment $0 < \sigma < 1$ of the real axis, and that $\zeta(0) \neq 0$.

Solution.

(a) Let K be a compact subset of $\{\mathrm{Re}(s) > 0\}$, with $|s| \le M$ and $\mathrm{Re}(s) \ge \delta > 0$ on K.

Step 1. Group the terms in pairs. Since $(2k-1)^{-s} - (2k)^{-s} = s\int_{2k-1}^{2k} x^{-s-1}\,dx$,

$$|(2k-1)^{-s} - (2k)^{-s}| \le |s|\,(2k-1)^{-\delta-1} \le M(2k-1)^{-1-\delta} \qquad (s \in K).$$

So the even partial sums $S_{2N}(s) = \sum_{k\le N}[(2k-1)^{-s} - (2k)^{-s}]$ converge uniformly on K.

Step 2. The odd partial sums satisfy $S_{2N+1} - S_{2N} = (2N+1)^{-s}$, which has modulus $\le (2N+1)^{-\delta} \to 0$ uniformly on K. Hence all partial sums converge uniformly on K.

Step 3. The partial sums are entire, so $\tilde{\zeta}$ is holomorphic in $\mathrm{Re}(s) > 0$ by Theorem 5.2.

(b) For $s > 1$ all series converge absolutely and may be rearranged:

$$\zeta(s) - \tilde{\zeta}(s) = 2\sum_{m\ge 1}(2m)^{-s} = 2^{1-s}\zeta(s).$$

(c)

Step 1. Both $\tilde{\zeta}$ and $(1 - 2^{1-s})\zeta(s)$ are holomorphic on the connected set $\{\mathrm{Re}(s) > 0\} \setminus \{1\}$ (Corollary 2.6). They agree on $(1, \infty)$ by (b), so they agree on that whole set (Theorem 4.8).

Step 2. For real $0 < \sigma < 1$, $x \mapsto x^{-\sigma}$ is strictly decreasing, so every bracket in $S_{2N}(\sigma)$ is positive and increasing in N. Hence $\tilde{\zeta}(\sigma) = \lim S_{2N}(\sigma) \ge 1 - 2^{-\sigma} > 0$. Also $1 - 2^{1-\sigma} < 0$. So $\zeta(\sigma) = \tilde{\zeta}(\sigma)/(1 - 2^{1-\sigma}) < 0$; in particular $\zeta(\sigma) \neq 0$.

Step 3. ($s = 0$.) The proof of Theorem 2.3 gives $\xi(s) = \frac{1}{s-1} - \frac{1}{s} + E(s)$ with E entire, so $s\,\xi(s) \to -1$ as $s \to 0$. Also $\frac{s}{2}\Gamma(\frac{s}{2}) = \Gamma(\frac{s}{2} + 1) \to 1$. Writing

$$\zeta(s) = \pi^{s/2}\frac{\xi(s)}{\Gamma(s/2)} = \pi^{s/2} \cdot s\,\xi(s) \cdot \frac{1}{2 \cdot \frac{s}{2}\Gamma(\frac{s}{2})},$$

we get $\zeta(0) = 1 \cdot (-1) \cdot \frac{1}{2} = -\frac{1}{2} \neq 0$. ∎

Exercise 7.7 (Reality of ξ)

Show that $\xi(s) = \pi^{-s/2}\Gamma(s/2)\zeta(s)$ is real when s is real ($s \neq 0, 1$) and when $\mathrm{Re}(s) = 1/2$.

Solution.

Step 1. For real $s > 1$, each of $\pi^{-s/2}$, $\Gamma(s/2) = \int_0^\infty e^{-u}u^{s/2-1}du$ and $\zeta(s) = \sum n^{-s}$ is real, so $\xi(s) \in \mathbb{R}$.

Step 2. $g(s) = \overline{\xi(\overline{s})}$ is holomorphic on $\Omega = \mathbb{C} \setminus \{0, 1\}$ (conjugate power series coefficients, as in Theorem 5.6), and $g = \xi$ on $(1, \infty)$ by Step 1; Theorem 4.8 gives $\xi(\overline{s}) = \overline{\xi(s)}$ on the connected Ω.

Step 3. If $s \in \mathbb{R} \setminus \{0, 1\}$, Step 2 gives $\xi(s) = \overline{\xi(s)}$, so $\xi(s)$ is real.

Step 4. If $\mathrm{Re}(s) = 1/2$, then $\overline{s} = 1 - s$. By Theorem 2.3, $\overline{\xi(s)} = \xi(\overline{s}) = \xi(1 - s) = \xi(s)$, so $\xi(s)$ is real. ∎

Exercise 7.12 (The n-th prime)

Let p_n denote the n-th prime. Use the prime number theorem to show that $p_n \sim n \log n$ as $n \to \infty$.

Solution.

Step 1. For $x \geq 3$ we have $\pi(x) \geq 1$ and $\log\log x > 0$. By the prime number theorem $\pi(x)\log x/x \to 1$, so

$$\log \pi(x) + \log\log x - \log x = \log \frac{\pi(x)\log x}{x} \to 0.$$

Step 2. Dividing by $\log x \to \infty$, and using $\log\log x / \log x \to 0$,

$$\frac{\log \pi(x)}{\log x} = \frac{\log \pi(x) + \log\log x - \log x}{\log x} + 1 - \frac{\log\log x}{\log x} \to 1.$$

So $\log \pi(x) \sim \log x$.

Step 3. Multiplying the two asymptotic relations,

$$\frac{\pi(x)\log \pi(x)}{x} = \frac{\pi(x)\log x}{x} \cdot \frac{\log \pi(x)}{\log x} \to 1.$$

Step 4. Take $x = p_n \to \infty$. Then $\pi(p_n) = n$, so $n \log n / p_n \to 1$, i.e. $p_n \sim n \log n$. ■